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REFERENCE LEVELS AND ACHIEVABLE DOSES IN MEDICAL AND DENTAL IMAGING: RECOMMENDATIONS FOR THE UNITED STATES

National Council on Radiation Protection & Measurements

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Reference Levels and Achievable Doses in Medical and Dental Imaging: Recommendations for the United States

Recommendations of the
NATIONAL COUNCIL ON RADIATION PROTECTION AND MEASUREMENTS

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Preface

The National Council on Radiation Protection and Measurements (NCRP) has a continuing interest in the operational safety and radiation protection of patients and medical staff in radiological diagnostic and therapeutic practices. Since 2000 NCRP issued Report No. 133 on *Radiation Protection for Procedures Performed Outside the Radiology Department* (2000); Report No. 147, *Structural Shielding Design for Medical X-Ray Imaging Facilities* (2004); Report No. 151, *Structural Shielding Design and Evaluation for Megavoltage X- and Gamma-Ray Radiotherapy Facilities* (2005); Report No. 155 on *Management of Radionuclide Therapy Patients* (2006); and Report No. 168 on *Radiation Dose Management for Fluoroscopically-Guided Interventional Medical Procedures* (2010). In addition, NCRP reports have been published on specialized topics related to radiation protection in medical procedures, including Report No. 140 on *Exposure Criteria for Medical Diagnostic Ultrasound: II. Criteria Based on All Known Mechanisms* (2002) and Report No. 145 on *Radiation Protection in Dentistry* (2003).

This Report represents an important addition to the existing series of NCRP reports on radiation safety and health protection in medicine. At the present time there are no consistent guidelines in the United States on the optimal radiation doses for diagnostic medical procedures. Other nations, including those in the European Union, have adopted a consistent set of data for guidance and management of radiation dose in medical imaging procedures. Diagnostic reference levels also should be adopted in the United States. This Report provides dosimetric data in a standardized format and the scientific basis supporting its use.

Diagnostic reference levels and achievable doses are dynamic values changing over time and with changes in technology. It is the responsibility of the professional medical imaging organizations, in cooperation with NCRP, to update these values as appropriate.

This Report is directed at medical imaging physicians, medical physicists, and staff working in medical imaging facilities using ionizing radiation. This Report was prepared by NCRP Scientific Committee (SC) 4-3 on Diagnostic Reference Levels and Achievable Doses, and Reference Levels in Medical Imaging: Recommendations for Applications in the United States. Serving on SC 4-3 were:

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John D. Boice, Jr.
President

Contents

1. Executive Summary

Diagnostic reference levels (DRLs), which are a form of investigation levels (ICRP, 1996), represent an important tool to optimize image quality and the radiation dose delivered to patients. The goal is to manage the dose to the patient to be commensurate with the medical purpose. By surveying the radiation doses associated with imaging examinations throughout the country, DRLs can be established (typically at the 75th percentile of the distribution), based on actual practice patterns. However, the survey data must be robust and representative of the practice (*i.e.*, statistically valid). It should be noted that radiation doses may be either too high or too low with regard to the image quality desired. Too low a dose, for example, may result in an inadequate image. Consequently, it will be necessary to consider both image quality and patient dose since if the image quality does not provide the necessary clinical information the patient has been exposed needlessly to radiation. In fact, the consequence of poor image quality goes beyond the radiation dose to the patient—a false negative diagnosis may lead to a negative impact in terms of patient care.

DRLs provide the first step in the optimization process. However, to encourage optimization for the 75 % of facilities below the DRL, an achievable dose is provided.

Achievable doses represent the median (50th percentile) of the dose distribution which means that 50 % of the facilities are operating below this level already. DRLs and achievable doses have been used by the Health Protection Agency (HPA) (formerly the National Radiological Protection Board) in the United Kingdom for more than 20 y. Over this period of time HPA observed a 55 % reduction in the 75th percentile dose (Hart *et al.*, 2007). This Report from the National Council on Radiation Protection and Measurements (NCRP) includes recommended achievable doses where sufficient data are available.

Image quality and patient dose, both appropriate for the clinical goals, are essential. There is a possibility that facilities with low radiation doses may have inadequate image quality (*e.g.*, noise levels may too high) which could reduce the clinical effectiveness of the examination. Hence, low radiation dose images could be detrimental to patient care. Likewise, facilities with high radiation doses must also ensure that their image quality is appropriate for the clinical task. Optimization in medical imaging is the process of achieving the appropriate balance between clinical image quality (*i.e.*, clinical effectiveness) and patient radiation dose. Optimization is essential to ensure that benefit of the x-ray examination significantly outweighs the potential risk from the radiation exposure. (Note: Clinical effectiveness or utility is of utmost importance but cannot be measured or quantified easily. Consequently, image quality is used as a surrogate for clinical effectiveness.)

Optimization requires a team with expertise in various areas. This team, often called the *clinical dose optimization team* (CDOT), should include imaging physicians (*i.e.*, radiologists, cardiologists, interventional specialists, and orthopedic specialists), a qualified medical physicist, radiographic technologists, as well as staff from other disciplines involved in medical imaging. It is the responsibility of the CDOT to review image quality, patient radiation doses, procedures, and imaging protocols and compare these to published national values. Whenever an institution's patient doses exceed DRLs or the image quality is not appropriate for the clinical examination, optimization is required.

In the United States, the most robust and representative survey data for ionizing radiation doses from medical imaging are provided by the Nationwide Evaluation of X-Ray Trends (NEXT) Program. The NEXT Program, started in the 1970s, is a cooperative effort of the U.S. Food and Drug Administration (FDA) and state radiation control offices through the Conference of Radiation Control Program Directors, Inc. (CRCPD), with additional funding provided by the American College of Radiology (ACR). Selected medical x-ray imaging examinations are evaluated periodically at randomly selected clinical imaging facilities. FDA is responsible for survey design, selection of the facilities, and publication of statistical summaries. The state radiation control program staff conducts the site visits, gathering comprehensive data on patient workloads, equipment inventory and features, and aspects of quality control and quality assurance. Surveyors also make measurements of radiation output from the imaging unit for selected examination using patient-equivalent phantoms and the routine clinical x-ray technique factors used by the facility. A similarly robust repository of comparable human dose data is not available in the United States. Consequently, this Report uses exclusively phantom-based survey data. The associated DRLs for radiography, fluoroscopy, and computed tomography (CT) are based on these data.

The use of phantoms for dose measurement [*e.g.*, incident air kerma for radiography and computed tomography dose index (*CTDI*) for CT] is advantageous for several reasons. Their use:

- eliminates the laborious alternative of collecting large amounts of data from patient examinations;
- standardizes the data collection process for multiple facilities;
- provides a means for facilities to implement their own data collection process for comparison to survey results as part of a dose management program; and
- permits the observation of trends in clinical practice over time as surveys are repeated in response to changes in clinical practice and technology.

Other, more recent survey data for CT doses have been made available through ACR's CT accreditation process. While these data are also based on phantom measurements, they suffer from a possible bias associated with the nature of accreditation programs. Sites seeking accreditation may have higher standards or may artificially lower their doses to comply with the accreditation standards. Thus, measured doses through the accreditation process may reflect temporary or recent adjustments to radiation dose and may not reflect true practice patterns across the United States.

The most recent NEXT survey for CT was performed in 2005 and 2006. The survey was comprised of 267 sites across 31 states. Given the depth of the data that were collected and the relatively limited resources for its analysis, the survey data from 2005 and 2006 are still being analyzed (FDA, 2010a). This Report is based on a subsample of 40 clinical sites randomly selected from the entire survey population.

Reference levels (RLs) are similar to DRLs in concept except that these are provided for other than diagnostic x-ray examinations (*e.g.*, interventional). RLs are derived directly from fluoroscopically-guided interventional (FGI) procedures, thus including factors such as equipment variation, skill of the interventionalist, complexity of the procedure and that of the examination including patient variables.

RLs for FGI procedures pose a unique challenge as compared with DRLs for diagnostic imaging procedures. Although DRLs in this Report are based on measurements with standardized phantoms, RLs for FGI procedures rely on patient-based data. Many factors confound determining RLs including the size (thickness) and clinical condition of the patient, the skill of the operating team, and the equipment used to perform the procedure. In 2003, the Radiation Doses in Interventional Radiology Procedures (RAD-IR) Study documented the radiation doses resulting from various FGI procedures throughout the United States (Miller *et al.*, 2003a;

2003b). Specifically, RLs were determined for 26 separate FGI procedures. This Report also provides guidance as to how RLs and substantial radiation dose levels (SRDLs) might be used in FGI procedures to guide process improvement efforts. As detailed in NCRP Report No. 168 (NCRP, 2010), SRDLs are values below which tissue reactions (deterministic effects) are highly unlikely and above which such injuries are possible.

Nuclear medicine procedures pose an additional challenge for establishing DRLs. Limited nuclear medicine survey data are available for the United States. Consequently, a survey[1] was performed of administered activities for nuclear medicine procedures throughout the United States. A range of administered activities was reported and compared to previously established minimum and maximum values recommended by the International Commission on Radiological Protection (ICRP, 1988). The results of the survey correspond well with ICRP published guidelines.

DRLs and achievable doses, and RLs are dynamic values changing over time and with changes in technology. NCRP examined the U.S. data reported by numerous groups and recognized that these various groups often used slightly different methods to gather and calculate the guidance values which are summarized in this Report. In some cases the data source is a sample of consecutive cases at an institution; at others it is a set of nonconsecutive measurements. Also, in some cases the 75th percentile is calculated using 30 cases, in others it is calculated using far more values. The method for defining procedures also lacks standardization. In other words, the data sources and measurement procedures are not standardized, and the methods are changing rapidly. Consequently, in the next 3 to 5 y it is incumbent on the professional societies, in cooperation with NCRP and others, to review the methodology used in determining guidance levels, standardize these methods wherever possible, select additional imaging examinations for analysis, update reported values, and add new values for the expanded array of procedures. This interval, while arbitrary, reflects a compromise between the pace of innovation and the available resources.

In summary, DRLs and achievable doses, and RLs pose a unique opportunity for medical imaging practitioners in the United States to optimize examination techniques with reductions in radiation dose while maintaining or improving image quality. The DRLs, achievable doses, and RLs are summarized in Tables 6.16, 6.17, and 7.1. Phantom-based survey data for radiography, fluoroscopy,

[1]Bushberg, J.T. (2010). Personal communication (University of California, Davis, Sacramento, California).

and CT were used due to their availability and reproducibility. Patient-based dose data for FGI procedures were used due to the complexity of patient-related factors that influence the accumulative radiation dose for specific FGI procedures. For nuclear medicine, a recently conducted survey of administered activities was performed to document practice patterns within the United States for nuclear medicine imaging procedures. This Report concludes with recommended DRLs and achievable doses for selected radiological imaging examinations.

2. Introduction

2.1 Historical Background

2.1.1 *Diagnostic Reference Levels and Reference Levels*

ICRP introduced the concept of DRLs in Publication 60 (ICRP, 1991), and subsequently recommended their use in Publication No.73 (ICRP, 1996). ICRP Publication No. 73 states:

> "The Commission now recommends the use of diagnostic reference levels for patients. These levels, which are a form of investigation level, apply to an easily measured quantity, usually absorbed dose in air, or in a tissue-equivalent material at the surface of a simple phantom or representative patient. In nuclear medicine, the quantity will usually be the administered activity. In both cases, the diagnostic reference level will be intended for use as a simple test for identifying situations where the levels of patient dose or administered activity are unusually high. If it is found that procedures are consistently causing the relevant diagnostic reference level to be exceeded, there should be a local review of the procedures and the equipment in order to determine whether the protection has been adequately optimized. If not, measures aimed at reduction of the doses should be taken."

Additional advice on DRLs was provided by ICRP (2001; 2007a). ICRP recognized the value of DRLs as a means to investigate and identify situations where the level of patient dose or administered activity is unusually high, relative to benchmark data. These values are easily measured quantities using a simple phantom for radiographic, fluoroscopic, and CT diagnostic imaging.

RLs are similar to DRLs in concept except that RLs are provided for other than diagnostic x-ray examinations (*e.g.*, interventional).

The overall impact of DRLs on medical imaging is primarily to reduce the radiation dose to patients and, secondarily, to healthcare workers. In addition, the goal is to maintain or improve image quality appropriate to the clinical requirements of the procedure through the process of optimization.

If DRLs (either patient- or phantom-based) or RLs are exceeded, "there should be a local review of procedures and equipment in order to determine whether the protection has been adequately optimized" (ICRP, 1991; 1996). However, *DRLs should not be viewed as absolute determinants of appropriate use of medical radiation*. Rather, they are targets to assist clinicians in reducing risks while achieving the notable benefits of imaging procedures. *DRLs are not intended for regulatory or commercial purposes, or to establish legal standards of care*. Dose reduction using DRLs requires practices to pay heed to the RLs and voluntarily adopt strategies to reduce doses below these levels.

The specific imaging examinations to be analyzed, and the values for these examinations that constitute DRLs, achievable doses, and RLs should be selected and reviewed by professional medical organizations, in cooperation with NCRP. It is incumbent upon these groups to update the reported values, and add new ones for an expanded array of procedures every 3 to 5 y. This interval, while arbitrary, reflects a compromise between the pace of innovation and the available resources. Some examinations or procedures tend to result in high patient radiation doses (*e.g.*, perfusion CT examinations) provide a challenge in developing DRLs due to the paucity of data. Interim DRLs can be developed from data available in the literature (Hausleiter *et al.*, 2009). However, limitations in these data should be acknowledged such as small sample size, data from a limited number of institutions, or lack of patient demographics.

When DRLs or RLs are exceeded, the facility's CDOT should investigate, in consultation with the appropriate imaging physician and a qualified medical physicist. Available options to reduce patient dose should be balanced with the image quality necessary for good clinical care, and with case- or patient-specific considerations (*e.g.*, patient size, weight, motion). If higher levels are justified, then the higher levels are deemed acceptable.

In 1997, the European Union (EU) issued its medical dose directive requiring all member states to establish criteria for when imaging examinations should be performed, and DRLs for controlling medical radiation. Within the directive, reference doses were transformed into regulatory requirements for DRLs. The DRLs were defined in terms of a standard sized patient for a specific examination, and there was no suggestion that they should be applied to individual patients. However, the directive also requires that local review be performed when administered radiation doses consistently exceed DRLs. Experience from HPA and the Royal College of Radiologists has been quite positive. HPA initiated the use of DRLs in the mid-1980s. In a 2005 review the authors noted a 55 % decrease in the 75th percentile values in 11 radiographic projections (Hart *et al.*, 2007).

In the United States, an American Association of Physicists in Medicine (AAPM) task group on DRLs reviewed the work and findings of the European Commission (EC); the HPA; the Royal College of Radiologists; the International Atomic Energy Agency (IAEA); CRCPD; NEXT; and state specific data from New Jersey, Michigan, and Pennsylvania. AAPM selected DRLs at the 75th to 80th percentile of relevant survey data with the intent to represent the state-of-the-practice, not the state-of-the-art (Gray *et al.*, 2005). ACR *Technical Standard for Diagnostic Medical Physics Performance Monitoring of Radiographic and Fluoroscopic Equipment*, specifies annual evaluations of radiation dose to patients and comparison of those doses to appropriate recommendations or guidelines by a qualified medical physicist (ACR, 2011a). It is expected that about 25 % of facilities will have dose levels for some procedures that exceed the relevant DRLs; these facilities should review their procedures and dose optimization will likely be needed.

DRLs provide insight into patient radiation doses resulting from situations where variations in the imaging equipment and techniques are the major factors influencing system performance. However, in many cases, one should also consider how these same patient doses might be affected by variations in operator performance, patient factors, and operational environments. Dynamic examinations such as fluoroscopic procedures provide examples where system performance is dependent on the equipment available, imaging protocols, operator skill, patient factors, and operational environments. In such cases, RLs should be used for comparison purposes as these take into account the variability introduced by operator skill, patient factors, etc.

In summary, DRLs, achievable doses, and RLs provide a means for the medical imaging community to compare dose levels across the United States for diagnostic imaging and guidance for interventional procedures. Imaging professionals should ensure that equipment and clinical practice are optimized to provide the necessary diagnostic information with radiation doses that are as low as reasonably achievable (ALARA). As DRLs, achievable doses, and RLs represent the actual state of the collective practice in the United States, practitioners are encouraged to apply these to their respective practices commensurate with their particular clinical imaging needs. This is intended to further reduce radiation dose, respectful of the need to maintain image quality for diagnostic purposes.

2.1.2 *Achievable Doses*

DRLs provide the first step in the optimization process. Facilities in which radiation doses are above the DRL should work to optimize their equipment and techniques to ensure that they are producing quality medical images at appropriate radiation doses. However, DRLs provide little information or incentive for the majority of facilities that are using radiation doses below the DRL for some or all of their examinations.

NRPB (1999) states "For those departments that are already well below reference dose levels, it is considered that the objective should be to reduce doses to those achievable by standard techniques and technologies in widespread use, without compromising adequate image quality. Indeed, in the longer term it should be possible to replace reference doses with achievable doses."

It is usually easy and inexpensive to optimize medical image systems (CRCPD, 1988). Particular attention should be directed to optimum x-ray tube filtration, kilovoltage (CRCPD, 2008), screen-film combination, and digital image receptors. As an example, patient entrance skin doses can be reduced by 25 % by increasing the half-value layer (by adding aluminum to the beam) from 2.3 to 3.2 mm aluminum without a perceptible change in image contrast (Gray *et al.*, 1983). [It should be noted that FDA increased the requirement for the half-value layer for new equipment from 2.3 to 2.9 mm aluminum at 80 kVp (FDA, 2010b)]. A 50 % or greater reduction in patient entrance dose, can be obtained by increasing the kilovoltage, with the appropriate reduction in milliampere-seconds, for an anterior-posterior (AP) lumbosacral spine or abdomen projection from 65 to 75 kVp (Gray *et al.*, 1983), with no perceptible change in image contrast.

Although no formal system for determining achievable doses exists at the present time, the concept of achievable dose is straightforward. If one selected the median dose from the NEXT data as an achievable dose, then 50 % of the facilities are already producing images below that dose. In fact, NRPB found that facilities whose techniques met the EC (1996a) recommendations were able to achieve lower patient doses. These facilities had a mean dose that was similar to the median dose of the entire population of facilities. Consequently, the median dose of the NEXT data should serve as the achievable dose. It should be stressed, however, that clinically appropriate image quality and dose must always be considered. If the dose can be lowered while maintaining adequate clinical image quality, then the dose should be lowered. It is not appropriate to sacrifice clinical image quality in order to lower patient dose. In addition, facilities with patient doses significantly below the achievable dose should consider initiating a review of clinical image quality to determine whether such low doses are being obtained at the sacrifice of image quality and medical benefit to the patient.

Optimization, the process of ensuring clinical effectiveness of the images while achieving the appropriate radiation dose to the patient, is central to both DRLs and achievable doses. Many facilities are producing images with appropriate image quality and doses near or below the achievable dose levels. However, it is essential to ensure that clinical image quality, the surrogate for clinical effectiveness, is maintained, or improved, at these lower patient radiation doses. It is not acceptable to compromise image quality to the point that it may have deleterious effects on clinical effectiveness in order to reduce patient radiation doses.

2.2 Phantom Versus Patient Doses

Imaging equipment manufacturers have slowly adopted the ability to provide dosimetric data on an individual patient basis. The U.S. Federal Performance Standard for fluoroscopic equipment has required since June 2006 that manufacturers provide air-kerma rate and cumulative air-kerma display at the operators' location (FDA, 2009). While DRLs have been based on phantom survey data in the United States, it is tempting to use patient-based dose data for DRLs (as is done in Europe and most of the rest of the world) in the future. However, patient-based data in the United States are not sufficiently robust to allow for adequate DRL determination at this time due to the paucity of data available. It is possible that as patient dose data accumulate in the collective medical imaging databases, phantom-based survey data may be supplemented with patient-based data at some point in the future. For this Report, phantom-based data will provide a robust and reproducible tool for dose and dose-rate assessment. (It should be noted that patient-based dose registries are being developed. For example, the ACR Dose Index Registry currently collects dose information for CT scans but will expand to other modalities and will start collecting data based on patient size.)

Phantom-based DRL data include variability due to the x-ray imaging equipment and facility-to-facility variation in equipment use (*e.g.*, use of different kilovoltage at different facilities for the same projection). Patient-based DRL data are preferred by many countries since they are a function of two major variables (*i.e.*, equipment variability and dose variation due to the imaging techniques selected by the operator under the direction of the imaging physician). Variations in patient doses due to the selection of radiographic techniques can be quite significant. However, when using patient-based data, the variability in patient size adds another level of complexity to evaluating doses for comparison to DRLs.

2.3 Approach for Use of Diagnostic Reference Levels and Achievable Doses, and Reference Levels in the United States

DRLs provide guidance for equipment and protocol performance under controlled circumstances. Ideally, all imaging examinations that use ionizing radiation under the umbrella of DRLs would be included. However, sufficient phantom-based or standardized patient data must exist for DRLs to be applicable and robust.

Achievable doses provide guidance in diagnostic imaging as to the dose levels which are achievable. They are a guide for those facilities meeting the optimization goals of DRLs to further optimize their patient radiation doses. However, it is essential for the optimization process to ensure that clinical image quality appropriate for the imaging examination is maintained, or improved, as patient radiation doses are lowered. It is never acceptable to compromise image quality to the point that it may have deleterious effects on clinical effectiveness in order to reduce patient radiation doses.

During and after FGI procedures, SRDLs provide another guidance tool (NCRP, 2010) since exceeding these values warrants follow-up for possible tissue reactions (deterministic effects).

RLs provide guidance for overall system performance under working conditions. Developing RLs that adequately describe all the possible variations encountered during clinical operations is a laudable goal but it will require defining all the factors which influence radiation use and collecting enough data to construct a large catalog of RLs. At present, RLs have only been proposed for a few types of FGI procedures (Miller *et al.*, 2009).

Once DRLs and achievable doses, and RLs are established, professional guidelines must be developed to assist the practicing community with the local evaluation of clinical dosimetry for proper comparison with these values. This process is critical to the adoption of DRLs and achievable doses, and RLs in clinical practice. Finally, DRLs and achievable doses, and RLs should undergo periodic evaluation to incorporate trends in clinical practice and implement the most current and clinically applicable dosimetric data available. NEXT surveys, performed by the staff of state radiation protection agencies, with the cooperation and support of FDA, CRCPD (Frankfort, Kentucky), and ACR (Reston, Virginia), provide the largest collection of survey data applicable to DRLs and achievable doses in the United States.

2.3.1 *Radiography and Fluoroscopy*

NEXT survey data are available for adult and pediatric radiography. For adults, NEXT data are available for the posterior-anterior

(PA) chest radiograph, the AP abdominal radiograph, the AP lumbosacral spine radiograph, and two dental radiographs (bitewing and lateral cephalometric). For pediatric patients, NEXT data are available for the AP chest radiograph. NEXT data are also available for adult fluoroscopic evaluation of the upper gastrointestinal (GI) tract. Although pediatric fluoroscopic data are not available through NEXT, facilities are encouraged to work with their medical physicist to document available indicators of patient dose (*e.g.*, incident air-kerma rate) and related technical values as a means to generate local benchmark data.

In 2002, digital radiographic imaging, both computed and digital radiography accounted for 32 % of medical imaging (for abdomen and lumbosacral spine) (Moyal, 2006). The proportion of digital imaging will continue to increase in the future. This may impact DRLs in that the incident air kerma for digital modalities is typically slightly higher (~10 % on average) than for screen-film imaging (Moyal, 2006).

The use of DRLs and optimization are even more important with the use of digital imaging. Most digital imaging systems allow for doses over a wide dynamic range with clinically-acceptable image quality. In contrast, screen-film systems have a limited dynamic range with films becoming dark and unusable if the radiation exposure was increased significantly. Technologists trained at different institutions may use techniques which result in significant variations in patient dose. In addition, with digital systems doses may increase over time (Vano *et al.*, 2007). Consequently, it is essential to compare patient doses to DRLs on a periodic basis (*e.g.*, at least once per year) for a wide range of radiographic projections.

Published studies demonstrate differences in patient dose for digital imaging as compared to screen-film imaging. Compagnone *et al.* (2006), Seibert *et al.* (1996), and Spelic *et al.* (2010) demonstrated higher patient doses for computed radiography compared to screen-film systems and lower doses for digital radiography compared with screen-film imaging for PA chest radiographs. Compagnone *et al.* also found similar differences for a number of other radiographic projections. With the exception of chest radiography, insufficient data exist to recommend DRLS that are imaging technology specific. The DRLs recommended in this Report for chest, abdomen and spine radiography are based on data from the randomly selected NEXT data at the time of the survey.

2.3.2 *Computed Tomography*

The ACR CT Accreditation Program began in 2002 and provides a robust collection of survey data of CT practices in the United

States, in addition to the NEXT CT data. ACR data are available for adult head and abdominal CT examinations. In addition to the ACR Accreditation Program, ACR has established a Dose Index Registry which is a source of patient CTDI data (ACR, 2011b). NEXT data are available for adult head, chest, and abdomen-pelvis, and for pediatric head and abdomen-pelvis CT examinations.

It is important to point out that the NEXT survey is a random sample of doses while the ACR data are biased by the sample of participants seeking accreditation. The NEXT data are obtained from 250 to 300 randomly selected facilities in the United States. The ACR data are obtained through their accreditation program which limits the facility data to those practices wishing to obtain accreditation (*i.e.*, facilities potentially interested in patient dose and image-quality optimization).

DRLs and achievable doses are provided in this Report for only five examinations. However, CT scanners contain tens to hundreds of protocols. Consequently, the DRLs and achievable doses should be used to ensure that all protocols result in reasonable patient radiation doses by reviewing the protocols and technique factors used.

2.3.3 *Fluoroscopically-Guided Interventional Procedures*

The development and use of RLs for FGI procedures differ from processes found in other areas of medical imaging.

In the United States, DRLs are based on standardized phantoms. Exceeding these DRLs signals a need to assess the imaging equipment and protocol. This approach has clear advantages since phantom-based DRLs describe a system with relatively few variables. Furthermore, since system evaluation is independent of clinical operations, the periodic checks can be scheduled to avoid interfering with clinical workflow.

Establishing and using RLs for FGI procedures requires a different approach because the system being evaluated is substantially more complex. In interventional imaging, the system consists of the operator, patient, equipment and environment (Bucholz and Duncan, 2008; Duncan and Evens, 2009; Miller *et al*., 2009; Sridhar and Duncan, 2008). Despite the advances in computer-based simulation of FGI procedures (Dawson and Gould, 2007), no phantoms are available that can recreate the variables encountered while performing image-guided procedures. These major variables fall into five categories: team skill, task difficulty, equipment capabilities, patient factors (*i.e.*, size, anatomy, vessel tortuosity), and environmental factors. Ideally, patients are treated by highly-skilled teams who employ the best available equipment and tightly-controlled environments to maximize the probability their particular procedure is successfully accomplished with the least possible amount of ionizing radiation. In the real world, operational environments vary between routine daytime procedures to emergent after-hours cases. In addition, issues with equipment capability and availability can negate the benefit of recruiting even the most highly-skilled personnel. Finally, human factors such as prior experience with a particular type of procedure, equipment setup, or other team members (*e.g.*, the inclusion of residents and fellows at academic institutions) can markedly influence system performance (Duncan and Evens, 2009; Ericsson, 2004; Reagans *et al.*, 2005; Teplitz, 1991). As a result, RLs in FGI procedures must be developed with the overall system in mind.

An understanding of RLs in FGI procedures must include an understanding of task difficulty. Task difficulty reflects procedure type as well as patient-specific variables. This means separate RLs must be developed for different procedures. While it is common to describe procedures in general, descriptive terms such as renal stent, biliary drainage, or transjugular intrahepatic portosystemic shunt creation, local variations in both nomenclature and practice confound the resulting datasets. Fortunately, in the United States, billing records provide a more standardized nomenclature for these procedures: the Current Procedural Terminology (CPT) codes and their indications, the International Statistical Classification of Diseases and Related Health Problems [commonly known as the ICD (ICD-9)] codes. The advantage of using this nomenclature is that hospital and other providers have created reliable and robust methods of capturing these descriptors. In addition, the codes are updated yearly (AMA, 2010). (Note: Updating the codes is also a disadvantage as coding terms and conventions can change periodically.)

2.3.4 *Nuclear Medicine*

Robust data for the administered activities for various nuclear medicine procedures are lacking. Thus, local self assessment, according to Society of Nuclear Medicine (SNM), American Society of Nuclear Cardiology, and ACR guidelines may be necessary. Relevant procedures in the adult include myocardial perfusion scintigraphy, hepatobiliary scintigraphy, positron-emission tomography (PET), and bone scintigraphy. According to NCRP Report No. 160 (NCRP, 2009), PET is the fastest growing type of study in nuclear medicine. In children, relevant procedures include bone scintigraphy and renal scintigraphy. Some data are available from a survey carried out in 2010, with limited results provided in this Report.[2]

[2]Bushberg, J.T. (2010). Personal communication (University of California, Davis, Sacramento, California).

2.3.5 *Types of Examinations Excluded*

Diagnostic x-ray projections or examinations that are obtained in certain clinical practices will not be addressed in this Report.

CT-guided invasive and interventional procedures, and cardiac procedures, both diagnostic and therapeutic; have not been included in this Report due to the lack of sufficiently robust data on radiation doses in the United States.

Since 1992, the dose per image for screening mammography has been regulated by the Mammography Quality Standards Act (MQSA, 1992) at 3 mGy for a craniocaudal view of the breast, simulated with a standard breast phantom. The regulatory process precludes wide variability in mammography doses.

Chiropractic and podiatric examinations are not specifically mentioned in this text. However, the DRLs in this Report apply equally to all medical imaging facilities, including mobile-based imaging providers and chiropractic facilities. (The NEXT lumbosacral spine radiography survey includes radiation doses for a sample of chiropractic sites.) Radiation doses and image quality should be optimized using the same approaches for all medical imaging facilities.

Cone-beam computed tomography (CBCT) is a new technology with limited dissemination in the U.S. healthcare system. As such, limited experience and data are available for establishing DRLs for this technology. Consequently, no DRLs are provided in this Report for CBCT.

2.4 Implementing Diagnostic Reference Levels and Achievable Doses, and Reference Levels in Clinical Practice

DRLs and achievable doses, and RLs provided in this Report are meant to provide feedback to the medical imaging team at each institution. This team includes the imaging physician, qualified medical physicist, and radiologic technologist. By periodically comparing their results to these DRLs and achievable doses, and RLs the team can learn how their doses compare to the available data from multi-institutional studies. Although the data sets used to create these DRLs and achievable doses, and RLs are imperfect, such comparisons are still informative. Exceeding the DRL should trigger an investigation into equipment or protocol (technique) factors to determine the cause of the higher doses and to optimize all aspects of the examination. Consistently exceeding the RLs for FGI procedures [*e.g.*, exceeding the reference level (RL) three consecutive times] should also trigger an investigation that evaluates what combination of equipment, protocols, personnel, patient or environmental factors caused the observed results. Once causes are identified and addressed, the team should continually reevaluate system performance to confirm that results fall below the DRL and achievable dose or RL. In this manner, DRLs and achievable doses or RLs serve as signals or "trigger tools" to indicate when and where process improvement efforts are needed (Carey and Lloyd, 2001; Carey and Stake, 2003).

Most providers of medical imaging services are aware of the ALARA principle, which states that radiation doses should *be as low as reasonably achievable, economic and social factors being taken into account*. Though traditionally applied to occupational doses, it is appropriate to apply this principle to patient doses, as well. It is essential to keep in mind that medical imaging studies are performed to affect patient care. Hence, a medical imaging procedure performed at lower dose is only "reasonable" if it answers the clinical question. In other words, a lower dose procedure that is inadequate to answer the clinical question delivers radiation dose to the patient without the requisite benefit and is fundamentally "not reasonable."

The process of self-assessment must be supported by a high-level institutional commitment to quality medical imaging and the appropriate delivery of radiation dose to patients needed to support the clinical management of each patient. The institutional commitment must include allocation of the necessary resources to accomplish these tasks. Essential resources include time for personnel to devote to the process, and time on imaging systems to test potential dose-reducing methods, where needed. Budgetary allocations may be needed to pay for services not performed by staff, or for replacement clinical coverage while staff members devote time to the self-assessment.

Armed with an institutional commitment, the team should be selected and charged with this important responsibility. Some facilities have formed *image quality – dose optimization* teams, so named to emphasize the essential balance to be maintained through the process. Others simply use the term *radiation dose optimization*, recognizing that this entails both dose and image-quality objectives. This Report uses the term *clinical dose optimization team* (CDOT).

CDOT must include the appropriate mix of expertise, including at a minimum: imaging physicians, medical imaging physicists, senior radiologic technologists, and imaging facilities management. Physicians from relevant specialty areas may include radiologists, cardiologists, dentists, neurovascular or interventional specialists, orthopedic specialists, emergency physicians, and other medical specialists who order and interpret medical images produced using ionizing radiation. Additional contributions from nursing and other disciplines may also be valuable, depending on the complexity of the institution and level of patient care provided.

CDOT should develop and implement a plan to assess the appropriateness of radiation doses delivered to patients for medical imaging. CDOT should develop priorities appropriate to the facility's needs, with initial attention directed to the studies most commonly performed and those studies that deliver the most significant doses to patients.

With the proliferation of computer-based imaging systems, traditional technique charts have largely been replaced by anatomic or procedural protocols that are available at the operator's console. It remains good practice for the properly trained operator to have the flexibility to alter protocol parameters to best image a particular patient. For example, the abdomen of larger patients should not be imaged with the same protocols used for small pediatric patients. However, the *default* protocol values should only be changed with appropriate review and approval. CDOT should investigate the status of default protocols over the range of imaging modalities. Where software capability allows, default protocols should be "locked down," requiring management approval for changes. Where software security is not available, institutional policy should require management approval before alteration of default protocols.

As CDOT reviews protocols, doses, and image quality, it is important to remember that dose reduction may not be the end result of optimization. Since achieving the right level of image quality is essential to the clinical goal of the examination, it is possible that doses from some examinations may increase as a result of the self-assessment. For example, through the decade of the 1990s, screen-film mammography doses (mean glandular doses) actually increased modestly (FDA, 2010c). Breast imaging professionals expressed the need for higher contrast mammograms and screen-film manufacturers responded with a new generation of mammography film that had higher contrast in the higher dose (darker) regions of the film. To take advantage of the increased contrast, x-ray doses were increased, resulting in improved image quality and increased mean glandular doses. The result was improved mammographic image quality providing the potential for earlier detection of breast cancer.

The process of implementing DRLs will be facility dependent. CDOT is charged with examination-specific assessment of doses and image-quality needs, comparison with relevant DRLs, suggestion of possible protocol alternatives, pilot testing of protocol alternatives, and iteration of the process until an optimized protocol is determined. CDOT should also periodically (*e.g.*, at least annually) review protocols and practices to ensure that protocols and techniques have not inadvertently changed periodically. Protocols for new equipment should be initially assessed before the equipment is used for patient examinations, and reassessed after a longer (*i.e.*, at three to six months) experience has been obtained. Finally, it is essential to communicate the results of the efforts of CDOT to imaging staff, to explain the process and benefits of change, and to encourage staff to raise concerns for specific protocols that may appear suboptimal. Facility management may also wish to consider how to communicate the process and results to referring physicians, patients, and local news media. Patients are increasingly aware of radiation dose from medical imaging procedures, and it may be beneficial to communicate this process to others who may ask "How do I know if I am getting too much radiation from this procedure?"

2.5 Future Methodology

NCRP Report No. 168 (NCRP, 2010) outlines the basics of statistical process control. These techniques are being increasingly applied to improve system performance in healthcare (Carey and Lloyd, 2001; Carey and Stake, 2003). The statistical process control hypothesizes that the observed variation in any process reflects statistical noise as well as causal factors (Deming, 2000). Other techniques, such as creating control charts where data are plotted as a time series, can be used to help identify causal and random variation. The statistical process control methodology will be used extensively in the future of optimization of image quality and patient radiation doses in medical imaging.

3. Phantoms for Determination of Diagnostic Reference Levels

Physical phantoms provide a convenient means of capturing dose data from a x-ray imaging system while ensuring that the data are representative of clinical applications. The development of radiographic phantoms dates back to the very early 1900s, and a comprehensive review can be found in the literature (ICRU, 1989; 1992a). The earliest phantoms were primarily directed at therapeutic dosimetry applications. Phantoms developed for the evaluation of medical x-ray imaging equipment became available in the mid-1900s. It is desirable that phantoms represent, as closely as possible, the x-ray attenuation characteristics of the anatomical structures they model. It is particularly critical in the practice of radiation therapy that appropriate phantom materials be used in order to ensure accurate calibration and dosimetry. The primary goal in medical imaging is to produce images of high clinical value using an optimized radiation dose to the patient.

Phantoms can be employed to estimate patient dose for a class of patients (*e.g.*, average-sized adults). A phantom also may be used to characterize image quality. Consequently, phantoms that are used to evaluate medical x-ray imaging equipment fall into one or both of two classes: dosimetry and image-quality assessment.

Dosimetry phantoms are designed to approximate the attenuating and scattering properties of humans specific to the imaging modality being investigated. Phantoms can be constructed of readily available materials including polymethyl-methacrylate (PMMA) [also known as acrylic, Lucite® (Lucite International, Inc., Cordova, Tennessee), or Plexiglas® (Altuglas International Arkema Inc., Bristol, Pennsylvania)], aluminum, and copper. The use of these materials also renders the phantoms simple to manufacture. However, depending on the modality, it is often convenient to work with other materials as well. While it is important to select materials which provide attenuation properties similar to real patients, the design criteria must also consider the particular modality being tested as well as the dosimetric endpoint. CT, for example, presents a very different beam-patient geometry than a simple radiographic projection.

Anthropomorphic phantoms are available that closely represent a variety of tissues and structures, permitting accurate dosimetry. If one wishes merely to estimate the average patient dose, then a simpler phantom is appropriate.

The use of standardized phantoms permits institutional tracking of patient doses from room to room and over time. Their use also permits direct comparison of local dose measurements with available multi-site data sets collected on the same type of phantom. ACR promoted the use of a standardized mammography phantom through its mammography accreditation program beginning in the late 1980s (Bassett *et al.*, 1993). The NEXT survey captures dosimetric data nationwide using standardized phantoms (CRCPD, 2010). Phantom standardization, including choice of phantom and the associated protocol for data collection, allows the acquisition of consistent data throughout the United States.

Phantoms also provide a platform for the assessment of diagnostic image quality under conditions that are typical for the modality. If the primary task is to determine an index for patient dose then a patient-equivalent phantom should be used in order to drive automatic exposure controlled equipment to radiation output levels that are patient-representative. Newer CT systems are also capable of adjusting radiation output depending on patient size, however, the standard phantoms presently used for estimating CT doses (*e.g.*, values for CTDI) (uniform cylindrical phantoms with diameters of 16 and 32 cm) are not patient equivalent. While such phantoms are often constructed for this specific purpose, image-quality test objects can also be incorporated into dosimetry phantoms to determine patient dose and evaluate image quality at the same time. The ACR mammography accreditation phantom incorporates image-quality test objects, simulating fibers, specks and masses (ACR, 1999). The benefit of such phantoms is that image quality can then be assessed under simulated clinical conditions (provided that the phantom is appropriate for the dosimetry task).

3.1 Phantoms for Radiographic Projections of the Chest, Abdomen, and Lumbosacral Spine

The quantity of dosimetric interest for DRLs in projection radiography is patient incident air kerma ($K_{a,i}$) free-in-air. This dosimetric quantity should be evaluated under the guidance of a qualified medical physicist. Patient-equivalent phantoms, for specific patient sizes, should be used to simulate tissue absorption when evaluating automatic exposure control driven x-ray equipment. The x-ray gantry and patient support device (radiographic table and Bucky) should be configured as they would be for clinical examinations. The setup geometry should be fully documented to permit any computations needed to determine the incident air kerma from the measurements. Beam quality (half-value layer) also should be determined for the estimation of effective dose, if desired.

The American National Standards Institute (ANSI) published a standard for performing sensitometry of medical film processors (ANSI, 1982). The standard included the specification for a radiographic phantom, referred to as the *basic test object*. This ANSI standard phantom was designed to provide x-ray attenuation similar to that of an anthropomorphic phantom. It consists of layers of PMMA and sheets of aluminum and could be modified for testing a variety of radiographic projections including chest (including an air gap), skull, pelvis, and extremities. Although the primary goal of this ANSI standard was to provide a means for characterizing the speed and contrast of screen-film-based radiographic systems, the phantom provided an economical means of testing radiographic systems under clinical conditions.

In the early 1980s the Center for Devices and Radiological Health (CDRH) developed phantoms similar to the ANSI phantoms in response to the increased use of automatic exposure control features on radiographic equipment. The "LucAl," or CDRH, chest phantom, composed of Lucite® (PMMA) and aluminum, was clinically validated in multi-site testing to provide attenuation equivalent to an adult chest. This phantom simulates an adult with a mass of ~74 kg and height of ~1.7 m, and a 23 cm PA chest dimension (Conway *et al.*, 1984). AAPM published Report No. 31, *Standardized Methods for Measuring Diagnostic X-Ray Exposures* (Chu *et al.*, 2005) which summarizes the radiographic properties of several commercially-available dosimetry phantoms for selected diagnostic x-ray projections. The ANSI standard phantom as configured for chest radiography was found to result in estimates for patient entrance air kerma (PA projection) that are 33 % higher than comparable measurements obtained using the CDRH chest phantom (Chu *et al.*, 2005).

Similar phantoms are available for diagnostic projections of the AP abdomen and lumbosacral spine. The ANSI standard abdomen phantom is a modified version of the ANSI standard chest phantom with the inclusion of an additional 50 mm of PMMA in place of the air gap. The CDRH phantom is a combined abdomen and lumbosacral spine phantom and has been clinically validated to represent a standard adult patient with an ~22 cm AP abdomen dimension (Conway *et al.*, 1990). The geometry for the CDRH abdomen and lumbosacral spine phantom is shown in Figure 3.1, including the large PMMA block and the strip on top of the phantom to simulate the increased absorption of the spine. AAPM Report No. 31 modified the standard ANSI abdomen configuration by adding a narrow 7 mm thick aluminum strip to model the spine (Chu *et al.*, 2005). This modified ANSI phantom was found to result in doses that were 15 % lower than those for the CDRH phantom (Chu *et al.*, 2005).

FDA developed a phantom for the survey of pediatric chest radiography which was used for the 1998 NEXT survey (Moyal, 2004). The survey found that this radiographic projection was primarily conducted using a manual x-ray technique rather than relying on automatic exposure control. The use of manual x-ray techniques for pediatric chest radiography precluded a rigorous validation of the phantom at clinical test sites. The CDRH pediatric phantom can be used for the evaluation of chest radiography for patients between the ages of 9 and 15 months. Presently the CDRH pediatric chest phantom is not commercially available.

3.2 Phantoms for Fluoroscopic Imaging

Fluoroscopy is a dynamic radiological imaging modality used to evaluate mobile structures, visualize contrast media, guide image recording (*e.g.*, GI examinations), or to guide interventional procedures. It involves the delivery of radiation to the patient over longer periods of time than are typical for projection radiography. A complete patient examination typically consists of both fluoroscopic imaging and one or more sequences of recorded images. Dosimetry is a complex task because each patient examination is unique regarding the distribution and total quantity of radiation delivered. The rate at which the radiation dose is delivered to the patient can be measured. Measurements of incident air-kerma rate are referenced to a location that approximates the skin entrance plane (*e.g.*, 1 cm above the table top for fixed-tube, under-table fluoroscopic systems). Contributions to patient dose from scout or overhead radiographs, and fluorographic (spot) image acquisitions should also be included in patient dosimetry for these procedures.

Contrast material is frequently used in fluoroscopy. It provides increased attenuation compared with surrounding soft tissue, with a corresponding increase in dose rate by the fluoroscopic system. During a GI study, for example, barium contrast material will drive the output of the fluoroscopic system to higher dose rates, often including an increase in kilovoltage (CRCPD, 2009). Dosimetry measurements should include the impact of contrast material on the patient radiation dose. The phantom includes a 1.6 mm copper sheet to simulate the presence of barium sulfate.

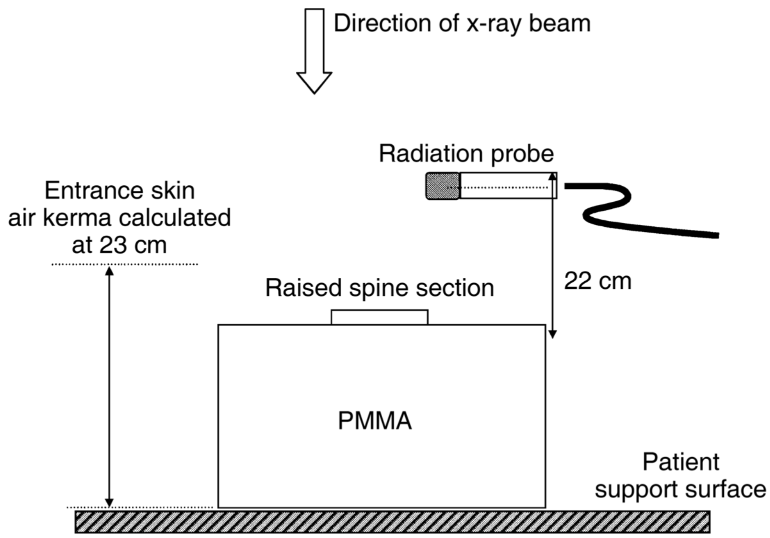

Fig. 3.1. Illustration of measurement geometry for determination of incident air kerma using the CDRH abdomen and lumbosacral spine phantom.

A dosimetry phantom is usually employed to drive the fluoroscopic system to radiation output rates that are typical for clinical examinations. The automatic dose rate control attempts to deliver a constant dose rate to the image receptor (image intensifier or flat-panel receptor). Dosimetry measurements acquired with a phantom should be reproducible. Consequently, an uniform phantom is recommended in order to avoid the dependence of the data on the positioning of the phantom with respect to the fluoroscopic gantry. For general testing of fluoroscopic systems, simple phantoms can be employed, such as copper or aluminum sheets. Such simple phantoms will generally not provide patient-equivalent attenuation across the range of beam energies encountered clinically. Patient-equivalent phantoms are recommended for proper characterization of patient dose during fluoroscopic examinations.

The Medical Imaging and Technology Alliance, a division of the National Electrical Manufacturers Association, developed a cardiovascular benchmarking phantom for fluoroscopy that can be configured representing several patient thicknesses, and includes sets of low-contrast and detail image-quality test objects. CDRH developed a fluoroscopy phantom that is based on the spine configuration of the CDRH combined abdomen and lumbosacral spine phantom (Suleiman *et al.*, 1997). This phantom was originally developed to evaluate standard fluoroscopic equipment used for upper GI studies. The CDRH phantom can be used to characterize both fluoroscopic and fluorographic (spot image) dose to the patient. Similar to projection radiography, the fluoroscopic and fluorographic half-value layer should be determined as needed for dosimetric calculations.

Table 3.1 summarizes the radiography and fluoroscopy phantoms discussed in this Report. It is recommended that facilities adhere, as closely as possible, to the measurement procedures and conditions under which the reference data were gathered. The phantoms discussed in the preceding sections can be used to characterize patient doses for radiographic projections of adult chest, abdomen, and lumbosacral spine, and general fluoroscopy. The facility staff should consult with a qualified medical physicist prior to collecting data to ensure that the survey provides sufficient detail to estimate radiation doses. In addition, staff should ensure that the equipment is configured for both the specific projections and patient sizes, and appropriate techniques (kilovolt peak, milliamperage, etc.) are used.

3.3 Phantoms for Computed Tomography

The dosimetric quantity of interest for computed tomography (CT) is the volumetric CTDI ($CTDI_{\text{vol}}$). Measurements are made using the standard CT phantoms which are cylindrical phantoms composed of PMMA with a 16 cm diameter (for adult and pediatric head examinations, and pediatric body examinations) or 32 cm diameter (for adult body examinations) (Figure 3.2). (These phantoms were not designed as patient-equivalent phantoms but to determine compliance with federal performance standards.) Many

TABLE 3.1—*Summary of radiographic and fluoroscopic phantoms.*

Phantom	Projection and Modality
Modified ANSI phantom	Radiographic chest, abdomen, lumbosacral spine (skull and extremities configuration also available)
CDRH chest phantoms: adult (PA) and pediatric (PA or AP)	Radiographic chest (adult and pediatric)
CDRH abdomen and lumbosacral spine phantom	Adult radiographic abdomen (AP) and lumbosacral spine (AP)
CDRH fluoroscopic phantom	Upper GI fluoroscopy (adult)

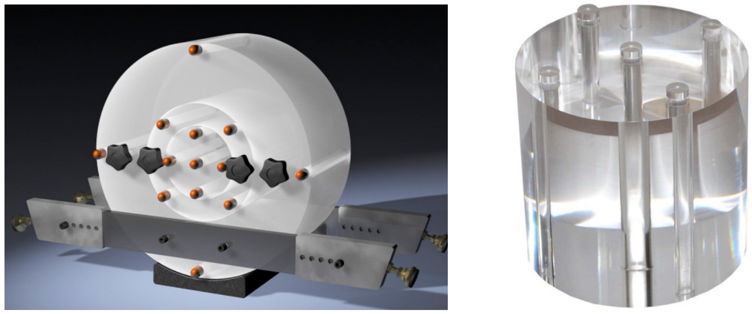

Fig. 3.2. (Left) CT dosimetry phantom. Black knobs are handles to make moving and aligning the phantom easier. Orange objects are PMMA rods to fill holes when ionization chamber is not in position (photo courtesy of CIRS, Inc., Norfolk, Virginia). (Right) close-up of 16 cm diameter PMMA head phantom.

CT systems have the ability to modulate the dose based on patient attenuation as the x-ray source rotates around the patient (*i.e.*, automatic exposure control). CT dosimetry continues to be based on the standard 16 and 32 cm diameter phantoms. However, these phantoms are not equivalent to a typical patient nor were they designed to evaluate automatic exposure control systems available today. Therefore, these phantoms should be used with a manually selected constant x-ray output. The technique factors (particularly the x-ray tube current) used for imaging a CT dosimetry phantom are typically those used for an average or standard patient. Thus, current DRLs for CT and the measurements for comparison with these DRLs are not entirely phantom-based, but are also affected by the selection of these technique factors from patient imaging examinations.

3.4 Phantoms for Fluoroscopic Equipment

The U.S. Code of Federal Regulations performance standard for fluoroscopic equipment specifies an upper limit for fluoroscopic air-kerma rate for prescribed imaging equipment geometry (FDA, 2009). It should be kept in mind that compliance of fluoroscopic equipment with this federal performance standard does not imply that equipment is optimized with regard to dose and image quality. In addition, FDA regulations apply only at the time of equipment purchase and installation, and place no requirement on the users of the equipment to ensure that the equipment continues to comply with federal regulations. However, there are requirements for equipment quality control, evaluation, and calibration in many states, but vary significantly from state to state.

Proper assessment of dose and image quality is the key to fluoroscopy system optimization. Fluoroscopic equipment manufactured after June 2006 must display, at the operator's location, the air-kerma rate and the cumulative air kerma to a specified reference point (FDA, 2009). The International Electrotechnical Commission provides a similar standard for fluoroscopic equipment and includes specification for the display of cumulative air-kerma area product (IEC, 2010). The dynamic display of doses can assist clinical staff in optimizing the practice of fluoroscopy. Cumulative doses can assist the medical physics community in assessing and managing patient dose. Most fluoroscopic equipment provides a display of cumulative fluoroscopy time for a complete examination. It is recommended that the fluoroscopic time not be the sole means of quantifying patient risk (Miller *et al*., 2012; NCRP, 2010; Stecker *et al*., 2009). Appropriate phantoms, such as those discussed in this section, should be used for proper comparison with recommended DRLs in fluoroscopy (for diagnostic, not interventional, procedures).

3.5 Phantoms for Fluoroscopically-Guided Interventional Procedures

State-of-the-art fluoroscopic equipment for fluoroscopically-guided interventional (FGI) procedures incorporates sophisticated technology for automatically adjusting the radiation output of the system based on both the attenuation characteristics of the patient and the clinical task. Such systems employ a C-arm that holds the image receptor and x-ray tube for complex angulations of the x-ray beam during procedures. A complete interventional procedure includes a combination of fluoroscopic imaging and digital recording of images at higher dose rates than used in fluoroscopy. While fluoroscopic systems must comply with state regulations for maximum radiation output rate when operated in available fluoroscopic modes, there is no radiation rate limit when the fluoroscopic equipment is operated in a digital acquisition mode (*i.e*., recording or cine mode). Consequently, it is important that contributions to patient dose from both fluoroscopic and digital acquisition modes of operation be determined.

The range of patient doses can be very large, even with respect to a single procedure (*e.g*., diagnostic coronary angiography) (Miller *et al*., 2009). Factors that contribute to this large range of patient doses include patient size, equipment performance, technique factors selected by the users, complexity of the clinical task, and clinician skill and experience.

The CDRH fluoroscopy phantom may also be used to assess dose rates for fluoroscopic equipment for interventional procedures

(CRCPD, 2009). Most fluoroscopic equipment manufacturers incorporate additional beam filtration that is inserted automatically into the beam during clinical procedures based on patient attenuation. Phantoms used for characterizing patient radiation doses should be patient-equivalent in order to ensure that such automated systems select realistic, clinical techniques for dose measurements.

4. Methods for Characterizing Patient Dose

Two approaches exist for the characterization of patient dose from radiological procedures: patient-based dosimetry and phantom-based dosimetry. Which method provides an outcome most representative of routine clinical practice partly depends on the imaging modality and dosimetric endpoint. For instance, in routine chest radiography, the measurement of incident air kerma using a patient-equivalent phantom is typically sufficient. For FGI procedures a primary clinical concern is the potential for tissue injury (deterministic skin injury) (FDA, 1994; Miller *et al.*, 2010). The determination of peak skin dose for individual patients is therefore an important dose measurement. A number of methodologies for estimating dose to the skin have been developed. One method is the direct measurement of skin dose (including backscattered radiation) using small dosimeters placed directly on the patient skin. These dosimeters are typically either thermoluminescent dosimeters (Shrimpton, 1994) or optically-stimulated luminescent dosimeters (Botter-Jensen *et al.*, 2003; Landauer, 2012). Radiochromic dosimetry film is also used to provide a two-dimensional mapping of dose distribution for individual patients. Most fluoroscopic equipment is configured with the display of rate and cumulative values for one or more dose indices including cumulative air kerma at a specified reference point ($K_{a,r}$) (required on all fluoroscopic equipment manufactured after June 2006), and cumulative air-kerma area product (P_{KA}). Electronic dose displays should be monitored periodically during fluoroscopic procedures as a means to manage patient dose.

Doses to individual patients are affected by several variables. For example, the dose for a specific radiographic projection will vary by factors of 10 or more depending on the patient size. In addition, radiologic technologists may select different radiographic technique factors, namely peak tube voltage, or beam quality, which can result in a factor of 10 or more difference in patient dose. Digital radiography also provides the opportunity for the technologist to select a range of techniques that affect patient doses, with little obvious impact on the image quality as seen by the imaging physician. Lower doses for digital radiography can result in noise in the image that may be apparent to the imaging physician. Conversely, the use of higher digital radiography doses may not be apparent to the imaging physician.

Phantom-based dosimetry is used to make measurements of a representative patient (*i.e.*, an average size patient). Phantom measurements are typically straightforward to make since relatively few measurements are required on x-ray equipment and can be made without interfering with patient examinations. These measurements eliminate several variables from the measurement process including patient size and the variability in dose introduced by individual radiologic technologists in selecting technique factors. Automatic exposure control systems in fluoroscopic x-ray imaging determine the radiation dose rate based on the attenuation of the patient, by controlling the peak tube voltage, tube current, filtration, or x-ray pulse width. Automatic exposure control systems in radiographic x-ray imaging determine the tube current-exposure time product based on the attenuation of the patient, for a preset peak tube potential. Consequently, a patient-representative phantom should be used to determine system response. The major advantages of this approach are the ability to collect dose data quickly for a representative patient, and to characterize doses across multiple facilities and rooms under standardized measurement conditions. In addition, image quality can be determined under simulated clinical conditions. The disadvantages include the initial effort required to develop such phantoms (or alternatively to fund their acquisition) and providing these phantoms for multi-site studies.

4.1 Units of Measure

4.1.1 *Fundamental Units*

The International Commission on Radiation Units and Measurements (ICRU) published recommendations in Report 60 (ICRU, 1998) and Report 85 (ICRU, 2011) for quantities and units of measurement for the characterization of ionizing radiation. ICRU Report 74 (ICRU, 2005) further discusses appropriate quantities and units pertaining to x-ray-based medical imaging. Two quantities, exposure and kerma, are commonly used to specify the quantity of radiation at a point of measurement. While these two quantities are often used interchangeably, they refer to fundamentally different processes involving the interaction of ionizing radiation with matter. Exposure refers to the quantity of electric charge of one polarity (negative or positive) produced in a volume of air when all liberated charge-carrying particles are stopped in air (ICRU, 1998). Exposure is a measure of the amount of ionization caused by the radiation and is defined as (ICRU, 2011):

$$X = \frac{\mathrm{d}q}{\mathrm{d}m}, \tag{4.1}$$

where dq is the absolute value of the mean total charge of ions of a single polarity produced when all the electrons and positrons liberated or created by photons incident on a mass dm of dry air are completely stopped in dry air.

The SI unit for exposure is coulomb per kilogram (C kg^{-1}). Previously, the special unit for exposure was roentgen (1 R = 2.58 × 10^{-4} C kg^{-1}). Dosimeters that provide a measurement of exposure usually employ an air-filled ionization chamber that is calibrated to a recognized standard (Johns and Cunningham, 1983; NIST, 2010).

The kinetic energy released per mass, or kerma, concerns the initial transfer of energy to matter. Kerma is more closely related to absorbed dose because it characterizes the initial transfer of energy by the radiation beam into the medium of interest. Kerma (K) is defined as (ICRU, 2011):

$$K = \frac{\mathrm{dE_{tr}}}{\mathrm{d}m}, \tag{4.2}$$

where $\mathrm{dE_{tr}}$ is the mean sum of the initial kinetic energies of all charged particles liberated in the element of mass dm. The SI unit of kerma is joule per kilogram (J kg^{-1}) with the special name gray (Gy) (1 Gy = 1 J kg^{-1}). Kerma can be quoted for any specified material at a point in free space or in an absorbing medium (*e.g.*, air kerma).

For medical imaging applications, air kerma is used. ICRU Report 74 (ICRU, 2005) expresses air kerma for monoenergetic x rays in the diagnostic energy range as:

$$K_\mathrm{a} = \Psi\left(\frac{\mu_\mathrm{tr}}{\rho}\right)_\mathrm{a}, \tag{4.3}$$

where Ψ is the x-ray beam energy fluence, and $(\mu_\mathrm{tr}/\rho)_\mathrm{a}$ is the mass energy transfer coefficient for air. If the beam contains a spectrum of x-ray energies, then air kerma should be determined as the sum of the contributions from each energy. The SI unit for air kerma is also J kg^{-1}, with the special name gray. Air kerma can be determined approximately (ICRU, 1992b) from the measurement of exposure using the relation:

$$K_a = 0.00876\ X. \tag{4.4}$$

Where K_a is air kerma free-in-air in gray and X is exposure in roentgen. This simple relationship holds for x-ray energies in the diagnostic range. Above these energies, exposure is not easily measured and the conversion factor becomes energy dependent (ICRU, 1992b).

Absorbed dose (D) characterizes the energy deposited in a medium by ionizing radiation. ICRU Report 60 (ICRU, 1998) defines absorbed dose as:

$$D = \frac{\mathrm{d}\bar{\varepsilon}}{\mathrm{d}m}, \tag{4.5}$$

where $\mathrm{d}\bar{\varepsilon}$ is the mean energy imparted by ionizing radiation to matter of mass $\mathrm{d}m$. The SI unit for absorbed dose is J kg^{-1} with the special name gray.

4.1.2 *Clinical Dosimetry*

For patient dosimetry, measurements and patient doses are often expressed in the United States in air kerma. The notation of ICRU Report 74 (ICRU, 2005) is used in this Report. Specifically, incident air kerma ($K_{a,i}$) designates the air-kerma incident on the skin-entrance plane of the patient or phantom without backscattered radiation. Entrance-surface air kerma ($K_{a,e}$) designates the air kerma at the point where the x-ray beam enters the patient or phantom and includes backscattered radiation.

For a discussion of the topic of uncertainty in diagnostic radiation dosimetry the reader is referred to the IAEA publication entitled *Dosimetry in Diagnostic Radiology: An International Code of Practice* (IAEA, 2007).

Backscattered radiation contributes to both patient dose and to field measurements of dose in diagnostic x-ray imaging. The inclusion of backscattered radiation in dose measurements is determined by the geometry of the experimental set-up. For example, the CDRH adult chest and adult abdomen and lumbosacral spine phantoms were designed to provide a geometry which minimizes backscattered radiation. For interventional fluoroscopic imaging equipment, IEC (2010) specifies radiation measurement geometries that minimize backscattered radiation contributions.

Incident air kerma and entrance-surface air kerma are both sufficient indicators of patient dose in projection radiography for tracking institutional and population trends with time. However, the quantity used is often determined by the preference of the investigator. For example, conversion coefficients are available in the literature for determining organ and tissue doses, and effective doses for selected diagnostic x-ray examinations (Hart *et al.*, 1994). That publication provides conversion coefficients that are normalized to entrance-surface dose, which includes backscattered radiation.

Figure 4.1 shows the measurement geometry for determination of fluoroscopic air-kerma rate. For a C-arm gantry the U.S. performance standard specifies a maximum fluoroscopic air-kerma rate at a fixed distance (30 cm) with respect to the input surface of the image receptor. Dose display features available on newer fluoroscopic equipment routinely reference fluoroscopic dose indices (rate and cumulative values for air kerma) at a point in space located with respect to the isocenter of the gantry. In particular, FDA (2009) and IEC (2010) both specify a reference location of 15 cm from the isocenter toward the x-ray tube for the display of fluoroscopic dose indices on C-arm type fluoroscopic systems. IEC (2010) refers to this location as the *patient entrance reference point*. The air kerma referenced to this location is referred to as *air kerma at the reference point* ($K_{a,r}$).

Air-kerma area product (P_{KA}) is a quantity that expresses the total amount of radiation received by a patient during a procedure. P_{KA} is defined as the integral of the air-kerma free-in-air (*i.e.*, in the absence of backscattered radiation) over the area of the x-ray beam in a plane perpendicular to the beam axis.

4.2 Computed Tomography Dosimetry

To be useful clinically, DRLs must be easy to measure using techniques that are adopted uniformly. In computed tomography (CT), most reported DRLs are derived from the CT dose index (*CTDI*). Until recently, most *CTDI* values were expressed as the weighted *CTDI* ($CTDI_w$), measured in a phantom with a 100 mm long ionization chamber as follows (Bongartz *et al.*, 1998; McCollough *et al.*, 2008; McNitt-Gray, 2002):

$$CTDI_w = \frac{1}{3}\,CTDI_{100,\mathrm{center}} + \frac{2}{3}\,CTDI_{100,\mathrm{edge}}\,. \tag{4.6}$$

With the widespread use of helical CT imaging, a more useful dose metric was developed ($CTDI_{vol}$) that considers the dose to a volume of tissue encompassed by a single rotation of the x-ray gantry, as a function of the pitch of the helix:

$$CTDI_{vol} = \frac{CTDI_w}{\mathrm{pitch}}\,, \tag{4.7}$$

where pitch is defined as the ratio of the patient couch advance to the nominal x-ray beam width.

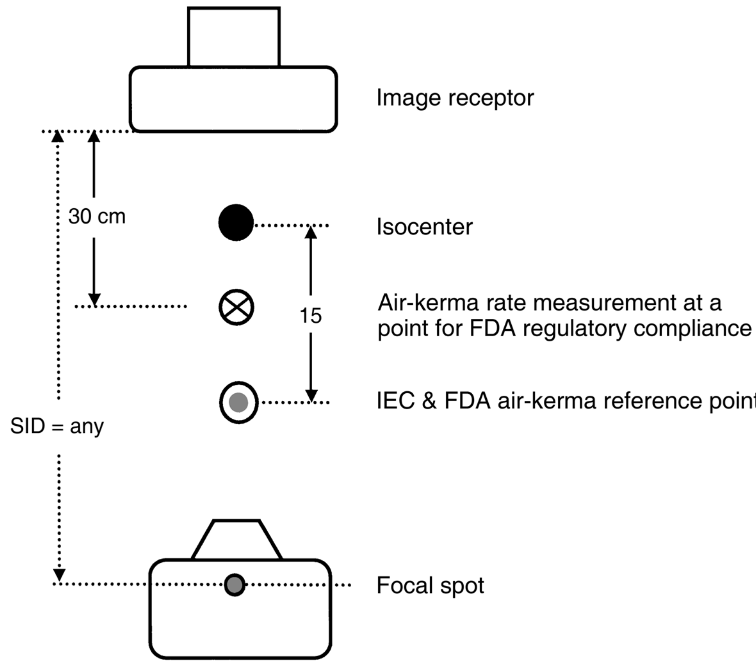

Fig. 4.1. Measurement geometry for fluoroscopic isocentric imaging systems (drawing not to scale).

$CTDI_{vol}$ is integrated over the scan length of a CT acquisition to obtain the dose-length product. The dose-length product can be used as a practical value for the estimation of effective dose. $CTDI_{vol}$, however, remains an important and useful metric in CT for comparing the performance of CT systems with regard to the utilization of radiation. As such, $CTDI_{vol}$ is regarded as the primary dose metric for CT DRLs in this Report. ICRU (2005) expresses similar quantities for CT in air kerma.

5. Data Sources

The Nationwide Evaluation of X-Ray Trends (NEXT) Program periodically surveys a nationwide sample of clinical facilities having equipment for a particular x-ray examination or projection (CRCPD, 2010). Recognizing the need to characterize patient radiation doses in medical x-ray imaging, in 1972 the FDA's Bureau of Radiological Health (now known as CDRH) collaborated with the CRCPD to initiate annual surveys of a sample of U.S. clinical facilities covering a number of commonly performed radiological examinations. State radiation control personnel are trained to gather data regarding examination workloads and equipment to make radiation measurements for the determination of patient doses and to evaluate image quality. Past surveys have included examinations (projections) of the adult chest, abdomen, and lumbosacral spine; pediatric chest, dental radiography, upper GI fluoroscopy, CT, and cardiac catheterization. Patient-equivalent phantoms are used to assist in determining radiation doses for systems with automatic exposure control. Results of these surveys are available from CRCPD (2010).

NEXT data have some limitations. A single projection (*e.g.*, PA chest projection or AP lumbosacral spine projection) is evaluated periodically resulting in updated data being available relatively infrequently. In addition, it has often taken several years for the data from the survey to be published. Consequently, the data may not represent current practice but rather the practice 3 to 5 y ago which becomes significant with rapidly changing technology (*e.g.*, changing from screen film to digital imaging).

In 1996 the state of New Jersey initiated a paradigm shift in their regulatory activities (Lipoti, 2008; Timins *et al.*, 2007). New regulations, implemented in 2001, place an emphasis on quality control, image quality, and patient dose. Their survey data provide an indication of the impact of these efforts on image quality and dose. After 4 y their data show that image quality has improved 12 to 32 % while patient doses have decreased from 38 to 70 % (depending on examination) (NJDEP, 2012). These data indicate the need for an integrated effort at optimization by medical imaging facilities, medical physicists, and regulatory organizations, and the need for an emphasis on both image quality and dose.

5.1 Adult Radiography and Fluoroscopy

NEXT survey results for the adult chest, and adult abdomen and lumbosacral spine radiography are available online (Moyal, 2006; Spelic, 2005). These NEXT surveys used phantoms representing a person with a mass of ~74 kg and height of ~1.7 m.

NEXT survey data are also available for the upper GI fluoroscopy examination. The most recent survey was conducted in 2003 at hospital and nonhospital facilities. The available patient dose information includes incident air-kerma rate, fluoroscopic technique factors (kilovolt peak, milliamperage), number of image acquisitions (*e.g.*, spot images), fluorographic technique factors (kilovolt peak, milliampere-seconds), and estimated skin entrance air kerma for fluorographic images.

IMV Medical Information Division, Inc. (Des Plaines, Illinois) conducts nationwide surveys of clinical facilities, capturing responses from a large population of facilities performing specific examinations (IMV, 2012). Recent benchmark reports from IMV include cardiac catheterization, CT, radiography and fluoroscopy, and digital radiography (computed and digital radiography). These surveys provide statistical information regarding examinations including workloads, types of imaging equipment, general site characteristics, and capital equipment budgeting. The surveys are not intended to characterize dosimetric aspects of the medical imaging practice, and do not provide any dose-related data (*e.g.*, CT or radiographic technique factors).

Patient size can affect the rate at which radiation is delivered during FGI procedures. Figures 5.1 and 5.2 show the effect of increasing attenuation on average air-kerma rates for routine fluoroscopic and cine recording modes of operation for cardiac fluoroscopy systems. Data are from the NEXT survey of cardiac catheterization conducted in 2008 to 2009 (Spelic, 2010). Note that although the maximum fluoroscopic dose rate (Figure 5.1) is restricted in compliance with the U.S. performance standard for fluoroscopic equipment, the performance standard does not restrict radiation output rate when in the cine recording (fluorographic) mode of operation (Figure 5.2) (FDA, 2010d).

5.2 Pediatric Chest Radiography

The NEXT Program conducted a survey of pediatric chest radiography in 1998 from a nationally-representative sample of U.S. clinical facilities (Moyal, 2004). A pediatric phantom was developed by CDRH to measure dose and image quality.

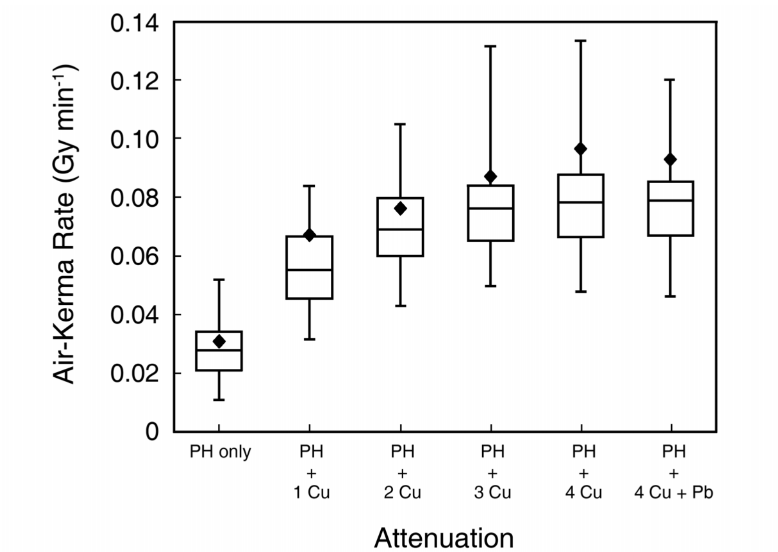

Fig. 5.1. Box plot of air-kerma rates observed in the NEXT survey of cardiac catheterization for the fluoroscopy mode of operation at a point ~30 cm from the image receptor. NEXT surveyors used an attenuation phantom roughly equivalent to a 22 cm AP adult abdomen. Attenuation was increased by adding layers of copper to the phantom configuration (CRCPD, 2009). The diamond indicates the mean air-kerma rate. The lower edge, central line, and upper edge of the box are the 25th, 50th, and 75th percentiles. The whiskers are the 5th and 95th percentiles. (PH = basic phantom. 1 Cu, 2 Cu, etc. indicates additional copper added to basic phantom.)

The survey indicated that almost 95 % of facilities used manual exposure techniques for pediatric chest examinations. Patient dose measurements do not require a phantom for manual technique. RLs in this Report are derived from the NEXT 1998 survey results for the pediatric chest projection for a 15 month old patient of mass ~10 kg. This survey also showed that the majority of pediatric chest examinations are conducted without an anti-scatter grid for this age group. RLs are provided for the PA projection separately for examinations with and without grids.

In this Report, DRLs for pediatric radiography will be discussed within each section and not in a separate pediatric chapter.

5.3 Pediatric Fluoroscopy

DRLs for pediatric fluoroscopic examinations are available in the European literature with a paucity of information in the North

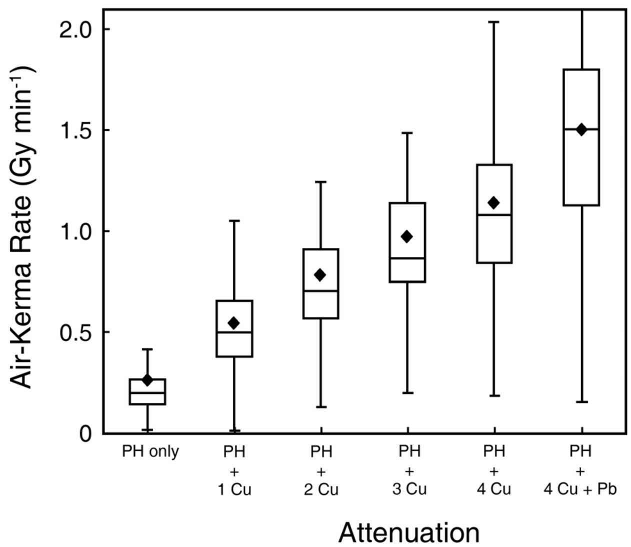

Fig. 5.2. Box plot of air-kerma rates observed in the NEXT survey of cardiac catheterization for the cine recording mode of operation. The measurement conditions and graphical display are identical to those for Figure 5.1. The mean fluoroscopic air-kerma rates from Figure 5.1 are reproduced (diamonds) for comparison. (PH = basic phantom. 1 Cu, 2 Cu, etc. indicates additional copper added to basic phantom.)

American literature. The availability in the European literature is due to an earlier awareness of dose and government mandates regarding dose levels in many European countries (Frush, 2009; Hart *et al*., 2007). Pediatric fluoroscopic doses in Europe are not based on surveys using phantoms (Ward *et al*., 2008).

5.4 Pediatric Computed Tomography

In 2001, an article in *USA Today* "linked" CT scans in children to development of cancer and captured the public's attention (Sternberg, 2001). Epidemiologic studies were initiated, and recently an association between repeated CT examinations in children and leukemia and brain cancer was suggested (Pearce *et al*., 2012). In response to these concerns, the Society for Pediatric Radiology, ACR, AAPM, and American Society of Radiologic Technologists formed the Alliance for Radiation Safety in Pediatric Imaging (ARSPI, 2010). Currently, more than 60 organizations are members of the Alliance. The Image Gently® campaign was launched in 2008 and is a direct result of the Alliance's efforts (ARSPI, 2010).

5.5 Digital Radiography

Digital radiographic imaging (both computed and digital radiography) is replacing screen-film projection radiography. The 2001 and 2002 NEXT surveys evaluated state-of-the-practice patient doses for adult chest, abdomen, and lumbosacral spine radiography. The sample of facilities included sites using digital imaging equipment, both computed and digital radiography. The 2001 NEXT survey found that ~8 % of U.S. facilities were using digital equipment for chest radiography, increasing to 11 % in 2002.

A more recent study (2005 to 2009) from the Michigan Department of Community Health (MDCH, 2012a; 2012b) found that 36 % of chest and 21 % of AP lumbosacral spine imaging equipment is digital.

As of May 2010, 65 % (5,663 out of 8,655) of FDA-certified mammography facilities use digital mammography equipment as compared with only 14 % (1,235 of 8,834 facilities) of facilities in October 2006 (FDA, 2010c).

The percentage of facilities using digital-based imaging technologies has likely increased substantially since these surveys were conducted. A benchmark report on radiographic equipment in U.S. hospitals by IMV in 2005 to 2006 found that the majority of hospitals have computed radiography equipment and 18 % have digital radiography equipment (IMV, 2006a).

DRLs are particularly important with digital radiography due to the potential for increased radiation dose per image. If the patient dose is increased for screen-film imaging systems, the film becomes darker until it reaches the point that it is not an usable image. With digital imaging, however, an increase in radiation dose does not result in a change in the image appearance on the display, except that in higher dose images the quantum noise is reduced. Consequently, the patient dose can be increased without the knowledge of the responsible imaging physician (Vano *et al*., 2007).

Data from the 2001 and 2002 NEXT surveys demonstrated 75th percentile values for incident air kerma that are higher for digital-based technologies (computed and digital radiography) than for film-based imaging (Table 5.1). The 75th percentiles for the 2001 survey (chest radiography) are different for computed radiography (0.19 mGy) and digital radiography (0.13 mGy). However, this is based on limited data (57 observations for computed radiography and 18 for digital radiography), and may not be representative for these technologies. The 2002 NEXT survey (abdomen and lumbosacral spine) also captured limited data regarding computed and digital radiography. Given the data limitations and the tendency for

TABLE 5.1—*Seventy-fifth percentile incident air kerma for screen-film versus digital imaging systems (in milligray).*

	Screen-Film	Computed and Digital Radiography
PA chest	0.15	0.18
AP abdomen	3.2	3.4
AP lumbosacral spine	3.9	4.3

these technologies to mature and improve with time, DRLs were not determined for each individual technology. The DRLs recommended in this Report for routine chest, abdomen and spine radiography are based on all of the data from the randomly selected population at the time of the survey.

5.6 Dental Radiography

5.6.1 *Intraoral Radiography*

The most common radiograph in dentistry is the posterior bitewing projection or intraoral radiograph. Traditionally, this intraoral view is made with direct-exposure film (*i.e.*, no intensifying screens are used). These films are used primarily to examine for interproximal caries (tooth decay) in the posterior teeth and to evaluate the supporting alveolar bone for signs of periodontal disease. Based on 2005 to 2009 survey data from the state of Michigan, ~25 % of dental facilities in that state now use digital radiography (MDCH, 2012a; 2012b), primarily direct digital (digital radiography) and, to a lesser extent, storage phosphor plate (computed radiography or photostimulable phosphor) technology. Storage phosphor plates requires about the same radiation dose as E-F-speed (high-speed) dental film while direct digital requires about half, or less, of the dose for E-F-speed film.

It should be appreciated, however, that the potential for dose reduction associated with use of direct-digital technology, either a charge-coupled device or complementary metal oxide semiconductor system, is offset by the tendency of dentists who have digital systems to take more images than those using film (Berkhout *et al.*, 2003). Most likely this results from the ease of making digital images. Furthermore, use of digital imaging with its broad latitude allows for overexposure without the visual feedback of a dark image provided by film.

The x-ray technique factors used for the posterior bitewing projection are usually the same as those used for maxillary and mandibular posterior periapical views made when more complete coverage of teeth and their supporting structures are indicated. Accordingly, the bitewing view is used for DRLs for intraoral dental radiography.

The use of E- or E-F-speed film results in a substantial reduction in patient dose compared to D-speed film. The median incident air kerma (1.30 mGy) for the E-speed film users is 76 % of the median incident air kerma (1.70 mGy) for the users of D-speed films (Moyal, 2007). This dose reduction, however, is somewhat less than has been found in laboratory studies that show the speed of the E-speed film [Ektaspeed Plus® (Eastmond Kodak, Rochester, New York)[3]] is about twice that of the D-speed film (ultra-speed) (Thunthy and Weinberg, 1995; Wakoh *et al.*, 1995). These findings show that many dentists who use fast film do not take full advantage of the opportunity to reduce patient dose (Looe *et al.*, 2006). [Ektaspeed® film has been replaced by Kodak with a slightly faster E-F-speed film called InSight® (Ludlow *et al.*, 2001a)].

Note also the wide range of doses used for both D- and E-speed films in Table 5.2. While the use of the faster E- or E-F-speed film generally reduces patient dose, there is considerable overlap of doses for the two different speed groups. This wide range of entrance doses results from variation in film-processing quality. Films are often underdeveloped due to low developer temperature, shortened development time, or improperly replenished developer solution[4] (Rushton *et al.*, 1999; Thornley *et al.*, 2006).

TABLE 5.2—*Michigan Department of Community Health incident air kerma for dental bitewing images 2005 to 2009 (in milligray).*[a]

	Digital	E-F-Speed Film	D-Speed Film
Machines tested	4,690	3,597	9,262
Median incident air kerma	0.8	1.3	2.2
75th percentile	1.5	1.8	3.0

[a]Original data in milliroentgen converted to milligray by using a conversion factor of 8.76 mGy R^{-1}.

[3]Kodak dental films are now manufactured by Carestream Health, Inc., Rochester, New York.

[4]Gray, J.E. (2010). Personal communication (Dental Image Quality and Dose, LLC, Steger, Illinois).

The Michigan Department of Community Health, Radiation Safety Section, publishes acceptable upper limits, known as *Standards of Care*. The website states "Although the Standard of Care limits are not regulations, they are considered by the Department to be upper limits which should not normally be exceeded for average-size patients" (MDCH, 2012a; 2012b). These values, derived from surveys made from 2005 to 2009, are shown in Table 5.3. Of dentists using film, most use D-speed film (72 %) while only 28 % use E-F-speed film. Digital imaging is used by 27 % of the dental facilities in Michigan.

5.6.2 *Cephalometric Radiography*

Cephalometric views are made most frequently by orthodontists for initial assessment of dental malocclusion or craniofacial disorders. They are also made for assessment of progress during the course of orthodontic treatment. Most of these radiographs are lateral views but PA views may be used for patients with left-right facial asymmetries. Cephalometric radiographs are 20 × 25 cm in size and cover the facial region including the base of the skull, the cervical spine, and the submandibular soft tissue. These images have traditionally been made using a screen-film imaging system but now many digital systems are in use. The images are similar to lateral skull views used in medical radiography except that it is not important to image the posterior portion of the skull in the lateral views. These images are also made with a graded filter over the anterior face to allow the nose, lips and chin to be readily visualized.

TABLE 5.3—*Michigan Department of Community Health acceptable upper limits for dental bitewing incident air kerma 2005 to 2009.*

Kilovolt Peak	Acceptable Upper Limit (mGy)[a]	
	E-F-Speed Film	D-Speed Film
50 – 59	2.8	5.0
60 – 69	2.0	3.9
70 – 79	1.5	3.2
80 – 89	1.1	2.0
90 – 99	1.1	2.0

[a]Original data in milliroentgen converted to milligray by using a conversion factor of 8.76 mGy R^{-1}.

Two competing dose measurement techniques are used for cephalometric imaging and for panoramic imaging. The earlier system is skin entrance dose although more recently the dose-area product, now referred to as air-kerma area product (P_{KA}). The results of these systems can be interconverted only approximately. The air-kerma area product has the advantage of reflecting the use of soft tissue filtration in the anterior face and other filtration of the posterior skull. Furthermore, measurements of air-kerma area product values do not require a patient or phantom to be present. Accordingly, these values are the preferred measure.

5.6.3 *Panoramic Radiography*

Determining patient dose from panoramic imaging is challenging. Some studies have measured the total dose to specific sites. Others have measured the dose-width product (DWP) or air-kerma area product. These methods use film, thermoluminescent, or optically-stimulated luminescent dosimeters; a pencil ionization chamber (Isoardi and Ropolo, 2003) in the case of DWP measurements; or a large area ionization chamber in the case of air-kerma area product measurements. The Institute of Physics and Engineering in Medicine argued that air-kerma area product is more appropriate than DWP as it better conforms to medical DRLs and because it more accurately reflects the patient dose (IPEM, 2004).

From 1997 to 1998, Winston and Shearer (2010) surveyed 121 dental facilities chosen at random from a database of over 1,000 registered facilities operating panoramic units in the western 22 counties of Pennsylvania. Using a phantom consisting of a skull covered with tissue-equivalent material the authors made panoramic dose measurements using the typical techniques of the facility. The authors found that the average intraoral air kerma was 0.30 mGy (range 0.6 to 0.91 mGy) while the average thyroid gland exposure was 0.74 mGy (range 0.18 to 2.19 mGy). These data show the wide variability of patient exposures (over 10 times) in clinical practice.

5.6.4 *Cone-Beam Computed Tomographic Imaging*

Cone-beam computed tomographic (CBCT) imaging is being used for many aspects of dental specialty care. This technology provides cross-sectional slices comparable to the bone window of a conventional CT examination, but at a significantly lower radiation dose. CBCT imaging is most commonly used for assessment of orofacial structures in orthodontic treatment planning; evaluation of the amount of alveolar bone support for implant treatment planning; assessing integrity of the osseous components of the temporomandibular joint for patients suffering from discomfort in these structures; and morphologic assessment of individual teeth with persistent sensitivity of unknown origin for signs of fracture, perforation, or periapical disease. The volume of tissues imaged ranges widely from as small as a cylinder 4 cm in diameter and 4 cm high to spherical volumes of 30 cm in diameter. Further information on CBCT can be found on the Image Gently® website (ARSPI, 2010) and in the American Academy of Oral and Maxillofacial Radiology executive opinion statement (Carter *et al*., 2008).

CBCT technology is changing rapidly with new systems being introduced and technical specifications changed frequently. The effective dose from CBCT examination varies from 11 to 1,073 μSv (Hirsch *et al*., 2008; Lofthag-Hansen *et al*., 2008; Ludlow and Ivanovic, 2008; Ludlow *et al*., 2003, 2006; Pauwels *et al*., 2012; Sedentexct, 2011). This wide variation results in large part from the size of the volume exposed but also from factors intrinsic to each machine such as whether or not there is an automatic dose control; the ability to collimate the beam; whether or not the x rays are pulsed in synchrony with the sensitivity of the receptor; the number of source, or basis, images collected; and other factors. Many machines offer the operator only limited means to vary patient dose. Development of DRLs for CBCT imaging will require air-kerma area product measurements of various manufacturer's machines in dental offices and radiographic laboratories. To date there are no surveys of doses for CBCT systems in clinical practice. Consequently, it is not possible to provide CBCT DRLs.

CBCT imaging is replacing panoramic imaging for many diagnostic tasks. The main advantage of CBCT imaging is high-quality, cross-sectional views and the flexibility to display images in a plane (axial, sagittal, coronal or oblique). The main concern of CBCT is that its effective dose is potentially much higher than for panoramic radiography. Accordingly, CBCT imaging should be reserved only for those clinical situations where panoramic imaging is inadequate and CBCT examination can be expected to add measurable value.

6. Recommended Diagnostic Reference Levels and Achievable Doses

Facility staff should consult with a qualified medical physicist regarding the measurement of phantom doses for the purpose of comparison of these doses to the DRLs provided in this Report. The qualified medical physicist can also suggest alternate methods such as documenting radiographic techniques during clinical examinations for a selected period of time and making subsequent air-kerma measurements from radiographic equipment using technique factors based on those clinical observations. The qualified medical physicist must make measurements so that the facility can determine the patient entrance dose from the technique factors which they record for each patient. Patient doses should be determined from all x-ray units used for specific projections, as doses can vary significantly from system to system (*e.g.*, by factors of 10 or more).

6.1 Radiographic and Fluoroscopic Examinations

Diagnostic radiographic and fluoroscopic examinations account for ~11 % of total collective effective dose to the population in the United States (NCRP, 2009). This relatively high proportion of the collective effective dose and the high number of procedures, particularly for selected radiographic projections, drives the need to optimize patient dose in this segment of the radiologic practice.

Anti-scatter grids can significantly improve image quality but generally require a higher dose (two to three times higher) to the patient. The NEXT surveys for 2001 and 2002 showed that over 90 % of clinical facilities now use grids routinely for adult chest, lumbosacral spine, and abdomen examinations.

CRCPD published a guide on patient doses including recommendations for selected diagnostic x-ray examinations (Winston *et al.*, 2003). It is worth noting that a number of states have regulatory guides or limits on patient dose for certain radiographic examinations. Some states provide different guidance based on the speed of the screen-film combination used. CRCPD is a good source of technical information on many aspects of diagnostic imaging (CRCPD, 2010).

DRLs are provided in this Report for common projections of three frequently performed adult radiographic examinations, chest, abdomen, and lumbosacral spine. The selection of these three projections was based on the relative procedure volume and the availability of survey data.

6.1.1 *Chest Examinations*

Chest radiography is the most frequently performed examination. Data for 2006 (NCRP, 2009) showed the U.S. annual workload for chest radiography (128.9 million) to be approximately double that for all of CT (62 million).

6.1.1.1 *U.S. Diagnostic Reference Levels for Chest Examinations*. Several professional organizations have recommended DRLs for various radiographic projections. ACR published recommendations for DRLs in 2008 based on actual patient studies collected at a number of clinical sites and on NEXT survey data available at the time (ACR, 2008a). The ACR DRL for a PA adult chest projection (0.22 mGy incident air kerma) is significantly higher than the 75th percentile value of 0.15 mGy recommended in this Report.

AAPM published DRLs for selected diagnostic x-ray examinations in 2005 (Gray *et al.*, 2005). The AAPM DRL for an adult PA chest projection (0.25 mGy) is also higher than the DRL recommended in this Report.

CRCPD published a guide on patient doses including recommendations for selected diagnostic x-ray examinations (Winston *et al.*, 2003). Values for maximum acceptable PA chest projection incident air kerma (published values for entrance skin exposure were converted to air kerma) free-in-air range from 0.05 to 0.44 mGy (depending on grid use and screen-film combination).

6.1.1.2 *Non-U.S. Diagnostic Reference Levels for Chest Examinations*. The European Commission also published recommendations for DRLs for a broad variety of radiological procedures (EC, 1999). The EC DRLs are based on air-kerma area product, an unit of measurement that is widely available on displays on radiographic equipment in EU Member States. It is difficult to compare DRLs from Europe and North America due to the differences in approach in the survey data (*i.e.*, air-kerma area product versus air kerma, and patient versus phantom measurements).

A large study in the United Kingdom of dose for multiple pediatric radiographic and fluoroscopic studies evaluated mean air-kerma area product and entrance surface dose, including backscatter (Kyriou *et al.*, 1996). Mean air-kerma area product for a frontal

(AP or PA) chest radiograph for children 5 y to <10 y of age is 34 mGy cm^2 (minimum 14 mGy cm^2 and maximum 95 mGy cm^2). Mean entrance surface dose for the same group is 60 µGy. A Finnish study evaluated mean and 75th percentile doses for different age and thickness patients (Kiljunen *et al.*, 2007). For patients with a mean thickness of 14.4 cm (approximately the size of a 5 y old), the mean air-kerma area product was 23 mGy cm^2 and the 75th percentile was 29 mGy cm^2. The mean entrance surface dose was 57 µGy and 75th percentile was 71 µGy.

6.1.1.3 *Recommended Diagnostic Reference Levels and Achievable Doses for Chest Examinations*. Seventy-fifth percentile values for distributions of patient incident air kerma from the NEXT surveys of 2001 (adult chest) were used to determine the NCRP DRLs for chest examinations (Table 6.1). Median values were used to determine the achievable doses. For these projections incident air kerma is specified free-in-air at 23 cm from the patient support surface.

The recommended DRL is also provided for pediatric chest radiography based on a phantom that approximates an average patient for doses collected during the 1998 NEXT survey (Moyal, 2004). This survey collected data for a reference patient 15 months of age with a mass of ~10 kg. The patient incident air kerma is specified free-in-air at 14.5 cm from the patient support surface. The 1998 NEXT survey found that only ~10 % of U.S. facilities perform pediatric chest radiography with a grid. Consequently, DRLs and achievable doses are provided for pediatric chest projections performed with and without a grid (Table 6.1).

The 1998 NEXT survey found that almost 95 % of the facilities surveyed use a manual technique for pediatric chest radiography.

6.1.2 *Abdomen and Lumbosacral Spine Examinations*

While chest radiography and mammography are the first and second most frequently performed radiographic or fluoroscopic examinations in the United States, abdominal and lumbosacral spine x-ray radiography are the third and fourth most common, respectively (NCRP, 2009).

The radiation dose associated with these examinations is directly linked to the number of projections that are acquired. For example, the lumbosacral spine x-ray examination may or may not include several projections, including AP, lateral, and oblique projections. Similarly, the abdominal x-ray examination may or may not include several projections, including AP supine, upright, and decubitus projections. For the purpose of this Report, DRLs for

TABLE 6.1—*Recommended DRLs and achievable doses for radiographic projections of the chest (incident air kerma, free-in-air).*

Projection	DRL (mGy)	Achievable Doses (mGy)
Adult PA chest (with grid)	0.15	0.11
Pediatric PA chest (without grid)	0.06	0.04
Pediatric PA chest (with grid)	0.12	0.07

these imaging examinations are based on the AP projections. The selection of this projection is based on the relative procedure volume and the availability of survey data.

6.1.2.1 *U.S. Diagnostic Reference Levels for Abdomen and Lumbosacral Spine Examinations.* ACR recommended DRL for an AP abdomen projection (5.3 mGy) is higher than the DRL recommended in this Report.

AAPM published DRLs for selected diagnostic x-ray examinations in 2005 (Gray *et al.*, 2005). The AAPM DRL for an AP abdomen (4.5 mGy), and AP lumbosacral spine (5 mGy) are higher than the DRLs recommended in this Report for these projections.

6.1.2.2 *Non-U.S. Diagnostic Reference Levels for Abdomen and Lumbosacral Spine Examinations.* EC also published recommendations for DRLs for a broad variety of radiological procedures (EC, 1996a; 1996b; 1999). EC DRLs are based on air-kerma area product, an unit of measurement that is widely available on displays on radiographic equipment in EU Member States. It is difficult to compare DRLs from Europe and North America due to the differences in approach in the survey data (*i.e.*, air-kerma area product versus incident air kerma, with and without backscatter, and patient versus phantom measurements).

6.1.2.3 *Recommended Diagnostic Reference Levels and Achievable Doses for Abdomen and Lumbosacral Spine Examinations.* Seventy-fifth percentile values for distributions of patient incident air kerma from the NEXT surveys of 2002 (adult abdomen and lumbosacral spine) were used to determine the NCRP DRLs for abdomen and lumbosacral spine examinations (Table 6.2). For these projections incident air kerma is specified free-in-air at 23 cm from the patient support surface.

TABLE 6.2—*Recommended DRLs and achievable doses for radiographic projections of the abdomen and lumbosacral spine (free-in-air).*

Projection	DRL (mGy)	Achievable Doses (mGy)
AP abdomen	3.4	2.4
AP lumbosacral spine	4.2	2.8

6.1.3 *Fluoroscopic Examinations*

Patient dose from fluoroscopic examinations is affected by the equipment capabilities and performance, thickness of the body part being imaged, complexity of the clinical task, and clinician operating the equipment. Dose is typically delivered *via* two equipment modes of operation: fluoroscopy and fluorographic-image acquisitions (*e.g.*, spot radiographs). While the particular clinical task may determine the extent of utilization for these two modes of operation, both should be optimized to provide high-quality imaging consistent with acceptable dose delivery to the patient. The fluoroscopy mode of operation is typically characterized by the fluoroscopic air-kerma rate. Similarly the air kerma per image acquisition characterizes the fluorographic acquisition mode of operation. These two values (fluoroscopy air-kerma rate and air kerma per image) comprise the recommended DRLs for diagnostic fluoroscopy.

IMV conducted a survey of radiography and fluoroscopy in 2004 to 2005 and found GI studies to constitute ~38 % of all patient visits to radiographic and fluoroscopic rooms in hospitals having 150 or more patient beds (IMV, 2006b).

Upper GI fluoroscopy was selected for inclusion in this Report because it continues to be a routinely performed examination. DRLs are specified for fluoroscopic air-kerma rate without and with contrast for a patient having (approximately) a 22 cm abdomen.

DRLs could be considered for other dosimetric indicators. Total fluoroscopy time is only an approximate indicator of the radiation delivered to the patient. However, fluoroscopy time provides no dosimetric information for digital image acquisitions, which can account for a significant fraction of total patient dose. Cumulative values for air-kerma area product and reference air kerma ($K_{a,r}$) are useful dose metrics providing information for all imaging components of a complete fluoroscopic examination. Moreover, cumulative air-kerma area product provides a good indication of overall stochastic risk, while cumulative reference air kerma is a good predictor for risk of tissue effects (Miller *et al.*, 2012; NCRP, 2010; Stecker *et al.*, 2009). At this time, however, there are no comprehensive data for the determination of DRLs for either air-kerma area product or reference air kerma in upper GI fluoroscopic examinations performed in the United States. It is not known what fraction of fluoroscopic equipment currently in clinical use in the United States is equipped with dose displays, although all new fluoroscopic equipment must provide reference air kerma to comply with the federal performance standard for fluoroscopic equipment (FDA, 2009).

A qualified medical physicist should assist with appropriate measurements for determination of incident air-kerma rate and incident air kerma per image acquisition.

6.1.3.1 *U.S. Diagnostic Reference Levels for Fluoroscopic Examinations*. ACR recommended a DRL for routine abdominal fluoroscopy based on NEXT survey results (ACR, 2008a). Their DRL for fluoroscopic air-kerma rate of 57 mGy min^{-1} is the same as that recommended by AAPM (Gray *et al.*, 2005). As with radiography, a phantom is used to measure the fluoroscopic air-kerma rate.

Oral contrast media may be administered to the patient during the examination which will increase the radiation output from the fluoroscopic system and, hence, the patient dose. Previous U.S. efforts to specify a DRL for fluoroscopy do not account for the influence of oral contrast media on air-kerma rate. This Report includes DRLs for fluoroscopic imaging where contrast is introduced (Table 6.3). A DRL for the fluorographic-image acquisition component is also included in this Report.

6.1.3.2 *Non-U.S. Diagnostic Reference Levels for Fluoroscopic Examinations*. EC also published recommendations for DRLs for a broad variety of radiological procedures (EC, 1996a; 1996b; 1999). EC DRLs are based on air-kerma area product, an unit of measurement that is widely available on displays on radiographic equipment in EU Member States. It is difficult to compare DRLs from Europe and North America due to the differences in approach in the survey data (*i.e.*, air-kerma area product versus incident air kerma, backscatter conditions, and patient versus phantom measurements).

6.1.3.3 *Recommended Diagnostic Reference Levels and Achievable Doses for Fluoroscopic Examinations*. The DRLs in Table 6.3 represent 75th percentile values for incident air kerma (referenced to a location 1 cm above the patient support surface for conventional radiographic and fluoroscopic systems with the x-ray source beneath the patient support) from the NEXT survey of upper GI

TABLE 6.3—*Recommended DRLs and achievable doses and dose rates for upper GI fluoroscopy examination components.*

Component	DRLs	Achievable Doses
Incident air-kerma rate without oral contrast media (mGy min^{-1})	54	40
Incident air-kerma rate with oral contrast media (mGy min^{-1})	80	72
Incident air kerma per fluorographic image without oral contrast media[a] (mGy)		
Film	3.9	2.5
Digital	1.5	0.9
Incident air kerma per fluorographic image with oral contrast media (mGy)		
Film	27.5	18.7
Digital	9.9	5.3

[a]The 75th percentile for the number of fluorographic images in a fluoroscopic examination is 16.

fluoroscopy and the achievable doses represent the median values (CRCPD, 2009). Radiation exposure measurements for fluoroscopy and fluorographic (screen-film and digital) imaging were made using a geometry representative of clinical conditions which includes some backscatter due to the phantom-dosimeter geometry (Figure 6.1).

6.2 Computed Tomography Examinations

Computed tomography (CT) accounts for the greatest contribution to the collective dose from medical imaging in the United States and is the second largest contributor of all (medical and non-medical) radiation sources. In 2006, CT accounted for only 17 % of the medical imaging procedures (67 million out of 395 million), but contributed 49 % of the collective dose from medical sources (1.47 out of 3 mSv per capita) (NCRP, 2009).

The radiation dose associated with CT examinations is directly linked to a number of factors. For example, if the examination is performed without and with contrast, the dose will be twice that for an examination without contrast. In addition, pitch of the scan

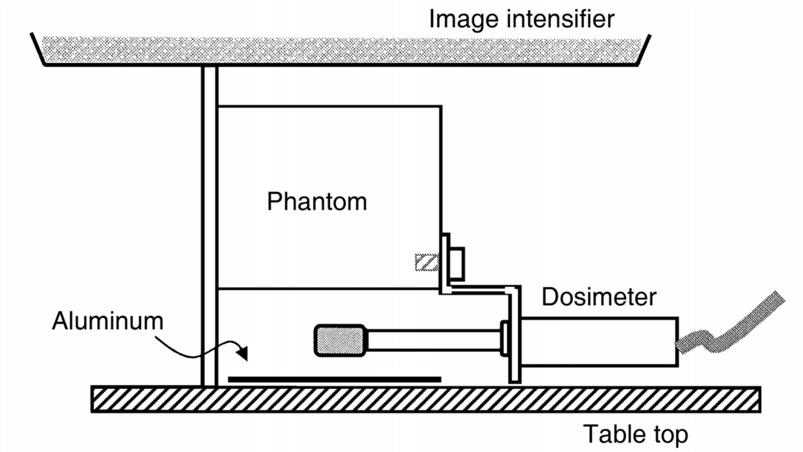

Fig. 6.1. Fluoroscopy phantom measurement geometry used during the 2003 NEXT survey of upper GI fluoroscopy for under-table x-ray tubes. Aluminum is present in the set-up only for measurements taken to determine beam quality.

acquisition, and the starting and endpoints of the scan, all impact patient dose. For the purpose of this Report, DRLs for CT applications are based on a single scan acquisition.

DRLs are provided for five frequently performed adult and pediatric CT examinations: adult head, chest, abdomen-pelvis, and pediatric (5 y old) head, abdomen-pelvis. The selection of these five examinations was based on the relative procedure volume and the availability of survey data.

6.2.1 *U.S. Diagnostic Reference Levels for Computed Tomography*

In 2002, ACR launched its CT Accreditation Program. The typical acquisition parameters for a site's adult head and abdomen, and pediatric abdomen examinations were used to calculate $CTDI_w$ and $CTDI_{vol}$, using the standard 16 cm PMMA phantom to measure the *CTDI* for routine head and pediatric body (5 y old) examinations and the 32 cm PMMA phantom to determine the *CTDI* for adult abdomen scans. Data from the first 3 y of that effort were examined, and DRLs that were specified for CT initially were modified (McCollough *et al.*, 2011). In 2002, DRLs ($CTDI_w$) were established for the adult head (60 mGy), the adult abdomen (35 mGy), and the pediatric (5 y old) abdomen (25 mGy); these were based on the U.K. national dose survey (Shrimpton *et al.*, 2005). By design, DRLs should be established such that ~25 % of practices exceed DRL values. But, since the accreditation program was initiated with these values, it was not clear how well these DRL values reflected current practice in the United States.

In 2008, the ACR CT Accreditation Program analyzed their experience with these DRLs for the first 3 y of their program (2002 to 2004). Between September 2002 and December 2004, 829 CT scanners were evaluated for accreditation by ACR (McCollough *et al.*, 2011). The ACR $CTDI_w$ values are shown in Tables 6.4, 6.5, and 6.6.

In 2004, 23.8, 2.3, and 6.9 % of sites reported $CTDI_w$ values above 2002 RLs for adult head, adult abdomen, and pediatric abdomen scans, respectively, compared to 49.6, 4.7, and 15 % in 2002. These results indicate that the initial DRLs that were set in 2002 did not match practice patterns in the United States. ~25 % of scanners should have fallen above a DRL that is based on the 75th percentile. But, in 2002, the adult head DRL was set too low as nearly 50 % of scanners were above the DRL, and the adult abdomen and pediatric abdomen were set too high because only 5 and 15 % of scanners, respectively, exceeded the DRL.

Because the DRL for adult head scans was set too low, practitioners appeared to lower their doses to the RL after the first year of the program (in 2003 and 2004) in order to meet this criterion. Several complaints were received from practitioners about poor image quality at this dose level. Graphical analysis suggested that the $CTDI_w$ data for adult head scans in 2003 and 2004 appeared to be artificially capped at 60 mGy and were not included in subsequent analysis. The 75th percentile was calculated for all three examinations, and ACR set new DRLs for 2008 at 75 mGy for head scans, 25 mGy for adult abdomen scans, and 20 mGy for pediatric abdomen scans (Table 6.7).

TABLE 6.4—*Adult head $CTDI_w$ (milligray) from the ACR CT Accreditation Program.*

	2002	2003	2004	2002 – 2004
Mean	66.7	58.5	55.8	59.1
Standard deviation	23.5	17.5	15.7	18.6
75th percentile	76.8	63.9	60.0	64.3
90th percentile	99.0	82.2	74.0	81.3

TABLE 6.5—*Adult body $CTDI_w$ (milligray) from the ACR CT Accreditation Program.*

	2002	2003	2004	2002 – 2004
Mean	18.7	19.2	17.0	18.4
Standard deviation	8.0	8.7	7.6	8.3
75th percentile	22.6	23.4	21.1	22.2
90th percentile	29.4	30.6	25.8	29.5

TABLE 6.6—*Pediatric abdomen $CTDI_w$ (milligray) over time from the ACR CT Accreditation Program.*

	2002	2003	2004	2002 – 2004
Mean	17.2	15.9	14.0	15.5
Standard deviation	9.7	8.6	7.0	8.4
75th percentile	20.6	20.5	18.4	20.0
90th percentile	26.6	25.6	23.4	24.9

TABLE 6.7—*ACR's CT DRLs.*

	2002 DRL ($CTDI_w$ or $CTDI_{vol}$)[a]	2008 DRL ($CTDI_{vol}$)
Adult head	60	75
Adult abdomen	35	25
Pediatric abdomen	25	20

[a]Assume use of pitch = 1 in converting $CTDI_w$ DRLs into $CTDI_{vol}$ DRLs.

The percentage of scanners exceeding the 2008 ACR DRLs (in terms of $CTDI_{vol}$) would have been 25.6, 15.9, and 26.4 % in 2002 and 9.1, 12.7, and 19.2 % in 2004 for the head, adult abdomen, and pediatric abdomen examinations, respectively, much more in keeping with the distribution of data that would be expected with DRLs that are tied to the 75th percentile of clinically-used values (Table 6.8).

While the DRLs for adult abdomen and pediatric abdomen scans were lowered between 2002 and 2008, the DRL for adult head scans was increased in 2008. A similar trend was observed in Europe where current $CTDI_{vol}$ DRLs for head CT examinations range from 60 mGy (EC) to 100 mGy (United Kingdom) with a median DRL of 75 mGy (Bongartz *et al.*, 2004; Shrimpton, 2004; Shrimpton *et al.*, 2005; SRPA, 2002). This trend may relate to the routine use of thinner 5 mm sections for cranial imaging as compared to the thicker 10 mm sections that were used previously.

6.2.2 *Non-U.S. Diagnostic Reference Levels for Computed Tomography*

In 1999, EC published recommended DRLs for adult CT examinations (Table 6.9). EC has also published DRLs for pediatric CT examinations (Table 6.10). However, some caution was advised when considering data that formed the basis of these adult and pediatric DRLs as some of it may have been derived from exam protocols rather than specific measurements on individual patients (IPEM, 2004). In 2009, EC published a reduction in the DRL for the adult abdomen to 15 mGy for acute pain, renal stones, and asbestosis, and 25 mGy for liver, pancreas, and adrenal glands (Bongartz *et al.*, 2004).

While EC recommended these DRLs for adult and pediatric CT examinations, many of the DRLs in use throughout the EU are based on their own surveys of patient doses (EAN, 2007). As such the absolute magnitude of the DRLs varies from country to country, and the values are not directly transferable to the more uniform phantom-based approach described in this Report. However, some similarities in trends are worth noting.

Although the original ACR CT DRLs were based on the levels recommended by EC in 1999, the ACR DRLs for adult abdomen and pediatric abdomen scans were lowered from 35 and 25 mGy to 25 and 20 mGy, respectively, between 2002 and 2008 (ACR, 2008b). Conversely, the DRL for adult head scans increased from 60 to 75 mGy during this same time frame. A similar trend was observed in Europe with an increase in head CT DRLs and a decrease in abdominal DRLs (Bongartz *et al.*, 2004; Shrimpton, 2004; Shrimpton *et al.*, 2005; SRPA, 2002).

TABLE 6.8—*Percentage of scanners above ACR's 2008 CT DRLs.*

	2002	2003	2004	2002 – 2004
Adult head	25.6	12.8	9.1	14.0
Adult abdomen	15.9	20.3	12.7	17.0
Pediatric abdomen	26.4	27.2	19.2	24.5

TABLE 6.9—*EC DRLs for adult CT examinations (Bongartz et al., 1998).*

Examination	$CTDI_w$ (mGy)
Brain	60
Face-sinus	35
Vertebral-paravertebral	70
Routine abdomen	35
Chest	30
Chest high resolution	35
Liver and spleen	35
Pelvis	35
Osseous pelvis	25

6.2.3 *Recommended Diagnostic Reference Levels and Achievable Doses for Computed Tomography*

Two sources of data were used for CT DRLs and achievable doses. Values for CT examinations of the adult head and abdomen-pelvis, and pediatric abdomen-pelvis were obtained from ACR data (ACR, 2011b). It should be noted that the DRL values for the adult head and abdomen-pelvis, and pediatric abdomen-pelvis from ACR and NEXT were similar so NCRP elected to use the ACR values.

Data from the 2005 to 2006 NEXT CT survey were used to derive DRLs and achievable doses for examinations of the adult chest and pediatric head (Spelic *et al.*, 2008). These data are publicly available for download from the FDA website (FDA, 2010a). A

TABLE 6.10—*EC DRLs for pediatric CT examinations (EC, 1996b).*

Examination	Patient Age (y)	$CTDI_w$ per slice or rotation (mGy)
Brain	<1	40
	5	60
	10	70
Chest (general)	<1	20
	5	30
	10	30
Chest (high-resolution CT)	<1	30
	5	40
	10	50
Upper abdomen	<1	20
	5	25
	10	30
Lower abdomen-pelvis	<1	20
	5	25
	10	30

report of the statistical analysis from this survey data was not available at the time of this Report. Consequently, a subset of survey data was evaluated to derive preliminary values for DRLs and achievable doses.

DRLs and achievable doses for adult chest and pediatric head examinations were derived from 75th percentile and median $CTDI_{vol}$ values, respectively, from the NEXT survey using a subsample of 40 clinical sites randomly selected from the entire surveyed population. These data were based on *CTDI* measurements made by the facility's qualified medical physicist using the standard 16 and 32 cm CT dosimetry phantoms.

DRLs for selected CT exams were derived from a preliminary analysis of data collected during the NEXT 2005 to 2006 survey of CT. Values for $CTDI_{vol}$ were derived from 75th percentile values representing a subsample of 40 clinical sites randomly selected from the entire surveyed population. CT scanning protocols were provided by each facility for an average size adult, and for two pediatric patient age groups, 1 y old or younger and 5 to 6 y old. Facilities were asked to provide a copy of the most recent medical physics survey report for the CT system. Doses were reported either for $CTDI_{100}$ (center and peripheral measurements in the standard 16 and 32 cm CT dosimetry phantoms), or for $CTDI_w$. These values for $CTDI_w$ from the medical physicist's reports were used to compute $CTDI_{vol}$ using the clinical scanning conditions routinely used by the facility. Values of $CTDI_{vol}$ for pediatric exams were determined using data for the 16 cm CT dosimetry phantom. If a pediatric exam was conducted at a different kilovoltage than that indicated in the medical physics report, the value for $CTDI_w$ was adjusted using data values derived from the ImPACT CT Patient Dosimetry Calculator (ImPACT, 2011) for the same CT scanner manufacturer and model. If a facility's medical physics report was incomplete (*e.g.*, missing data for $CTDI_w$ pertaining to body exams), then values for $CTDI_w$ from the ImPACT scan calculator were used.

Table 6.11 provides NCRP recommended DRLs and achievable doses for adult head and abdomen-pelvis, and pediatric abdomen-pelvis examinations as recommended by ACR (2011b). NCRP DRLs and achievable doses for adult chest and pediatric head examinations are the 75th percentile and median values, respectively, of the 2005 to 2006 NEXT survey subsample.

TABLE 6.11—*NCRP recommended DRLs and achievable doses ($CTDI_{vol}$) for adult and pediatric CT (in milligray).*

Procedure	DRLs	Achievable Doses
Adult head[a]	75	57
Adult chest[b]	21	14
Adult abdomen-pelvis[a]	25	17
Pediatric head[b] (5 y old)	40	31
Pediatric abdomen-pelvis[a] (5 y old)	20	14

[a]Based on ACR (2011b) data.
[b]Based on preliminary results of 2005 to 2006 NEXT CT survey.

6.3 Fluoroscopically-Guided Interventional Procedures

While fluoroscopically-guided interventional (FGI) procedures are less frequent than diagnostic fluoroscopy procedures and CT examinations, many interventional procedures have increased three- to fourfold during the last decade (AMA, 2010). Furthermore, the complexity of these procedures often leads to high patient doses. Concerns about radiation risks will likely lead to expanded use of alternative guidance modes such as ultrasound, magnetic resonance imaging, and electromagnetic tracking. For the foreseeable future, however, ionizing radiation will almost certainly remain the primary means of guiding devices and visualizing anatomic structures during a wide array of medical procedures. As a result, there is a clear need for dose optimization and a continuous assessment of the radiation risks associated with these procedures.

Since patient doses in FGI procedures are dependent on the equipment and operating environment, the skill of the team performing the procedure, as well as on patient and lesion specific variables (Balter *et al.*, 2008; Bucholz and Duncan, 2008; Duncan and Evens, 2009; IAEA, 2009; ICRP, 2007b; Miller *et al.*, 2009; Sridhar and Duncan, 2008), RLs reflect data from a substantially more complex system (Figure 6.2). Only the first two variables can be optimized; patient and lesion variables are not generally amenable to optimization. As a result, and as detailed in the introduction and in NCRP Report No. 168 (NCRP, 2010), the development of RLs for FGI procedures differs from the processes used to create DRLs in other areas of radiology. DRLs can be based on measurements with standardized phantoms but RLs in FGI procedures must be based on data obtained from clinical cases (Miller *et al.*, 2009; NCRP, 2010).

The distributions of patient doses for most types of FGI procedures are very wide, because the dose for each instance of a procedure is strongly dependent on individual clinical circumstances. Patient doses for a single procedure performed at a single institution may vary by two or even three orders of magnitude (Tsalafoutas *et al.*, 2006). A potential approach to minimizing the effect of patient and lesion factors is to include the "complexity" of the procedure in the analysis (Balter *et al.*, 2008; IAEA, 2009; ICRP, 2007b). At present, with the exception of some interventional cardiology procedures, complexity cannot be quantified for FGI procedures, so this adjustment is not yet possible for most FGI procedures (Balter *et al.*, 2008; IAEA, 2009).

When comparing data from routine clinical work to these RLs, it is important to recognize that RLs are set at the 75th percentile of patient dose from a sample of data for multiple institutions and

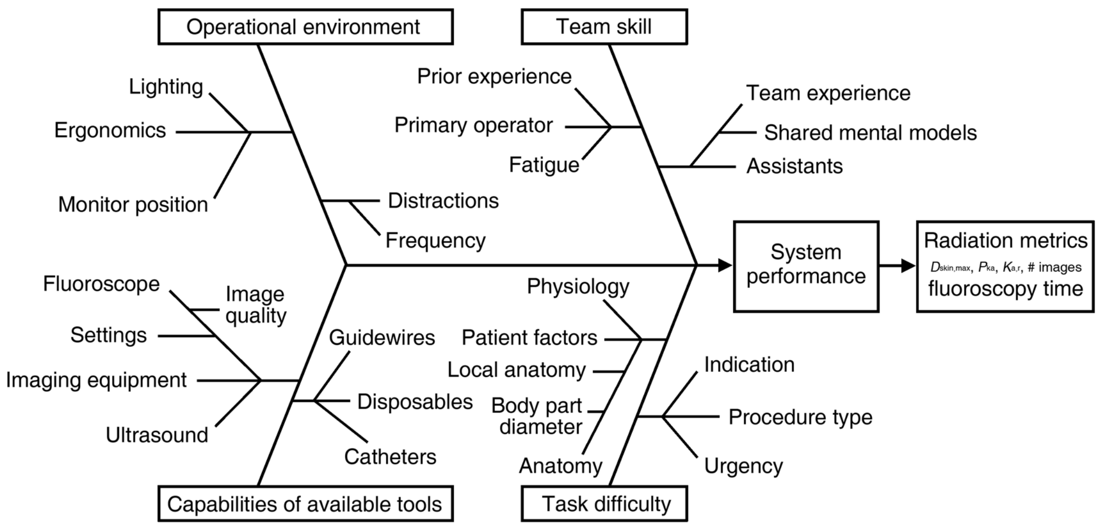

Fig. 6.2. Partial listing of factors which influence system performance and radiation use during FGI procedures. While DRLs utilize phantoms to minimize variation, RLs are far more variable because they are derived from the results of patient procedures in the clinical, operational environments.

as a result, one can expect that in an average practice, one of four patient's dose will exceed the RL. As ICRP has stated, RLs do not signify the difference between acceptable and unacceptable performance, but rather are meant to provide guidance for improving performance (ICRP, 2007b). They are neither dose limits nor thresholds that define competent performance of the operator or the equipment.

RLs are not intended as a trigger to prompt investigation of individual cases of a procedure. An alternative mechanism, the SRDL is used for this purpose (Section 6.3.3).

RLs are quality indicators for stochastic risk that help to guide the use of scarce investigative resources. Used as described in Section 6.3.3, they help to identify interventional procedures where patient radiation doses may not be optimized. If the RL for a specific FGI procedure is routinely exceeded, an investigation of the fluoroscopic equipment, clinical protocols, and operator performance may be warranted.

6.3.1 *U.S. Diagnostic Reference Levels for Fluoroscopically-Guided Interventional Procedures*

Miller *et al*. proposed initial RLs for a number of interventional radiology FGI procedures (Miller *et al*., 2009). These values were based on the Radiation Doses in Interventional Radiology Procedures (RAD-IR) Study, which is the only large, multi-center series of radiation dose data available for the United States as of 2011 (Balter *et al*., 2004; 2008; Miller *et al*., 2003a; 2003b). That study captured P_{KA}, $K_{a,r}$, fluoroscopy time, and number of fluorographic images from 2,142 cases that covered 21 different FGI procedures. Those data were recently reanalyzed to determine the 75th percentile for each of these four metrics in the 21 procedures. In four procedures, subgroups were analyzed. This yielded proposed RLs for 26 separate procedures. No comparable study that includes multiple centers from the United States exists for cardiac procedures.

6.3.2 *Non-U.S. Reference Levels for Fluoroscopically-Guided Interventional Procedures*

Miller *et al*. compared their proposed RLs to those reported for other populations (Miller *et al*., 2009). The authors noted that U.S. RLs were comparable to European levels for fluoroscopy time. RLs tended to permit a larger number of images, higher $K_{a,r}$ and P_{KA}. The authors postulated that this difference might reflect the larger body habitus of the U.S. population, the tendency to acquire more images in the United States, differences in procedure definitions, or other causes. For cardiac procedures, a series of investigators have examined radiation use during cardiac procedures and a variety of different RLs have been proposed. NCRP Report No. 168 (NCRP, 2010) included data from the United States and other countries. The 75th percentiles from Tables 4.2a through 4.2c in that report are provided in Table 6.12 as potential RLs for cardiac procedures.

6.3.3 *Recommended RLs for Fluoroscopically-Guided Interventional Procedures*

The NCRP-recommended RLs for FGI procedures (Table 6.13) are based on the 75th percentile of the RAD-IR Study data (Miller *et al.*, 2009). The lack of a multicenter dataset for cardiac procedures performed in the United States prevents NCRP from recommending RLs for cardiac procedures. Interested parties are encouraged to use the non-U.S. data in Section 6.3.2 to guide their practice. Since radiation metrics are now commonly being incorporated into radiology reports and billing records, CPT codes are included to help identify the relevant procedures. While coding terms and conventions change periodically, the coding terminology in Table 6.13 is current as of February 2012.

6.3.4 *Recommendations on Analyzing Data from Fluoroscopically-Guided Interventional Procedures*

NCRP Report No. 168 recommends that patient dose data be recorded in the medical record at the conclusion of each FGI procedure (NCRP, 2010). That report provides guidance on the use of the SRDLs for analyzing data from individual FGI procedures and on the use of RLs for aggregating and analyzing data from multiple FGI procedures. Since this Report is meant to supplement those recommendations, key concepts from NCRP Report No. 168 are reiterated here.

TABLE 6.12—*Potential non-U.S. RLs for cardiac procedures.*

Procedure	P_{KA} (Gy cm^2)	$K_{a,r}$ (Gy)	Fluoroscopy Time (min)
Diagnostic coronary angiography	49.4	1.0	8.7
Coronary angioplasty	98.4	3.0	19.8
Combined coronary angiography and angioplasty	138.3	2.7	23.6

TABLE 6.13—*Recommended RLs in FGI procedures (Miller et al., 2009).*

Procedure Description (CPT search strategy[a])	Air Kerma at the Reference Point (Gy)	Air-Kerma Area Product (Gy cm^2)	Fluoroscopy Time (min)	Number of Fluorographic Images
Transjugular intrahepatic portosystemic shunt creation (CPT 37182)	3.00	525	60	300
Biliary drainage (CPT 47510 or 47511)	1.40	100	30	20
Nephrostomy				
Drainage (CPT 50394)	0.40	40	15	12
Stone removal (CPT 50395)	0.70	60	25	14
Pulmonary angiography (CPT 36013, 36014, or 36015)	0.50	110	10	215
Inferior vena cava filter placement (CPT 37191)	0.25	60	4	40
Renal (CPT 36251 unilateral or 36252 bilateral)				
Without stent	2.00	200	20	210
With stent (plus CPT 37205)	2.30	250	30	200

Iliac angioplasty				
Without stent (CPT 37220)	1.25	250	20	300
With stent (plus CPT 37221)	1.90	300	25	350
Bronchial artery embolization (CPT 37215-8, 75726, 37204)	2.00	240	50	450
Hepatic chemoembolization (CPT 37245-8, 75726, 37204, 96420)	1.90	400	25	300
Uterine fibroid embolization (CPT 37210)	3.60	450	36	450
Other tumor embolization (no suitable descriptor beyond CPT 37204[b])	2.60	390	35	325
GI hemorrhage localization and treatment (CPT 37245-8, 75726, and either 37204 or 37202)	3.80	520	35	425
Embolization in the head (CPT 61624 or 61626 with one or more of the following: 75660, 75662, 75665, 75671, 75676, 75680, or 75685)				
For AVM[c]	6.00	550	135	1,500
For aneurysm	4.75	360	90	1,350
For tumor	6.20	550	200	1,700

Table 6.13—(*continued*)

Procedure Description (CPT search strategy[a])	Air Kerma at the Reference Point (Gy)	Air-Kerma Area Product (Gy cm^2)	Fluoroscopy Time (min)	Number of Fluorographic Images
Vertebroplasty (CPT 22520 thoracic or 22521 lumbar)	2.00	120	21	120
Pelvic artery embolization for trauma or tumor (CPT 36245-8, 75736, 37204)	2.50	550	35	550
Embolization in the spine for AVM or tumor (CPT 61624 and 75705)	8.00	950	130	1,500

[a]Search strategies based on CPT codes for particular procedures have been added to the procedure descriptions. Since most interventional procedures are described using multiple codes and radiation metrics are reported for the entire procedure, there are only a few cases (transjugular intrahepatic portosystemic shunt, uterine fibroid embolization, and inferior vena cava filter placement filter placement) where it is possible to match CPT codes with RLs. In most cases, the listed codes will be a subset of the codes used to describe a particular procedure and the listed codes or code combinations provide a means of identifying procedures of interest in a practice's billing records. The listed metrics (reference dose, kerma area product, fluoroscopy time, and number of images) from those procedures can then be compared to these RLs.

[b]Cases of "other tumor embolization" could be identified using CPT 37204 but the result would likely include nontumor embolizations. If needed, ICD-9 codes could be used to help identify specific cases.

[c]ICD-9 codes can also be used to distinguish between different types of lesions treated by embolization.

6.3.4.1 *Recommendations for Analyzing Data from Individual FGI Procedures*. The SRDL is used to manage risk of tissue effects. The purpose is to ensure that all radiation-related skin injuries are detected as soon as possible, and that none are ignored. NCRP Report No. 168 (NCRP, 2010) suggested SRDL values of $D_{\text{skin,max}}$, $K_{\text{a,r}}$, P_{KA}, and fluoroscopy time are 3 Gy, 5 Gy, 500 Gy cm^2, and 60 min, respectively. In individual cases where the observed metric exceeds the SRDL, NCRP recommends that the interventionalist place a note in the medical record justifying the radiation dose used. Whenever an SRDL is exceeded, NCRP also recommends that the patient and any caregivers should be informed about possible tissue effects (deterministic effects) and the need for follow-up. The SRDL of 3 Gy for $D_{\text{skin,max}}$ is substantially lower than the 15 Gy $D_{\text{skin,max}}$ which The Joint Commission considers a reviewable sentinel event. However, The Joint Commission recommends that radiation doses to the same body area be summed over a 6 to 12 month period (TJC, 2006; 2011).

6.3.4.2 *Recommendations for Aggregating and Analyzing Data from FGI Procedures*. NCRP Report No. 168 (NCRP, 2010) recommends that data from all instances of a specific FGI procedure should be combined into a facility data set (FDS) that is then compared to an advisory data set (ADS). The ADS for an FGI procedure is generated by obtaining dosimetric data for the specific procedure from a large number of facilities. The ADS includes data for all instances of the procedure at each of these facilities, rather than a selected sample of the data. The dose distributions of the ADS and FDS are then compared. Specific attention is directed to the fraction of a facility's cases that exceed the SRDL.

NCRP Report No. 168 (NCRP, 2010) recommends comparing the 50th percentile (median value) of the FDS to the 75th percentile of the ADS. The ADS's 75th percentile corresponds to the RLs in this Report. If the 50th percentile of the FDS exceeds the RL, an investigation should be undertaken. Investigations may also be desirable if the local median is below the 25th or above the 50th percentile (median) of the ADS. Radiation use below the 25th percentile might be attributable to incomplete procedures, inadequate image quality, or superior dose management. Radiation use above the 75th percentile might reflect poor equipment and settings, suboptimal procedure performance, or high clinical complexity. This portion of the analysis is analogous to the analytical process used with DRLs. The only difference is that the data used for RLs are derived from the entire patient population, rather than a sample of the population as is used for DRLs.

If an investigation is indicated, the fluoroscopy equipment should be investigated first. If the equipment is functioning properly and within specification, procedure protocols and operator technique should be examined (NRPB, 1990; Vano and Gonzalez, 2001; Wall, 2001). This sequence is recommended because equipment evaluation is relatively easy and operator evaluation is difficult.

The analysis can be extended to individual operators or to individual FGI-procedure rooms by comparing operator- or room-specific data to a published ADS, the local FDS, or to other pooled distributions of data for multiple facilities. Care should be taken in such an analysis to account for statistical interactions (*e.g.*, statistical confounding between the operator and the FGI-procedure room). This subgroup analysis can also be used for SRDL investigations.

6.3.4.3 *Additional Recommendations for Analyzing Data from Multiple FGI Procedures*. The techniques outlined above depend on collection and publication of ADS for a large number of FGI procedures. While FGI procedures are frequently described using text descriptions such as hepatic chemoembolization or percutaneous nephrostomy, a more robust method of identifying common procedures sets will be needed to develop a comprehensive ADS catalog. As demonstrated in Table 6.12, CPT coding patterns can be used to identify common procedure types. Combining CPT codes with other data elements in the administrative dataset (ICD-09/10 codes, patient age, and date of service) could lead to an even more granular ADS catalog. In the future, an even more granular procedure description could be achieved using the RadLex taxonomy (ACR, 2011b). Efforts to develop the necessary infrastructure needed to support creation of an ADS catalog, construct RLs, and facilitate data analysis are underway as of 2011 (Duncan *et al*., 2011; FDA, 2010e).

6.4 Dental Radiography

6.4.1 *Dental Intraoral Radiographic Examinations*

The most recent U.S. survey of patient doses from posterior bitewing views was conducted in 1999 and reported in 2007 (Moyal, 2007). This NEXT survey examined the radiological practices in 350 dental offices (94 % general practice) across the United States and included film type, dose in air, and frequency of examinations. In 1999 the most common film used in the United States was Kodak Ultraspeed® film, a D-speed film (73 % of offices). The next most common was Kodak Ektaspeed® film, an E-speed film (17 %

of offices). This situation is essentially unchanged in 2009.[5,6] No mention is made in the NEXT report of the use of digital imaging systems.

The NEXT survey found that the median number of radiographs made per patient was four and this was also the most frequent number of images made per patient (mode). These data indicate that four bitewing views was the most common intraoral dental radiographic examination. The entrance skin dose was measured and related to the film speed used (Table 6.14).

6.4.1.1 *U.S. Diagnostic Reference Levels for Intraoral Examinations*. The current intraoral dental DRL in the United States, recommended by AAPM, is 2.3 mGy for E-F-speed film and 3.5 mGy for D-speed film (Gray *et al*., 2005).

6.4.1.2 *Non-U.S. Diagnostic Reference Levels for Intraoral Examinations*. A study of dental doses in the United Kingdom from 1995 to 1998 also found reduced patient dose from dentists using E- rather than D-speed film (Napier, 1999). Furthermore, mean skin-entrance dose was reduced by using 60 to 70 kVp rather than 45 to 55 kVp. From these data, NRPB in 1999 recommended a DRL dose of 4 mGy for an adult mandibular molar intraoral radiograph (Napier, 1999). These lower doses, compared to earlier surveys in the United Kingdom, were obtained because of reduced use of low kilovolt peak machines and greater adoption of faster speed film, and, perhaps, the greater use of rectangular collimation (NRPB, 2001). A report of a joint working party recommended a DRL dose of 2.1 mGy (IPEM, 2004). More recent survey data in the United Kingdom, 2002 to 2004, showed that the 75th percentile dose has fallen to 2.4 mGy (Gulson *et al*., 2007).

A study of dental clinical practices in Germany also found wide variation in patient doses (Poppe *et al*., 2007a). The authors found that use of fast film and high kilovolt peak does not always correlate with lower patient doses. The discrepancy may be the consequence of poor film-processing techniques in some offices. The authors proposed a DRL of 1.67 mGy for posterior bitewing views.

In Greece, entrance surface dose showed significant variation, from 0.30 to 16.09 mGy with a 75th percentile value of 3.37 mGy (Hatziioannou *et al*., 2005). These authors recommended a DRL of 3.5 for all intraoral equipment currently operating in Greece and

[5]Guffey, B. (2009). Personal communication (Dental Systems Group, Carestream Health, Inc., Rochester, New York).

[6]D-speed film sales were 72 % of dental film sales in 2009.

TABLE 6.14—*Entrance skin air kerma for dental posterior bitewing views by film speed (in milligray) from NEXT data.*[a]

Speed	N	Mean	Standard Deviation	Minimum	25th Percentile	Median	75th Percentile	Maximum
D	233	1.70	0.91	0.074	1.05	1.63	2.29	5.55
E	53	1.30	0.64	0.074	0.96	1.16	1.61	2.91

[a] Original data in milliroentgen converted to milligray by using a conversion factor of 8.76 mGy R^{-1}.

a value of 2.8 for new installations. A DRL of 3.5 mGy is also recommended in Spain (Gonzalez *et al.*, 2001) while legislation in Belgium recommends a DRL of 2.5 mGy for a maxillary molar and a maximal limit of 4 mGy (FANC, 2008a).

6.4.1.3 *Recommended Diagnostic Reference Levels and Achievable Doses for Intraoral Examinations*. NCRP recommends adopting the NEXT E-speed film 75th percentile entrance skin dose of 1.6 mGy as the DRL for dental intraoral periapical and bitewing radiography. The NEXT survey demonstrated that a decade ago this goal was readily achievable with proper radiographic technique. This value (1.6 mGy) is also well above the mean value of 1.3 mGy for users of E-F-speed film and the median value of 0.8 mGy for users of digital systems in the Michigan surveys (MDCH, 2012a; 2012b). Recommendation of a single DRL for all intraoral imaging is consistent with the principle that patient dose should follow the ALARA principle.

NCRP recommends an achievable dose of 1.2 mGy for intraoral radiography. This is the median dose for E-F film in the Michigan survey and is higher than for digital systems.

This DRL value is intended to be one standard, regardless of the type of imaging receptor used by the dentist; whether D- or E-F-speed film, or storage phosphor, charge-coupled device, complementary metal oxide semiconductor digital systems. It is recognized, and intended, that meeting this standard will most likely require dentists in the United States who use D-speed film to convert to E-F-speed film. Such a conversion requires only reducing the exposure time or milliamperage by half. The kilovolt peak and processing conditions are otherwise unaffected. This conversion carries immediate patient benefits without loss of diagnostic quality (Ludlow *et al.*, 2001b).

It is the position of the American Dental Association that dentists should use E-F-speed film (ADA, 2006). This position is also held by NCRP (2003) and the American Academy of Oral and Maxillofacial Radiology (White *et al.*, 2001). E- and F-speed films are also used widely in Europe and recommended by the European guidelines on radiation protection in dental imaging (EC, 2004).

6.4.2 *Dental Cephalometric Radiographic Examinations*

In 1999, the NEXT survey examined doses from lateral cephalometric radiography (Table 6.15). Skin doses for cephalometric radiographs are much lower than for intraoral imaging. This is because of the use of intensifying screens with cephalometric imaging rather than direct-exposure film for intraoral images. In the

TABLE 6.15— *Incident air kerma (free-in-air) for lateral cephalometric radiography (in milligray) from NEXT survey data.*[a]

N	Mean	Standard Deviation	Minimum	25th Percentile	Median	75th Percentile	Maximum
20	0.15	0.12	0.012	0.068	0.13	0.20	0.43

[a]Original values in milliroentgen were converted to milligray by using a conversion factor of 8.76 mGy R^{-1}.

NEXT survey the large majority of orthodontic offices used rare-earth screen-film combinations while one-third of the offices evaluated used slow screen-film systems. Most offices did not use grids. Both the use of rare-earth screens and absence of grids reduce patient dose.

6.4.2.1 *U.S. Diagnostic Reference Levels for Cephalometric Examinations*. Recent cephalometric data are available from Michigan (MDCH, 2012a; 2012b). From 2005 to 2009, 670 units were tested using screen-film imaging and 152 using digital imaging. For both receptor types the median dose was 0.09 mGy. The 85th percentile was 0.14 mGy for screen-film imaging and 0.16 mGy for digital imaging.

6.4.2.2 *Non-U.S. Diagnostic Reference Levels for Cephalometric Examinations*. Various DRLs for lateral cephalometric radiography have been recommended in Europe. One study collected data on 523 patients at 78 sites (Gonzalez *et al.*, 2004). The authors also reviewed the existing world literature available at that time. On the basis of their study the authors recommended that the rounded 75th percentile dose of 0.4 mGy be used as the DRL. A study in Germany found a 75th percentile air-kerma area product value of 26.4 and 32.6 mGy cm^2 for children and adults respectively (Looe *et al.*, 2007). These data reflect a population of orthodontists using primarily 400-speed screen-film combinations. In Belgium, the recommended air-kerma area product DRL is 40 mGy cm^2 (FANC, 2008b). The European guidelines on radiation protection in dental imaging do not recommend a DRL for cephalometric radiography (EC, 2004).

6.4.2.3 *Recommended Diagnostic Reference Levels for Cephalometric Examinations*. NCRP recommends adopting the Michigan 85th percentile entrance skin dose of 0.14 mGy as the DRL for cephalometric radiography. In terms of air-kerma area product, NCRP recommends adopting the 75th percentile air-kerma area product of 26.4 and 32.6 mGy cm^2 for children and adults, respectively, as the air-kerma area product DRL for cephalometric radiography (Looe *et al.*, 2007).

NCRP recommends an achievable dose of 0.09 mGy for cephalometric radiography. This is the median dose found in the Michigan survey for both film and digital systems. In terms of air-kerma area product values, NCRP recommends an achievable dose of 14 and 17 mGy cm^2 for children and adults, respectively. These are also the median values from a study in Germany (Looe *et al.*, 2007).

6.4.3 *Dental Panoramic Radiographic Examinations*

Panoramic imaging is a common dental radiographic examination that covers all the tooth-bearing areas in a single image. In this examination the x-ray source rotates approximately two-thirds around the patient's head with the patient seated or standing and the head stabilized. The image receptor is a screen-film combination, storage phosphor plate (computed radiography), or a charge-coupled device sensor (digital radiography). These examinations are used for an overall assessment of the teeth and jaw such as in general dentistry, orthodontics, oral surgery, or for assessment of large disease processes that extend beyond the size of an intraoral film (*e.g.*, dentigerous cysts). While the patient dose from panoramic images using screen-film, computed radiography, and digital radiography modalities are comparable, each is much less than from a conventional, intraoral full-mouth examination.

6.4.3.1 *U.S. Diagnostic Reference Levels for Panoramic Examinations*. There are no DRLs for panoramic imaging in the United States.

6.4.3.2 *Non-U.S. Diagnostic Reference Levels for Panoramic Examinations*. A survey in Spain found a mean occipital skin dose of 0.53 mGy and a 75th percentile value of 0.66 mGy (Gonzalez *et al.*, 2001). A survey in Finland found a mean dose-area product of 94 mGy cm^2 with a range of 34 to 254 mGy cm^2 (Rantanen, 2000). Several studies have been conducted in the United Kingdom. A 1999 study found a mean DWP of 57.4 mGy mm with a 75th percentile value of 66.7 mGy mm and a range of 1.7 to 328 mGy mm (Napier, 1999). A study in the United Kingdom found a mean air kerma-area product of 113 mGy cm^2 and a mean DWP of 65.2 mGy mm with a 75th percentile of 75.8 mGy mm (Williams and Montgomery, 2000). Napier (1999) found a mean DWP of 66.7 mGy mm in England in 1995 to 1998 and recommended a DRL of 65 mGy mm. A more recent study in the United Kingdom conducted from 2002 to 2004 found that the mean DWP was reduced to 52 mGy mm (Gulson *et al.*, 2007). However, the authors highlighted the fact that this was not the case for all dental panoramic equipment as digital systems showed an increase in DWP approaching the 75th percentile dose of 65 mGy mm. In Greece, a survey found air-kerma area product 75th percentile values of 117, 97, and 77 mGy cm^2 for men, women, and children, respectively (Tierris *et al.*, 2004). A study in Germany found 75th percentile air-kerma area product values of 101, 87, 84, and 75 mGy cm^2, for large adults, adult males, adult females, and children, respectively, and proposed using these values as DRLs (Poppe *et al.*, 2007b). In Belgium, a DRL of 75 mGy mm (DWP) or 140 mGy cm^2 (air-kerma area product) for an adult man is recommended (FANC, 2008c).

Although there are virtues to both air-kerma area product and DWP (Doyle *et al.*, 2006; Gulson *et al.*, 2007; Helmrot and Alm Carlsson, 2005; IPEM, 2004), each of these measurements has limitations and thus may not serve as a good proxy of patient dose. For instance, making measurements in the absence of a patient does not reflect the benefits of machines that use automatic dose controls. Because of the difficulties in quantifying patient dose during panoramic radiography, the European guidelines on radiation protection in dental imaging decided against establishing a DRL for panoramic radiography until better methods are established (EC, 2004).

6.4.3.3 *Recommended Diagnostic Reference Levels for Panoramic Examinations*. Seventy-fifth percentile air-kerma area product for panoramic radiography have only been reported in Greece and Germany. For adult males these 75th percentile values are 97 and 84 mGy cm^2 for Greece and Germany, respectively. Accordingly, NCRP recommends adopting an air-kerma area product of 100 mGy cm^2 for the panoramic DRL. This value is readily achieved in contemporary practices and would serve to identify situations where patients are receiving significantly higher radiation doses.

For panoramic radiography, NCRP recommends an achievable dose-area product of 76 mGy cm^2. This is the median dose-area product for adult males in Germany (Poppe *et al.*, 2007b).

6.5 Nuclear Medicine Procedures

Determining RLs for commonly performed nuclear medicine studies is challenging due to limited available survey data, the number of radiopharmaceuticals used, and variability in how studies are performed. Mean effective doses from common nuclear medicine studies range from 0.3 to 20 mSv (Mettler *et al.*, 2008). RLs are expressed in administered activities (megabecquerel or millicurie) and not absorbed doses for diagnostic nuclear medicine studies.

In 2003, a survey from SNM evaluating utilization and the types of studies performed was sent to 4,425 nuclear medicine facilities throughout the United States performing general nuclear medicine studies and nuclear cardiac studies. The hospitals included community, university, private and government. Some hospitals and outpatient imaging centers performed a combination of general and nuclear cardiac imaging, and some only performed general and some only nuclear cardiac studies.

There were 983 respondents. Fifty-eight percent of the respondents identified their facility as hospital type of which 67 % of the hospitals were community based, 23 % private, 8 % government, and 2 % university based. Ninety-six percent of hospital-based facilities performed cardiac nuclear medicine procedures, 18 % performed PET, and 64 % performed nuclear medicine therapy. For nonhospital-based facilities, 36 % performed cardiac imaging only, 13 % performed general nuclear medicine only, 7 % performed PET, and 12 % performed nuclear medicine therapy (Merlino, 2004).

Ninety-eight percent of nonhospital-based facilities had radiopharmaceuticals commercially prepared compared to 76 % of hospital-based facilities. For the hospital-based facilities there was an average of 1,253 inpatient procedures and 2,318 outpatient procedures of which bone scans were the most common types of studies performed. For nonhospital-based facilities the average number of outpatient procedures was 2,093 of which the most common study performed was cardiac (Merlino, 2004).

6.5.1 *U.S. Reference Levels for Nuclear Medicine Examinations Compared with SNM Minimum and Maximum Doses*

Guidelines for recommended minimum and maximum doses are available from SNM, which has procedure guides for many of the commonly performed nuclear medicine studies, and ACR. ICRP Publication 53 entitled *Radiation Dose to Patients from Radiopharmaceuticals* and two addenda, ICRP Publications 80 and 106, are also good references providing information on biokinetic models, absorbed doses, and effective doses for commonly used radiopharmaceuticals (ICRP, 1988; 1998; 2008). There are limited survey data available in the United States for comparison with the SNM recommendations. In 2010, a survey was performed of routine, minimum and maximum administered radiopharmaceutical doses for nuclear medicine procedures. This survey was sent to selected nuclear medicine departments at academic centers with nine respondents. A range of administered activities was reported and compared to previously established minimum and maximum values recommended by SNM. The results are summarized in Table 6.16 and compared to the recommended administered activity range from ICRP where available.

In children, the recommended dose for diagnostic studies should be adjusted based on weight or body surface area. However, the administered radiopharmaceutical activities in children can vary

TABLE 6.16—*2010 survey data: Diagnostic studies with comparison with SNM recommended minimum and maximum administered activity.*

Radiopharmaceutical[a]	Activity	*n*	Minimum		Maximum		Mean		Median		75th percentile		SNM	
			MBq	mCi	MBq	mCi	MBq	mCi	MBq	mCi	MBq	mCi	MBq	mCi
^{99m}Tc-dimercaptosuccinic acid (DMSA)	Routine	9	111	3.00	370	10	189	5.11	185	5.00				
	Minimum	7	3.70	0.10	333	9.00	107	2.90	93	2.50	189	5.11		
	Maximum	7	163	4.40	407	11	252	6.81	185	5.00	289	7.81		
^{99m}Tc-mercaptoacetyl triglycine (MAG3)	Routine	9	111	3.00	370	10	261	7.06	278	7.50				
	Minimum	7	11.10	0.30	333	9.00	142	3.83	37	1.00	283	7.65		
	Maximum	7	111	3.00	407	11	325	8.79	370	10	379	10		
^{99m}Tc-methylene diphosphate (MDP)	Routine	8	740	20	925	25	833	23	833	23				
	Minimum	6	370	10	833	23	623	17	666	18	848	23	740	20
	Maximum	8	888	24	1,480	40	1,083	29	1,064	29	1,185	32	1,110	37

TABLE 6.16—(*continued*)

Radiopharmaceutical[a]	Activity	*n*	Minimum		Maximum		Mean		Median		75th percentile		SNM	
			MBq	mCi	MBq	mCi	MBq	mCi	MBq	mCi	MBq	mCi	MBq	mCi
^{99m}Tc-mebrofenin or disofenin (hepatobiliary scans)	Routine	9	111	3.00	296	8.00	189	5.11	185	5.00				
	Minimum	7	19	0.50	200	5.40	105	2.83	89	2.40	189	5.12	56	1.50
	Maximum	7	133	3.60	370	10	234	6.31	222	6.00	282	7.62	185	5.00
^{123}I-metaiodobenzylguanidine (MIBG)	Routine	9	185	5.00	370	10	288	7.78	370	10				
	Minimum	6	19	0.50	333	9.00	157	4.25	111	3.00	300	8.07		
	Maximum	7	222	6.00	407	11	354	9.57	370	10	391	11		
^{99m}Tc-labeled solids (GI emptying)	Routine	8	19	0.50	37	1.00	32	0.88	37	1.00				
	Minimum	6	3.70	0.10	33	0.90	20	0.55	18	0.50	31	0.86	19	0.50
	Maximum	6	19	0.50	74	2.00	43	1.15	41	1.10	50	1.34	37	1.00

^{123}I (NaI) (for the thyroid)	Routine	6	7.40	0.20	15	0.40	9.34	0.25	7.68	0.21				
	Minimum	8	3.70	0.10	7.40	0.20	5.52	0.15	5.44	0.15	7.77	0.21	7.40	0.20
	Maximum	6	7.40	0.20	22	0.60	13	0.34	12	0.32	26	0.71	19	0.50
^{99m}Tc-exametazime (HMPAO)	Routine	6	740	20	1,110	30	863	23	740	20				
	Minimum	6	370	10	999	27	617	17	555	15	733	21	370	10
	Maximum	6	740	20	1,480	40	1,079	29	1,073	29	1,193	32	1,110	30
^{99m}Tc-ethylcysteinate dimer (ECD)	Routine	4	740	20	1,110	30	925	25	925	25				
	Minimum	2	370	10	999	27	685	19	685	19	887	24	370	10
	Maximum	2	1,221	33	1,480	40	1,351	37	1,351	37	1,294	37	1,110	30
^{99m}Tc-sestamibi (rest)	Routine	3	555	15	740	20	617	17	555	15				
	Minimum	4	148	4.00	592	16	402	11	435	12	519	14		
	Maximum	4	610	17	1,665	45	1,022	28	907	25	1,153	31	1,480	40

TABLE 6.16—(*continued*)

Radiopharmaceutical[a]	Activity	*n*	Minimum		Maximum		Mean		Median		75th percentile		SNM	
			MBq	mCi	MBq	mCi	MBq	mCi	MBq	mCi	MBq	mCi	MBq	mCi
^{99m}Tc-sestamibi (stress)	Routine	4	740	20	1,480	40	1,017	28	925	25				
	Minimum	4	148	4.00	1,332	36	611	17	481	13	945	26	—	—
	Maximum	4	888	24	1,665	45	1,277	35	1,277	35	1,452	39	1,480	40
^{99m}Tc-tetrofosmin (rest)	Routine	4	370	10	925	25	648	18	648	18				
	Minimum	3	148	4.00	592	16	370	10	370	10	590	16	—	—
	Maximum	4	444	12	1,665	45	981	27	907	25	1,089	29	1,480	40
^{99m}Tc-tetrofosmin (stress)	Routine	4	370	10	925	25	740	20	833	23				
	Minimum	3	148	4.00	592	16	370	10	370	10	1,007	27	—	—
	Maximum	4	370	10	1,776	48	1,314	36	1,295	35	1,459	39	1,480	40

^{201}Tl chloride (stress/rest)	Routine	7	93	2.50	185	5.00	127	3.43	111	3.00				
	Minimum	4	37	1.00	133	3.60	89	2.40	93	2.50	133	3.60	74	2.00
	Maximum	4	111	3.00	185	5.00	156	4.23	165	4.45	172	4.64	148	4.00
^{99m}Tc-macroaggregated albumin (MAA)	Routine	9	111	3.00	222	6.00	168	4.56	148	4.00				
	Minimum	9	19	0.50	200	5.40	99	2.68	111	3.00	147	3.98	37	1.00
	Maximum	8	111	3.00	244	6.60	199	5.39	222	6.00	226	6.12	111	3.00
^{99m}Tc-labeled RBC (for GI bleeding)	Routine	9	555	15	1,110	30	842	23	740	20				
	Minimum	7	37	1.00	999	27	608	16	740	20	832	22	740	20
	Maximum	7	740	20	1,480	40	1,142	31	1,110	30	1,199	32	925	25
^{99m}Tc-labeled RBC (for MUGA)	Routine	9	740	20	1,110	30	904	24	925	25				
	Minimum	7	370	10	925	25	719	19	740	20	916	25	740	20
	Maximum	8	370	10	1,850	50	1,152	31	1,110	30	1,301	35	925	25

TABLE 6.16—(*continued*)

Radiopharmaceutical[a]	Activity	n	Minimum		Maximum		Mean		Median		75th percentile		SNM	
			MBq	mCi	MBq	mCi	MBq	mCi	MBq	mCi	MBq	mCi	MBq	mCi
^{67}Ga (for inflammatory disease)	Routine	8	185	5.00	370	10	236	6.38	185	5.00				
	Minimum	5	9.25	0.25	333	9.00	154	4.15	148	4.00	266	7.18	148	4.00
	Maximum	5	185	5.00	555	15	315	8.50	222	6.00	371	10.02	222	6.00
^{67}Ga (for tumor imaging)	Routine	7	222	6.00	370	10	338	9.14	370	10				
	Minimum	5	19	0.50	333	9.00	218	5.90	296	8.00	330	8.93	148	4.00
	Maximum	5	370	10	555	15	437	12	407	11	450	12	222	6.00
^{18}F-fluorodeoxyglucose (FDG)	Routine	8	370	10	555	15	518	14	555	15				
	Minimum	6	269	8.00	500	14	367	10	333	9.00	461	12	148	4.00
	Maximum	7	555	15	814	22	669	18	666	18	710	19	740	20

^{99m}Tc-sulfur colloid (liver/spleen)	Routine	9	148	4.00	370	10	201	5.44	185	5.00				
	Minimum	5	74	2.00	333	9.00	162	4.40	148	4.00	213	5.77	148	4.00
	Maximum	7	148	4.00	407	11	264	7.14	278	7.50	293	7.92	222	6.00
^{99m}Tc-DTPA (renal)	Routine	6	111	3.00	740	20	389	11	370	10				
	Minimum	6	19	0.50	444	12	157	4.25	56	1.50	407	11		
	Maximum	6	111	3.00	666	18	413	12	481	13	587	16		

[a]DTPA = diethylenetriame-pentaacetic acid
ECD = ethylcysteinate dimer
FDG = fluorodeoxyglucose
HMPAO = hexamethylpropyleneamine oxime
MAA = macroaggregated albumin
MAG3 = mercaptoacetyltriglycine
MDP = methylene diphosphate
MIBG = metaiodobenzylguanidine
MUGA = multi gated acquisition scan
RBC = red blood cells

greatly. This was shown in a 2008 survey of 13 pediatric hospitals in North America (Treves *et al.*, 2008) evaluating the routine, minimum and maximum administered radiopharmaceutical administered activities of commonly performed diagnostic studies (Table 6.17). In the pediatric survey for some radiopharmaceuticals the minimum administered activities differed up to a factor of 20 and the maximum administered activities up to a factor of 10 (Table 6.17). The administered activity used in children may be more variable as many radiopharmaceutical manufacturers do not provide guidance on pediatric administered activity, although they do provide routine recommended administered activities for adults (Treves *et al.*, 2008).

6.5.2 *Recommended Reference Levels for Diagnostic Nuclear Medicine Examinations*

6.5.2.1 *Recommended Reference Levels for Adult Nuclear Medicine Examinations*. The 2010 survey data showed that there is a large range of minimum and maximum administered activities for many of the commonly performed diagnostic nuclear medicine studies.

The minimum administered activities were higher than the SNM minimum administered activity recommendations for the majority of the nuclear medicine studies including brain, hepatobiliary, and perfusion lung studies.

The maximum administered activities from the survey were also higher for almost all of the nuclear medicine studies compared with the SNM maximum administered activity recommendation. Although the maximum administered activities were higher than the SNM recommendations, except for high-dose cardiac studies, perfusion lung scans, and gallium scans (rarely performed as gallium scans have been replaced with PET for oncologic imaging), this difference was usually minor. The largest magnitude of administered activity discrepancy was seen for perfusion lung scans performed with ^{99m}Tc-MAA (macroaggregated albumin) where the 75th percentile value of 226 MBq (6.12 mCi) was twice the SNM recommended maximum administered activity of 111 MBq (3 mCi). The highest maximum administered activities were seen for cardiac studies with ^{99m}Tc-sestamibi and ^{99m}Tc-tetrofosmin which were as high as 1,776 MBq (48 mCi) on the survey compared to the SNM recommendation of 1,480 MBq (40 mCi). This may be due to differences in equipment with some gamma cameras requiring higher administered activities of radiopharmaceutical in order to produce diagnostic-quality images and the increasing obesity in the U.S. adult population, which can require higher administered activities. For cardiac studies, additional recommendations and

TABLE 6.17—*2008 survey data of 13 pediatric hospitals: Radiopharmaceutical administered activity for children (Treves et al., 2008).*

Radiopharmaceutical[a]	Activity	*n*	Minimum		Maximum		Median		Mean	
			MBq	mCi	MBq	mCi	MBq	mCi	MBq	mCi
^{99m}Tc-dimercaptosuccinic acid (DMSA)	Activity kg^{-1}	8	1.11	0.03	3.70	0.10	2.22	0.06	2.35	0.06
	Minimum	11	5.55	0.15	74	2.00	19	0.50	26	0.71
	Maximum	11	74	2.00	222	6.00	185	5.00	151	4.09
^{99m}Tc-mercaptoacetyltriglycine (MAG3)	Activity kg^{-1}	8	1.85	0.05	10	0.28	5.55	0.15	5.69	0.15
	Minimum	12	19	0.50	148	4.00	37	1.00	54	1.46
	Maximum	13	111	3.00	370	10	370	10	279	7.36
^{99m}Tc-methylene diphosphate (MDP)	Activity kg^{-1}	8	7.40	0.20	13	0.36	11	0.30	11	0.29
	Minimum	13	22	0.60	185	5.00	93	2.50	100	2.70
	Maximum	13	666	18	925	25	740	20	820	22

TABLE 6.17—(*continued*)

Radiopharmaceutical[a]	Activity	*n*	Minimum		Maximum		Median		Mean	
			MBq	mCi	MBq	mCi	MBq	mCi	MBq	mCi
^{99m}Tc-diisopropyl iminodiacetic acid (DISIDA)	Activity kg^{-1}	7	1.85	0.05	3.70	0.10	2.78	0.075	2.97	0.08
	Minimum	13	15	0.40	74	2.00	37	1.00	36	0.98
	Maximum	13	93	2.50	370	10	185	5.00	201	5.42
^{123}I-metaiodobenzylguan idine (MIBG)	Activity kg^{-1}	7	5.18	0.14	7.40	0.20	5.55	0.15	5.45	0.15
	Minimum	11	37	1.00	185	5.00	37	1.00	76	2.05
	Maximum	13	296	8.00	370	10	370	10	360	10
^{99m}Tc-NaTcO$_4$ (for Meckel's diverticulum)	Activity kg^{-1}	7	1.63	0.04	5.92	0.16	5.18	0.14	4.46	0.12
	Minimum	12	7.40	0.20	148	4.00	37	1.00	49	1.33
	Maximum	12	111	3.00	555	15	370	10	348	9.42

^{123}I (NaI) (for the thyroid)	Activity kg^{-1}	4	0.06	0.002	0.22	0.006	0.10	0.0028	0.12	0.0033
	Minimum	10	0.56	0.560	11	0.30	3.70	0.10	3.76	0.10
	Maximum	11	3.70	0.10	20	0.54	8.14	0.22	9.45	0.26
^{99m}Tc-ethylcysteinate dimer (ECD) or ^{99m}Tc-exametazime (HMPAO)	Activity kg^{-1}	7	1.85	0.05	16	0.43	11	0.29	11	0.29
	Minimum	11	19	0.50	370	10	185	5.00	175	4.73
	Maximum	12	370	0.15	1,110	30	740	20	811	22
^{99m}Tc-sestamibi (MIBI)	Activity kg^{-1}	7	5.70	5.70	19	0.50	13	0.35	12	0.32
	Minimum	10	37	1.00	518	14	148	4.00	211	5.70
	Maximum	12	370	10	1,110	30	777	21	792	21
^{99m}Tc-macroaggregated albumin (MAA)	Activity kg^{-1}	7	1.11	0.03	4.88	0.13	1.85	0.05	2.31	0.06
	Minimum	12	7.40	0.20	37	1.00	19	0.50	21	0.58
	Maximum	12	111	3.00	222	6.00	139	3.75	144	3.89

TABLE 6.17—(*continued*)

Radiopharmaceutical[a]	Activity	*n*	Minimum		Maximum		Median		Mean	
			MBq	mCi	MBq	mCi	MBq	mCi	MBq	mCi
^{99m}Tc-Ultratag (Mallinckrodt) (for GI bleeding)	Activity kg^{-1}	6	3.70	0.10	11	0.30	8.33	0.23	7.92	0.21
	Minimum	9	37	1.00	148	4.00	74	2.00	78	2.11
	Maximum	11	185	5.00	740	20	740	20	616	17
^{99m}Tc-denatured RBCs (for the spleen)	Activity kg^{-1}	2	1.85	0.05	2.59	0.07	2.22	0.06	2.22	0.06
	Minimum	6	19	0.50	93	2.50	37	1.00	40	1.08
	Maximum	6	74	2.00	740	20	111	3.00	216	5.83
^{99m}Tc-Ultratag (Mallinckrodt) (for multiple gated acquisitions)	Activity kg^{-1}	5	7.40	0.20	15	0.40	8.14	0.22	9.69	0.26
	Minimum	10	44	1.20	370	10	102	2.75	132	3.57
	Maximum	11	555	15	925	25	740	20	733	20

^{67}Ga (for inflammatory disease)	Activity kg^{-1}	5	1.48	0.04	2.59	0.07	1.85	0.05	1.90	0.05
	Minimum	4	9.25	0.25	74	2.00	28	0.75	35	0.94
	Maximum	6	111	3.00	185	5.00	167	4.50	154	4.17
^{67}Ga (for tumor imaging)	Activity kg^{-1}	5	2.96	0.08	5.25	0.14	4.07	0.11	3.90	0.11
	Minimum	10	9.25	0.25	111	3.00	37	1.00	41	1.11
	Maximum	12	222	6.00	370	10	333	9.00	313	8.46
^{18}F-fluorodeoxyglucose (FDG)	Activity kg^{-1}	6	5.18	0.14	7.40	0.20	5.37	0.15	5.67	0.15
	Minimum	6	19	0.50	74	2.00	37	1.00	46	1.25
	Maximum	8	370	10	555	15	407	11	430	12

[a]DISIDA = diisopropyl iminodiacetic acid
DMSA = dimercaptosuccinic acid
ECD = ethylcysteinate dimer
FDG = fluorodeoxyglucose
MAG3 = mercaptoacetyltriglycine
MDP = methylene diphosphate
MIBI = sestamibi
RBC = red blood cells

guidelines on appropriate utilization and dosing are also available from the American Society of Nuclear Cardiology (Cerqueira *et al.*, 2010; Henzlova *et al.*, 2006).

For the studies where the maximum administered activities exceeds the SNM maximum administered activities, the 75th percentile values for the maximum administered activities from the survey are closer to the SNM maximum administered activity. Seventy-fifth percentile maximum administered activities are recommended as RLs for the maximum administered activity for the diagnostic studies listed in the survey. However, as can be seen from the survey, the maximum administered activities used by the respondents are higher than the SNM recommendations and the 75th percentile RL recommendations.

For nuclear medicine, the 75th percentile maximum RLs should be used as guidelines to limit unnecessary radiation dose as long as diagnostic-quality nuclear medicine studies are obtained, but not as absolute limits. If diagnostic-quality nuclear medicine studies are not obtainable with these RLs due to higher dose requirements of the particular gamma cameras being used or patient weight, then the NCRP recommended RLs may need to be exceeded.

6.5.2.2 *Recommended Reference Levels for Pediatric Nuclear Medicine Examinations*. The survey of 13 pediatric hospitals in North America (Treves *et al.*, 2008) showed considerably more variability in the minimum and maximum administered activities for diagnostic nuclear medicine studies compared with the 2010 adult survey.

This issue was addressed by a working group from SNM through the Pediatric Imaging Council, Society for Pediatric Radiology, and ACR when consensus guidelines for administered radiopharmaceutical activities in children and adolescents were released in October 2010 for 11 commonly performed pediatric nuclear medicine studies (Gelfand *et al.*, 2011). (These guidelines are a part of the Image Gently® *Go with the Guidelines* initiative of the Alliance for Radiation Safety in Pediatric Imaging.)

These guidelines differ from the European Association of Nuclear Medicine Pediatric Dose Card (Lassmann *et al.*, 2007) as the administered activities in these guidelines are slightly lower for infants and small children, administered activities for ^{99m}Tc-DMSA (dimercaptosuccinic acid) and ^{18}F-FDG (fluorodeoxyglucose) are considerably lower, and administered activities for orally-administered ^{99m}Tc-labeled radiopharmaceuticals and for radionuclide cystography provide a range of administered activities for each type of study rather than an administered activity per kilogram.

These consensus recommendations which provide guidance for commonly performed pediatric nuclear medicine studies are recommended by NCRP. Administered activities can be further decreased if diagnostic-quality studies are obtainable from the equipment utilized. In some circumstances, if diagnostic-quality nuclear medicine studies are not obtainable with these administered activities due to a higher dose requirement of the particular gamma cameras being utilized or patient weight, then the consensus recommendation may need to be exceeded.

7. Summary of Diagnostic Reference Levels and Achievable Doses

Diagnostic reference levels (DRLs), which are a form of investigation levels (ICRP, 1996) represent an important tool in medical imaging as part of a process to optimize the radiation dose delivered to their patients. The goal is to produce images of comparable or improved image quality while reducing the radiation dose to the patient through optimization.

As noted, DRLs provide little incentive for optimization for the 75 % of the facilities with doses below the DRL for a particular examination. The achievable dose provides a dose level which is readily achievable by 50 % of the facilities when the median dose is chosen as the achievable dose. Consequently, NCRP recommends achievable doses as the median dose of the NEXT survey data or other survey data on which DRLs are based. A summary of the DRLs and achievable doses is provided in Table 7.1.

Due to the complexity of the RLs the reader is referred to Table 6.12 for FGI procedures, Table 6.16 for adult nuclear medicine studies, and Table 6.17 for pediatric nuclear medicine studies,

It should be stressed, that if DRLs are exceeded, "there should be a local review of procedures and equipment in order to determine whether the protection has been adequately optimized" (ICRP, 1991; 1996). However, ***DRLs should not be viewed as absolute determinants of appropriate use of medical radiation***. Rather, they are supplements to professional judgment which must ultimately consider the benefits and risks of ionizing radiation for medical imaging. ***DRLs are not intended for regulatory or commercial purposes or to establish legal standards of care***. In addition, DRLs are not to be applied to the doses for individual patients. It is essential to ensure that the appropriate clinical information is available in the image throughout the optimization process. ***Optimization must take into account both patient dose and clinical utility, based on image quality***.

TABLE 7.1—*NCRP recommended DRLs and achievable doses (in milligray, unless noted).*

	DRL	Achievable Dose
Adult PA chest (with grid)	0.15	0.11
Pediatric chest (without grid)	0.06	0.04
Pediatric chest (with grid)	0.12	0.07
AP abdomen	3.4	2.4
AP lumbosacral spine	4.2	2.8
Upper GI fluoroscopy (without oral contrast media)	56 mGy min^{-1}	40 mGy min^{-1}
Upper GI fluoroscopy (with oral contrast media)	81 mGy min^{-1}	72 mGy min^{-1}
Fluorographic image (without contrast)		
Film	3.9	2.5
Digital	1.5	0.9
Fluorographic image (with contrast)		
Film	27.5	18.7
Digital	9.9	5.3
Adult head CT ($CTDI_{vol}$)	75	57
Adult chest CT ($CTDI_{vol}$)	21	14
Adult abdomen-pelvis CT ($CTDI_{vol}$)	25	17
Pediatric head CT (5 y old) ($CTDI_{vol}$)	40	31
Pediatric abdomen-pelvis CT (5 y old) ($CTDI_{vol}$)	20	14
Dental posterior bitewing	1.6	1.2
Dental lateral cephalogram (adult)	0.14 mGy or 32 mGy cm^2	0.09 mGy or 17 mGy cm^2
Dental lateral cephalogram (pediatric)	26 mGy cm^2	14 mGy cm^2
Dental panoramic	100 mGy cm^2	76 mGy cm^2

Glossary

achievable dose: A dose which serves as a goal for optimization efforts. This dose is achievable by standard techniques and technologies in widespread use, while maintaining clinical image quality adequate for the diagnostic purpose. The achievable dose is typically set at the median value of the dose distribution.

advisory data set (ADS**):** In this Report, ADS is the full data set for a measured dose-related radiation field quantity (*i.e.*, $K_{a,r}$ or P_{KA}) that is used to evaluate the local dose distribution [*i.e.*, the facility data set (FDS)] for a group of patients for a particular type of fluoroscopically-guided interventional (FGI) procedure, in order to better manage patient doses at the local facility. An ADS for an FGI procedure is obtained by collecting data for a large number of procedures from each of many facilities. The entire data set (typically a lognormal distribution) is used and includes sufficient data to characterize the entire distribution, rather than just the 10th and 75th percentile values typically used for DRLs. The percentages of cases in ADS and FDS that exceed the substantial radiation dose level selected for FDS, and the values of $K_{a,r}$ or P_{KA} that define the potentially-high radiation dose procedure category are also used in the evaluation (see *facility data set*, *potentially-high radiation dose procedure*, and *substantial radiation dose level*).

air kerma: (see *FDA compliance air-kerma rate*, *kerma*, *total air kerma at the reference point*).

air-kerma area product (P_{KA}**):** The integral of the air-kerma free-in-air (*i.e.*, in the absence of backscatter) over the area of the x-ray beam in a plane perpendicular to the beam axis.

air kerma at the reference point ($K_{a,r}$**):** The air kerma at a point in space located at a fixed distance from the focal spot expressed in gray. For isocentric fluoroscopes (C-arms), the reference point lies on the central axis of the x-ray beam, 15 cm on the x-ray tube side of isocenter (IEC, 2000; 2010). The location of the reference point relative to the x-ray gantry does not change when the source-to-image-receptor distance is changed [Referred to as reference air kerma by IEC (2000; 2004; 2010).]

alveolar bone: The bone of the maxilla (upper jaw) or mandible (lower jaw) that supports the teeth.

anti-scatter grid: A device placed between the patient and image receptor to reduce the incidence of scattered radiation and increase the contrast in the image. The use of an anti-scatter grid results in an increased dose to the patient.

bitewing projection: A radiographic view that shows the crown of both the upper and lower teeth in a region on the same image. This view is used to detect interproximal tooth decay or alveolar bone loss associated with periodontal disease.

cephalometric radiograph: A lateral radiograph of the head, including the mandible, used to make measurements. Filters are often used to allow visualization of the soft tissue such as the nose and lips.

clinical dose optimization team (CDOT**):**. A group with a mix of expertise including imaging physicians, medical imaging physicists, senior radiologic technologists, and radiology management. The group works to optimize image quality and patient radiation doses.

computed radiography: A radiographic imaging technique in which an image is captured by a photostimulable phosphor plate and converted into a digital image.

craniofacial: Of or involving both the cranium and the face.

$CTDI_{100}$: The computed tomography dose index (CTDI) measured in a standardized phantom and integrated over the 100 mm length of the computed tomography pencil ionization chamber.

$CTDI_{vol}$: Represents the dose within the scan volume from a particular scan protocol for a standardized phantom. It is the $CTDI_w$ divided by the pitch.

$CTDI_w$: The weighted CTDI measure which is a combination of one-third of the $CTDI_{100}$ at the center and two-thirds of the $CTDI_{100}$ at the periphery of a standardized phantom.

deterministic effects: (see *tissue effects*).

diagnostic reference level (DRL**):** A radiation dose level serving as an investigational level. When doses exceed the DRL the reasons for the higher doses should be investigated. A process known as optimization is used to assure that the image quality is adequate for the clinical task and that the patient doses are appropriate.

digital radiography: A radiographic technique in which an image is captured by a solid-state detector and converted into a digital image.

direct-exposure film: Film which is exposed directly by x rays without the use of intensifying screens to produce a radiographic film image.

dose-area product: The product of the entrance skin dose and the cross-sectional area of the x-ray beam.

dose-length product: The product of the $CTDI_{vol}$ and the length of the scanned area in computed tomography.

dose-width product (DWP**):** The product of the dose at the secondary collimator without the patient present and the beam width.

effective dose: The effective dose is calculated by multiplying the actual dose to a specific organ times a risk weighting factor (each organ's relative radiosensitivity to developing cancer) and summing over the organs exposed. The effective dose represents the dose that the entire body would receive (uniformly) to result in the same risk of developing cancer.

E-F-speed film: A direct-exposure dental imaging film that is an E-speed film when hand processed and an F-speed film when machine processed. The designation of E/F-speed film is also used.

entrance surface air kerma ($K_{a,e}$): Designates the air-kerma incident on the patient including backscatter.

facility data set (FDS): In this Report, FDS consists of all of the data for a measured dose-related radiation field quantity (*i.e.*, $K_{a,r}$ or P_{KA}) for all cases of the specified procedure performed at an individual facility. A FDS is compared against the advisory data set (ADS) for the fluoroscopically-guided interventional procedure in order to better manage patient doses at the local facility. The percentages of cases in the FDS and ADS that exceed the substantial radiation dose level selected for the FDS, and the values of $K_{a,r}$ or P_{KA} that define the potentially-high radiation dose procedure category are also used in the evaluation (see *advisory data set*, *potentially high radiation dose procedure*, and *substantial radiation dose level*).

FDA compliance air-kerma rate: An U.S. Food and Drug Administration (FDA) compliance measurement for air-kerma rate. For C-arm systems the measurement is made 30 cm in front of the image receptor at any source-to-image-receptor distance. Therefore, the location of the reference point for this measurement will change relative to the x-ray gantry when the source-to-image-receptor distance is changed.

fluorographic-image acquisition: A single x-ray image obtained during a fluoroscopic examination to document pathology.

fluoroscopically-guided interventional (FGI) procedure: An interventional diagnostic or therapeutic procedure performed *via* percutaneous or other access routes, usually with local anesthesia or intravenous sedation, which uses external ionizing radiation in the form of fluoroscopy to localize or characterize a lesion, diagnostic site, or treatment site; monitor the procedure; and control and document therapy.

free-in-air: Radiation exposure measurements made without scattering media in the vicinity of the ionization chamber or radiation measurement device.

half-value layer: A measure of beam quality or hardness. The half-value layer, specified in millimeters of aluminum, is the amount of material which reduces the original beam intensity by one-half.

incident air kerma ($K_{a,i}$): Designates the air-kerma incident on the patient without backscatter.

interproximal caries: Interproximal refers to the surfaces of teeth that are in contact with adjacent teeth. Caries is the technical term for tooth decay.

investigation level: A level of radiation exposure which, when exceeded, will initiate a review or investigation of the reasons or need for that level of radiation exposure.

kerma: The sum of the initial kinetic energies of all the charged particles liberated by uncharged particles per unit mass of specified material. The SI unit for kerma is J kg^{-1}. The special name for kerma is the gray (Gy). Kerma can be quoted for any specified material at a point in free space or in an absorbing medium.

local self assessment: Evaluation of patient doses in order to obtain benchmark data when such data is not available in the literature or nationally.

malocclusion: Faulty contact between the upper and lower teeth when the jaw is closed.

optimization: Optimization in medical imaging is the balancing of the amount of ionizing radiation and image quality. As the amount of radiation increases the image quality typically improves. One must minimize the patient radiation dose while assuring that the image provides sufficient quality (information) to meet the clinical need. Optimization involves both the imaging systems (through testing and quality control) and imaging techniques such as kilovoltage and milliampere-seconds. Although this process implies optimization with the present imaging systems, one may choose to upgrade or replace imaging equipment where appropriate and when resources are available.

orofacial: Relating to the mouth and face.

orthodontic treatment: Dental treatment that corrects irregularities of the teeth relative to the surrounding anatomy.

osseous components: Components composed of, containing, or resembling bone.

periapical view: A radiographic view made of the teeth that shows the entire length of the tooth, from crown to end of the root. These are made of the upper teeth in the maxilla or of the lower teeth in the mandible. The teeth may be in the anterior or posterior regions.

periodontal disease: An inflammatory process of the gingiva (gums) that destroys alveolar bone and may result in tooth loss.

pitch: The amount of table increment per 360 degree rotation divided by the nominal scan width, when other scan factors are held constant.

positron emission tomography (PET)**:** A nuclear medicine imaging technique which produces functional images of the body using short-lived radioactive materials. The system detects pairs of gamma rays emitted indirectly by a positron-emitting radionuclide.

qualified medical physicist: An individual who is competent to practice independently one or more of the subfields in medical physics. The American College of Radiology (ACR) considers certification, continuing education, and experience in the appropriate subfield(s) to demonstrate that an individual is competent to practice one or more of the subfields in medical physics and to be a qualified medical physicist. ACR recommends that the individual be certified in the appropriate subfield(s) by the American Board of Radiology, the American Board of Medical Physics, or the Canadian College of Physicists in Medicine.

reference level (RL)**:** A radiation dose level when if exceeded results in an evaluation of the reasons for the higher than typical doses. RL is similar in concept to DRL but can be applied to areas other than diagnostic imaging (*e.g.*, interventional radiology).

spot image: A film (or digital) image acquired during a fluoroscopic examination to document pathology.

state-of-the-art data: Survey data from facilities representative of the best practices in terms of image quality and patient dose.

state-of-the-practice data: Survey data representative of today's practice (*i.e.*, usually a statistical sample including all types of facilities).

submandibular soft tissue: Soft tissue situated in the region below the lower jaw.

substantial radiation dose level (SRDL**):** An appropriately selected reference value used to trigger additional dose-management actions during a procedure and medical follow-up for a radiation level that might produce a clinically-relevant injury in an average patient. There is no implication that radiation levels above an SRDL will always cause an injury or that radiation levels below an SRDL will never cause an injury. The quantities and their SRDLs recommended are in Table 4.7 of NCRP Report No. 168 (NCRP, 2010)

tissue effects (deterministic effects): Effects that occur in all individuals who receive greater than a threshold dose; the severity of the effect varies with the dose above the threshold. Examples are radiation-induced cataracts (lens of the eye) and radiation-induced erythema (skin).

temporomandibular joint: The jaw joint formed by the mandible (lower jaw bone) articulating with the temporal (temple and side) bone of the skull.

workload: The number of medical imaging examinations carried out in a specific period of time (*e.g.*, the number of chest radiographs carried out per week or per year).

Abbreviations and Acronyms

ADS	advisory data set
ALARA	as low as reasonably achievable (principle)
AP	anterior-posterior
CBCT	cone-beam computed tomography
CDOT	clinical dose optimization team
CPT	Current Procedural Terminology (codes)
CT	computed tomography
CTDI	computed tomography dose index
$CTDI_{100}$	computed tomography dose index using a 10 cm (100 mm) long ionization chamber
$CTDI_{vol}$	volumetric computed tomography dose index based on the dose to the irradiated volume (*i.e.*, volume $CTDI_w$) and pitch
$CTDI_w$	weighted *CTDI*
DRL	diagnostic reference level
DWP	dose-width product
FDS	facility data set
FGI	fluoroscopically-guided interventional (procedure)
GI	gastrointestinal
ICD	International Statistical Classification of Diseases and Related Health Problems (codes)
K	kerma
$K_{a,e}$	entrance surface air kerma
$K_{a,i}$	incident air kerma
$K_{a,r}$	air kerma at the reference point
kerma	kinetic energy released in a mass
NEXT	Nationwide Evaluation of X-Ray Trends (Program)
PA	posterior-anterior
PET	positron-emission tomography
P_{KA}	air-kerma area product
PMMA	polymethyl-methacrylate (also known as acrylic, Lucite®, or Plexiglas®)
RAD-IR	Radiation Doses in Interventional Radiology Procedures (Study)
RL	reference level
SRDL	substantial radiation dose level

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The NCRP

The National Council on Radiation Protection and Measurements is a nonprofit corporation chartered by Congress in 1964 to:

1. Collect, analyze, develop and disseminate in the public interest information and recommendations about (a) protection against radiation and (b) radiation measurements, quantities and units, particularly those concerned with radiation protection.
2. Provide a means by which organizations concerned with the scientific and related aspects of radiation protection and of radiation quantities, units and measurements may cooperate for effective utilization of their combined resources, and to stimulate the work of such organizations.
3. Develop basic concepts about radiation quantities, units and measurements, about the application of these concepts, and about radiation protection.
4. Cooperate with the International Commission on Radiological Protection, the International Commission on Radiation Units and Measurements, and other national and international organizations, governmental and private, concerned with radiation quantities, units and measurements and with radiation protection.

The Council is the successor to the unincorporated association of scientists known as the National Committee on Radiation Protection and Measurements and was formed to carry on the work begun by the Committee in 1929.

The participants in the Council's work are the Council members and members of scientific and administrative committees. Council members are selected solely on the basis of their scientific expertise and serve as individuals, not as representatives of any particular organization. The scientific committees, composed of experts having detailed knowledge and competence in the particular area of the committee's interest, draft proposed recommendations. These are then submitted to the full membership of the Council for careful review and approval before being published.

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James F. Crow (1981) *How Well Can We Assess Genetic Risk? Not Very*
Harold O. Wyckoff (1980) *From "Quantity of Radiation" and "Dose" to "Exposure" and "Absorbed Dose"—An Historical Review*

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Currently, the following committees are actively engaged in formulating recommendations:

Program Area Committee 1: Basic Criteria, Epidemiology, Radiobiology, and Risk

SC 1-15 Radiation Safety in NASA Lunar Missions'
SC 1-20 Biological Effectiveness of Photons as a Function of Energy
SC 1-21 Multiplatform National Approach for Providing Guidance on Integrating Basic Science and Epidemiological Studies on Low-Dose Radiation Biological and Health Effects

Program Area Committee 2: Operational Radiation Safety

Program Area Committee 3: Nuclear and Radiological Security and Safety

Program Area Committee 4: Radiation Protection in Medicine

SC 4-4 Risks of Ionizing Radiation to the Developing Embryo, Fetus and Nursing Infant

Program Area Committee 5: Environmental Radiation and Radioactive Waste Issues

SC 5-1 Approach to Optimizing Decision Making for Late-Phase Recovery from Nuclear or Radiological Terrorism Incidents

Program Area Committee 6: Radiation Measurements and Dosimetry

SC 6-8 Operation TOMODACHI Radiation Dose Assessment Peer Review

In recognition of its responsibility to facilitate and stimulate cooperation among organizations concerned with the scientific and related aspects of radiation protection and measurement, the Council has created a category of NCRP Collaborating Organizations. Organizations or groups of organizations that are national or international in scope and are concerned with scientific problems involving radiation quantities, units, measurements and effects, or radiation protection may be admitted to collaborating status by the Council. Collaborating Organizations provide a means by which NCRP can gain input into its activities from a wider segment of society. At the same time, the relationships with the Collaborating Organizations facilitate wider dissemination of information about the Council's activities, interests and concerns. Collaborating Organizations have the opportunity to comment on draft reports (at the time that these are submitted to the members of the Council). This is intended to capitalize on the fact that Collaborating Organizations are in an excellent position to both contribute to the identification of what needs to be treated in NCRP reports and to identify problems that might result from proposed recommendations. The present Collaborating Organizations with which NCRP maintains liaison are as follows:

American Academy of Dermatology
American Academy of Environmental Engineers

American Academy of Health Physics
American Academy of Orthopaedic Surgeons
American Association of Physicists in Medicine
American Bracytherapy Society
American College of Cardiology
American College of Medical Physics
American College of Nuclear Physicians
American College of Occupational and Environmental Medicine
American College of Radiology
American Conference of Governmental Industrial Hygienists
American Dental Association
American Industrial Hygiene Association
American Institute of Ultrasound in Medicine
American Medical Association
American Nuclear Society
American Pharmaceutical Association
American Podiatric Medical Association
American Public Health Association
American Radium Society
American Roentgen Ray Society
American Society for Radiation Oncology
American Society of Emergency Radiology
American Society of Health-System Pharmacists
American Society of Nuclear Cardiology
American Society of Radiologic Technologists
American Thyroid Association
Association of Educators in Imaging and Radiological Sciences
Association of University Radiologists
Bioelectromagnetics Society
Campus Radiation Safety Officers
College of American Pathologists
Conference of Radiation Control Program Directors, Inc.
Council on Radionuclides and Radiopharmaceuticals
Defense Threat Reduction Agency
Electric Power Research Institute
Federal Aviation Administration
Federal Communications Commission
Federal Emergency Management Agency
Genetics Society of America
Health Physics Society
Institute of Electrical and Electronics Engineers, Inc.
Institute of Nuclear Power Operations
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International Society of Exposure Science
National Aeronautics and Space Administration
National Association of Environmental Professionals
National Center for Environmental Health/Agency for Toxic Substances
National Electrical Manufacturers Association
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Nuclear Energy Institute
Office of Science and Technology Policy
Paper, Allied-Industrial, Chemical and Energy Workers International Union
Product Stewardship Institute
Radiation Research Society
Radiological Society of North America
Society for Cardiovascular Angiography and Interventions
Society for Pediatric Radiology
Society for Risk Analysis
Society of Cardiovascular Computed Tomography
Society of Chairmen of Academic Radiology Departments
Society of Interventional Radiology
Society of Nuclear Medicine
Society of Radiologists in Ultrasound
Society of Skeletal Radiology
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U.S. Department of Labor
U.S. Department of Transportation
U.S. Environmental Protection Agency
U.S. Navy
U.S. Nuclear Regulatory Commission
U.S. Public Health Service
Utility Workers Union of America

NCRP has found its relationships with these organizations to be extremely valuable to continued progress in its program.

Another aspect of the cooperative efforts of NCRP relates to the Special Liaison relationships established with various governmental organizations that have an interest in radiation protection and measurements. This liaison relationship provides: (1) an opportunity for participating organizations to designate an individual to provide liaison between the organization and NCRP; (2) that the individual designated will receive copies of draft NCRP reports (at the time that these are submitted to the members of the Council) with an invitation to comment, but not vote; and (3) that new NCRP efforts might be discussed with liaison individuals as appropriate, so that they might have an opportunity to make suggestions on new studies and related matters. The following organizations participate in the Special Liaison Program:

Australian Radiation Laboratory
Bundesamt fur Strahlenschutz (Germany)
Canadian Association of Medical Radiation Technologists
Canadian Nuclear Safety Commission
Central Laboratory for Radiological Protection (Poland)
China Institute for Radiation Protection
Commissariat a l'Energie Atomique (France)

Commonwealth Scientific Instrumentation Research Organization (Australia)
European Commission
Heads of the European Radiological Protection Competent Authorities
Health Council of the Netherlands
Health Protection Agency
International Commission on Non-Ionizing Radiation Protection
International Commission on Radiation Units and Measurements
International Commission on Radiological Protection
International Radiation Protection Association
Japanese Nuclear Safety Commission
Japan Radiation Council
Korea Institute of Nuclear Safety
Russian Scientific Commission on Radiation Protection
South African Forum for Radiation Protection
World Association of Nuclear Operators
World Health Organization, Radiation and Environmental Health

NCRP values highly the participation of these organizations in the Special Liaison Program.

The Council also benefits significantly from the relationships established pursuant to the Corporate Sponsor's Program. The program facilitates the interchange of information and ideas and corporate sponsors provide valuable fiscal support for the Council's program. This developing program currently includes the following Corporate Sponsors:

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NCRP Reports

No.	Title
8	*Control and Removal of Radioactive Contamination in Laboratories* (1951)
22	*Maximum Permissible Body Burdens and Maximum Permissible Concentrations of Radionuclides in Air and in Water for Occupational Exposure* (1959) [includes Addendum 1 issued in August 1963]
25	*Measurement of Absorbed Dose of Neutrons, and of Mixtures of Neutrons and Gamma Rays* (1961)
27	*Stopping Powers for Use with Cavity Chambers* (1961)
30	*Safe Handling of Radioactive Materials* (1964)
32	*Radiation Protection in Educational Institutions* (1966)
35	*Dental X-Ray Protection* (1970)
36	*Radiation Protection in Veterinary Medicine* (1970)
37	*Precautions in the Management of Patients Who Have Received Therapeutic Amounts of Radionuclides* (1970)
38	*Protection Against Neutron Radiation* (1971)
40	*Protection Against Radiation from Brachytherapy Sources* (1972)
41	*Specification of Gamma-Ray Brachytherapy Sources* (1974)
42	*Radiological Factors Affecting Decision-Making in a Nuclear Attack* (1974)
44	*Krypton-85 in the Atmosphere—Accumulation, Biological Significance, and Control Technology* (1975)
46	*Alpha-Emitting Particles in Lungs* (1975)
47	*Tritium Measurement Techniques* (1976)
49	*Structural Shielding Design and Evaluation for Medical Use of X Rays and Gamma Rays of Energies Up to 10 MeV* (1976)

50 *Environmental Radiation Measurements* (1976)
52 *Cesium-137 from the Environment to Man: Metabolism and Dose* (1977)
54 *Medical Radiation Exposure of Pregnant and Potentially Pregnant Women* (1977)
55 *Protection of the Thyroid Gland in the Event of Releases of Radioiodine* (1977)
57 *Instrumentation and Monitoring Methods for Radiation Protection* (1978)
58 *A Handbook of Radioactivity Measurements Procedures*, 2nd ed. (1985)
60 *Physical, Chemical, and Biological Properties of Radiocerium Relevant to Radiation Protection Guidelines* (1978)
61 *Radiation Safety Training Criteria for Industrial Radiography* (1978)
62 *Tritium in the Environment* (1979)
63 *Tritium and Other Radionuclide Labeled Organic Compounds Incorporated in Genetic Material* (1979)
64 *Influence of Dose and Its Distribution in Time on Dose-Response Relationships for Low-LET Radiations* (1980)
65 *Management of Persons Accidentally Contaminated with Radionuclides* (1980)
67 *Radiofrequency Electromagnetic Fields—Properties, Quantities and Units, Biophysical Interaction, and Measurements* (1981)
68 *Radiation Protection in Pediatric Radiology* (1981)
69 *Dosimetry of X-Ray and Gamma-Ray Beams for Radiation Therapy in the Energy Range 10 keV to 50 MeV* (1981)
70 *Nuclear Medicine—Factors Influencing the Choice and Use of Radionuclides in Diagnosis and Therapy* (1982)
72 *Radiation Protection and Measurement for Low-Voltage Neutron Generators* (1983)
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76 *Radiological Assessment: Predicting the Transport, Bioaccumulation, and Uptake by Man of Radionuclides Released to the Environment* (1984)
77 *Exposures from the Uranium Series with Emphasis on Radon and Its Daughters* (1984)
78 *Evaluation of Occupational and Environmental Exposures to Radon and Radon Daughters in the United States* (1984)
79 *Neutron Contamination from Medical Electron Accelerators* (1984)
80 *Induction of Thyroid Cancer by Ionizing Radiation* (1985)
81 *Carbon-14 in the Environment* (1985)
82 *SI Units in Radiation Protection and Measurements* (1985)
83 *The Experimental Basis for Absorbed-Dose Calculations in Medical Uses of Radionuclides* (1985)

84 *General Concepts for the Dosimetry of Internally Deposited Radionuclides* (1985)
86 *Biological Effects and Exposure Criteria for Radiofrequency Electromagnetic Fields* (1986)
87 *Use of Bioassay Procedures for Assessment of Internal Radionuclide Deposition* (1987)
88 *Radiation Alarms and Access Control Systems* (1986)
89 *Genetic Effects from Internally Deposited Radionuclides* (1987)
90 *Neptunium: Radiation Protection Guidelines* (1988)
92 *Public Radiation Exposure from Nuclear Power Generation in the United States* (1987)
93 *Ionizing Radiation Exposure of the Population of the United States* (1987)
94 *Exposure of the Population in the United States and Canada from Natural Background Radiation* (1987)
95 *Radiation Exposure of the U.S. Population from Consumer Products and Miscellaneous Sources* (1987)
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99 *Quality Assurance for Diagnostic Imaging* (1988)
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102 *Medical X-Ray, Electron Beam and Gamma-Ray Protection for Energies Up to 50 MeV (Equipment Design, Performance and Use)* (1989)
103 *Control of Radon in Houses* (1989)
104 *The Relative Biological Effectiveness of Radiations of Different Quality* (1990)
105 *Radiation Protection for Medical and Allied Health Personnel* (1989)
106 *Limit for Exposure to "Hot Particles" on the Skin* (1989)
107 *Implementation of the Principle of As Low As Reasonably Achievable (ALARA) for Medical and Dental Personnel* (1990)
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112 *Calibration of Survey Instruments Used in Radiation Protection for the Assessment of Ionizing Radiation Fields and Radioactive Surface Contamination* (1991)
113 *Exposure Criteria for Medical Diagnostic Ultrasound: I. Criteria Based on Thermal Mechanisms* (1992)
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115 *Risk Estimates for Radiation Protection* (1993)
116 *Limitation of Exposure to Ionizing Radiation* (1993)
117 *Research Needs for Radiation Protection* (1993)
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119 *A Practical Guide to the Determination of Human Exposure to Radiofrequency Fields* (1993)
120 *Dose Control at Nuclear Power Plants* (1994)
121 *Principles and Application of Collective Dose in Radiation Protection* (1995)
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123 *Screening Models for Releases of Radionuclides to Atmosphere, Surface Water, and Ground* (1996)
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131 *Scientific Basis for Evaluating the Risks to Populations from Space Applications of Plutonium* (2001)
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137 *Fluence-Based and Microdosimetric Event-Based Methods for Radiation Protection in Space* (2001)
138 *Management of Terrorist Events Involving Radioactive Material* (2001)
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146 *Approaches to Risk Management in Remediation of Radioactively Contaminated Sites* (2004)

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150 *Extrapolation of Radiation-Induced Cancer Risks from Nonhuman Experimental Systems to Humans* (2005)
151 *Structural Shielding Design and Evaluation for Megavoltage X- and Gamma-Ray Radiotherapy Facilities* (2005)
152 *Performance Assessment of Near-Surface Facilities for Disposal of Low-Level Radioactive Waste* (2005)
153 *Information Needed to Make Radiation Protection Recommendations for Space Missions Beyond Low-Earth Orbit* (2006)
154 *Cesium-137 in the Environment: Radioecology and Approaches to Assessment and Management* (2006)
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168 *Radiation Dose Management for Fluoroscopically-Guided Interventional Medical Procedures* (2010)
169 *Design of Effective Radiological Effluent Monitoring and Environmental Surveillance Programs* (2010)
170 *Second Primary Cancers and Cardiovascular Disease After Radiation Therapy* (2011)
171 *Uncertainties in the Estimation of Radiation Risks and Probability of Disease Causation* (2012)
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NCRP Commentaries

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1	*Krypton-85 in the Atmosphere—With Specific Reference to the Public Health Significance of the Proposed Controlled Release at Three Mile Island* (1980)
4	*Guidelines for the Release of Waste Water from Nuclear Facilities with Special Reference to the Public Health Significance of the Proposed Release of Treated Waste Waters at Three Mile Island* (1987)
5	*Review of the Publication, Living Without Landfills* (1989)
6	*Radon Exposure of the U.S. Population—Status of the Problem (1991)*
7	*Misadministration of Radioactive Material in Medicine—Scientific Background* (1991)
8	*Uncertainty in NCRP Screening Models Relating to Atmospheric Transport, Deposition and Uptake by Humans* (1993)
9	*Considerations Regarding the Unintended Radiation Exposure of the Embryo, Fetus or Nursing Child* (1994)
10	*Advising the Public about Radiation Emergencies: A Document for Public Comment* (1994)
11	*Dose Limits for Individuals Who Receive Exposure from Radionuclide Therapy Patients* (1995)
12	*Radiation Exposure and High-Altitude Flight* (1995)
13	*An Introduction to Efficacy in Diagnostic Radiology and Nuclear Medicine (Justification of Medical Radiation Exposure)* (1995)
14	*A Guide for Uncertainty Analysis in Dose and Risk Assessments Related to Environmental Contamination* (1996)
15	*Evaluating the Reliability of Biokinetic and Dosimetric Models and Parameters Used to Assess Individual Doses for Risk Assessment Purposes* (1998)
16	*Screening of Humans for Security Purposes Using Ionizing Radiation Scanning Systems* (2003)
17	*Pulsed Fast Neutron Analysis System Used in Security Surveillance* (2003)
18	*Biological Effects of Modulated Radiofrequency Fields* (2003)
19	*Key Elements of Preparing Emergency Responders for Nuclear and Radiological Terrorism* (2005)
20	*Radiation Protection and Measurement Issues Related to Cargo Scanning with Accelerator-Produced High-Energy X Rays* (2007)
21	*Radiation Protection in the Application of Active Detection Technologies* (2011)
22	*Radiological Health Protection Issues Associated With Use of Active Detection Technology Systems for Detection of Radioactive Threat Materials* (2011)

Proceedings of the Annual Meeting

No.	Title
1	*Perceptions of Risk,* Proceedings of the Fifteenth Annual Meeting held on March 14-15, 1979 (including Taylor Lecture No. 3) (1980)

3	*Critical Issues in Setting Radiation Dose Limits*, Proceedings of the Seventeenth Annual Meeting held on April 8-9, 1981 (including Taylor Lecture No. 5) (1982)
4	*Radiation Protection and New Medical Diagnostic Approaches,* Proceedings of the Eighteenth Annual Meeting held on April 6-7, 1982 (including Taylor Lecture No. 6) (1983)
5	*Environmental Radioactivity,* Proceedings of the Nineteenth Annual Meeting held on April 6-7, 1983 (including Taylor Lecture No. 7) (1983)
6	*Some Issues Important in Developing Basic Radiation Protection Recommendations*, Proceedings of the Twentieth Annual Meeting held on April 4-5, 1984 (including Taylor Lecture No. 8) (1985)
7	*Radioactive Waste*, Proceedings of the Twenty-First Annual Meeting held on April 3-4, 1985 (including Taylor Lecture No. 9)(1986)
8	*Nonionizing Electromagnetic Radiations and Ultrasound,* Proceedings of the Twenty-Second Annual Meeting held on April 2-3, 1986 (including Taylor Lecture No. 10) (1988)
9	*New Dosimetry at Hiroshima and Nagasaki and Its Implications for Risk Estimates*, Proceedings of the Twenty-Third Annual Meeting held on April 8-9, 1987 (including Taylor Lecture No. 11) (1988)
10	*Radon*, Proceedings of the Twenty-Fourth Annual Meeting held on March 30-31, 1988 (including Taylor Lecture No. 12) (1989)
11	*Radiation Protection Today—The NCRP at Sixty Years*, Proceedings of the Twenty-Fifth Annual Meeting held on April 5-6, 1989 (including Taylor Lecture No. 13) (1990)
12	*Health and Ecological Implications of Radioactively Contaminated Environments*, Proceedings of the Twenty-Sixth Annual Meeting held on April 4-5, 1990 (including Taylor Lecture No. 14) (1991)
13	*Genes, Cancer and Radiation Protection,* Proceedings of the Twenty-Seventh Annual Meeting held on April 3-4, 1991 (including Taylor Lecture No. 15) (1992)
14	*Radiation Protection in Medicine,* Proceedings of the Twenty-Eighth Annual Meeting held on April 1-2, 1992 (including Taylor Lecture No. 16) (1993)
15	*Radiation Science and Societal Decision Making,* Proceedings of the Twenty-Ninth Annual Meeting held on April 7-8, 1993 (including Taylor Lecture No. 17) (1994)
16	*Extremely-Low-Frequency Electromagnetic Fields: Issues in Biological Effects and Public Health*, Proceedings of the Thirtieth Annual Meeting held on April 6-7, 1994 (not published).
17	*Environmental Dose Reconstruction and Risk Implications,* Proceedings of the Thirty-First Annual Meeting held on April 12-13, 1995 (including Taylor Lecture No. 19) (1996)
18	*Implications of New Data on Radiation Cancer Risk*, Proceedings of the Thirty-Second Annual Meeting held on April 3-4, 1996 (including Taylor Lecture No. 20) (1997)
19	*The Effects of Pre- and Postconception Exposure to Radiation*, Proceedings of the Thirty-Third Annual Meeting held on April 2-3, 1997, Teratology **59**, 181–317 (1999)

20 *Cosmic Radiation Exposure of Airline Crews, Passengers and Astronauts*, Proceedings of the Thirty-Fourth Annual Meeting held on April 1-2, 1998, Health Phys. **79**, 466–613 (2000)
21 *Radiation Protection in Medicine: Contemporary Issues*, Proceedings of the Thirty-Fifth Annual Meeting held on April 7-8, 1999 (including Taylor Lecture No. 23) (1999)
22 *Ionizing Radiation Science and Protection in the 21st Century*, Proceedings of the Thirty-Sixth Annual Meeting held on April 5-6, 2000, Health Phys. **80**, 317–402 (2001)
23 *Fallout from Atmospheric Nuclear Tests—Impact on Science and Society*, Proceedings of the Thirty-Seventh Annual Meeting held on April 4-5, 2001, Health Phys. **82**, 573–748 (2002)
24 *Where the New Biology Meets Epidemiology: Impact on Radiation Risk Estimates*, Proceedings of the Thirty-Eighth Annual Meeting held on April 10-11, 2002, Health Phys. **85**, 1–108 (2003)
25 *Radiation Protection at the Beginning of the 21st Century–A Look Forward*, Proceedings of the Thirty-Ninth Annual Meeting held on April 9–10, 2003, Health Phys. **87**, 237–319 (2004)
26 *Advances in Consequence Management for Radiological Terrorism Events*, Proceedings of the Fortieth Annual Meeting held on April 14–15, 2004, Health Phys. **89**, 415–588 (2005)
27 *Managing the Disposition of Low-Activity Radioactive Materials*, Proceedings of the Forty-First Annual Meeting held on March 30–31, 2005, Health Phys. **91**, 413–536 (2006)
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29 *Advances in Radiation Protection in Medicine*, Proceedings of the Forty-Third Annual Meeting held on April 16-17, 2007, Health Phys. **95**, 461–686 (2008)
30 *Low Dose and Low Dose-Rate Radiation Effects and Models*, Proceedings of the Forty-Fourth Annual Meeting held on April 14–15, 2008, Health Phys. **97**, 373–541 (2009)
31 *Future of Nuclear Power Worldwide – Health, Safety, and Environment*, Proceedings of the Forty-Fifth Annual Meeting held on March 2–3, 2009, Health Phys. **100**(1), 2–112 (2011)
32 *Communication of Radiation Benefits and Risks in Decision Making*, Proceedings of the Forty-Sixth Annual Meeting held March 8–9, 2010, Health Phys. **101**(5), 497–629 (2011)

Lauriston S. Taylor Lectures

No. Title

1 *The Squares of the Natural Numbers in Radiation Protection by* Herbert M. Parker (1977)
2 *Why be Quantitative about Radiation Risk Estimates?* by Sir Edward Pochin (1978)
3 *Radiation Protection—Concepts and Trade Offs* by Hymer L. Friedell (1979) [available also in *Perceptions of Risk*, see above]
4 *From "Quantity of Radiation" and "Dose" to "Exposure" and "Absorbed Dose"—An Historical Review* by Harold O. Wyckoff (1980)

5 *How Well Can We Assess Genetic Risk? Not Very* by James F. Crow (1981) [available also in *Critical Issues in Setting Radiation Dose Limits*, see above]

6 *Ethics, Trade-offs and Medical Radiation* by Eugene L. Saenger (1982) [available also in *Radiation Protection and New Medical Diagnostic Approaches*, see above]

7 *The Human Environment—Past, Present and Future* by Merril Eisenbud (1983) [available also in *Environmental Radioactivity*, see above]

8 *Limitation and Assessment in Radiation Protection* by Harald H. Rossi (1984) [available also in *Some Issues Important in Developing Basic Radiation Protection Recommendations*, see above]

9 *Truth (and Beauty) in Radiation Measurement* by John H. Harley (1985) [available also in *Radioactive Waste*, see above]

10 *Biological Effects of Non-ionizing Radiations: Cellular Properties and Interactions* by Herman P. Schwan (1987) [available also in *Nonionizing Electromagnetic Radiations and Ultrasound*, see above]

11 *How to be Quantitative about Radiation Risk Estimates* by Seymour Jablon (1988) [available also in *New Dosimetry at Hiroshima and Nagasaki and its Implications for Risk Estimates*, see above]

12 *How Safe is Safe Enough?* by Bo Lindell (1988) [available also in *Radon*, see above]

13 *Radiobiology and Radiation Protection: The Past Century and Prospects for the Future* by Arthur C. Upton (1989) [available also in *Radiation Protection Today*, see above]

14 *Radiation Protection and the Internal Emitter Saga* by J. Newell Stannard (1990) [available also in *Health and Ecological Implications of Radioactively Contaminated Environments*, see above]

15 *When is a Dose Not a Dose?* by Victor P. Bond (1992) [available also in *Genes, Cancer and Radiation Protection*, see above]

16 *Dose and Risk in Diagnostic Radiology: How Big? How Little?* by Edward W. Webster (1992) [available also in *Radiation Protection in Medicine*, see above]

17 *Science, Radiation Protection and the NCRP* by Warren K. Sinclair (1993) [available also in *Radiation Science and Societal Decision Making,* see above]

18 *Mice, Myths and Men* by R.J. Michael Fry (1995)

19 *Certainty and Uncertainty in Radiation Research* by Albrecht M. Kellerer. Health Phys. **69**, 446–453 (1995)

20 *70 Years of Radiation Genetics: Fruit Flies, Mice and Humans* by Seymour Abrahamson. Health Phys. **71**, 624–633 (1996)

21 *Radionuclides in the Body: Meeting the Challenge* by William J. Bair. Health Phys. **73**, 423–432 (1997)

22 *From Chimney Sweeps to Astronauts: Cancer Risks in the Work Place* by Eric J. Hall. Health Phys. **75**, 357–366 (1998)

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24 *Administered Radioactivity: Unde Venimus Quoque Imus* by S. James Adelstein. Health Phys. **80**, 317–324 (2001)

25 *Assuring the Safety of Medical Diagnostic Ultrasound* by Wesley L. Nyborg. Health Phys. **82**, 578–587 (2002)

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28 *Radiation Protection in the Aftermath of a Terrorist Attack Involving Exposure to Ionizing Radiation* by Abel J. Gonzalez, Health Phys. **89**, 418–446 (2005)

29 *Nontargeted Effects of Radiation: Implications for Low Dose Exposures* by John B. Little, Health Phys. **91**, 416–426 (2006)

30 *Fifty Years of Scientific Research: The Importance of Scholarship and the Influence of Politics and Controversy* by Robert L. Brent, Health Phys. **93**, 348–379 (2007)

31 *The Quest for Therapeutic Actinide Chelators* by Patricia W. Durbin, Health Phys. **95**, 465–492 (2008)

32 *Yucca Mountain Radiation Standards, Dose/Risk Assessments, Thinking Outside the Box, Evaluations, and Recommendations* by Dade W. Moeller, Health Phys. **97**, 376–391 (2009)

33 *Radiation Epidemiology-the Golden Age and Future Challenges* by John D. Boice, Jr., Health Phys. **100**(1), 59–76 (2011)

34 *Radiation Protection and Public Policy in an Uncertain World* by Charles E. Land, Health Phys. **101**(5), 499–508 (2011)

Symposium Proceedings

No. Title

1 *The Control of Exposure of the Public to Ionizing Radiation in the Event of Accident or Attack*, Proceedings of a Symposium held April 27-29, 1981 (1982)

2 *Radioactive and Mixed Waste—Risk as a Basis for Waste Classification,* Proceedings of a Symposium held November 9, 1994 (1995)

3 *Acceptability of Risk from Radiation—Application to Human Space Flight,* Proceedings of a Symposium held May 29, 1996 (1997)

4 *21st Century Biodosimetry: Quantifying the Past and Predicting the Future*, Proceedings of a Symposium held February 22, 2001, Radiat. Prot. Dosim. **97**(1), (2001)

5 *National Conference on Dose Reduction in CT, with an Emphasis on Pediatric Patients*, Summary of a Symposium held November 6-7, 2002, Am. J. Roentgenol. **181**(2), 321–339 (2003)

NCRP Statements

No. Title

1 "Blood Counts, Statement of the National Committee on Radiation Protection," Radiology **63**, 428 (1954)

2	"Statements on Maximum Permissible Dose from Television Receivers and Maximum Permissible Dose to the Skin of the Whole Body," Am. J. Roentgenol., Radium Ther. and Nucl. Med. **84**, 152 (1960) and Radiology **75**, 122 (1960)
3	*X-Ray Protection Standards for Home Television Receivers, Interim Statement of the National Council on Radiation Protection and Measurements* (1968)
4	*Specification of Units of Natural Uranium and Natural Thorium, Statement of the National Council on Radiation Protection and Measurements* (1973)
5	*NCRP Statement on Dose Limit for Neutrons* (1980)
6	*Control of Air Emissions of Radionuclides* (1984)
7	*The Probability That a Particular Malignancy May Have Been Caused by a Specified Irradiation* (1992)
8	*The Application of ALARA for Occupational Exposures* (1999)
9	*Extension of the Skin Dose Limit for Hot Particles to Other External Sources of Skin Irradiation* (2001)
10	*Recent Applications of the NCRP Public Dose Limit Recommendation for Ionizing Radiation* (2004)

Other Documents

The following documents were published outside of the NCRP report, commentary and statement series:

Somatic Radiation Dose for the General Population, Report of the Ad Hoc Committee of the National Council on Radiation Protection and Measurements, 6 May 1959, Science **131** (3399), February 19, 482–486 (1960)

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Residential Radon Exposure and Lung Cancer Risk: Commentary on Cohen's County-Based Study, Health Phys. **87**(6), 656–658 (2004)